

LTE SERIES · STUDY GUIDE

LTE and LTE-Advanced, explained

A visual guide for telecom engineers: architecture, air interface, procedures, VoLTE and LTE-Advanced.

Telecom Knowledge Hub

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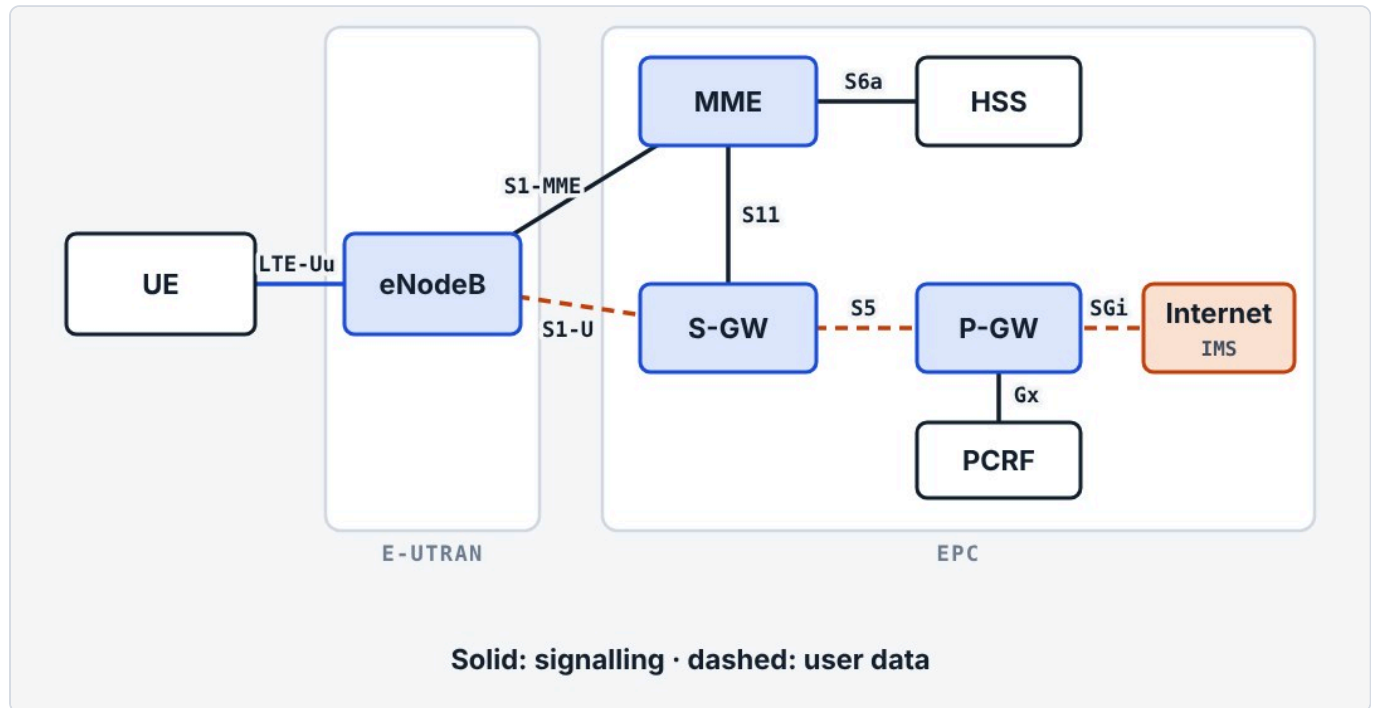
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EPS Architecture

The nodes of an LTE network and what each one does: eNodeB, MME, S-GW, P-GW, HSS and PCRF.

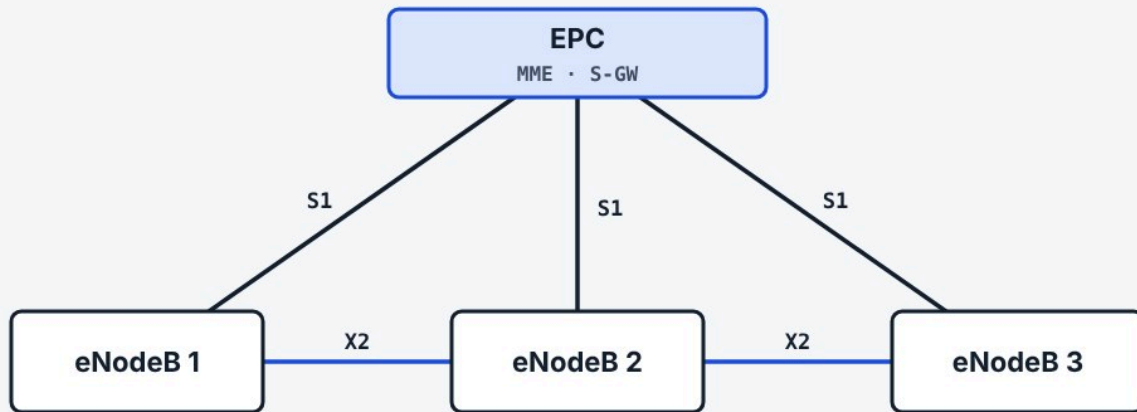
1.1 The big picture



An LTE network has two parts. The radio network, E-UTRAN, is made of eNodeBs only. The core network, EPC, handles mobility, sessions and the connection to the internet. Together they form the Evolved Packet System, EPS. Everything is packet switched: there is no circuit-switched domain.

TS 23.401 §4.2 · TS 36.300 §4

1.2 E-UTRAN: only eNodeBs

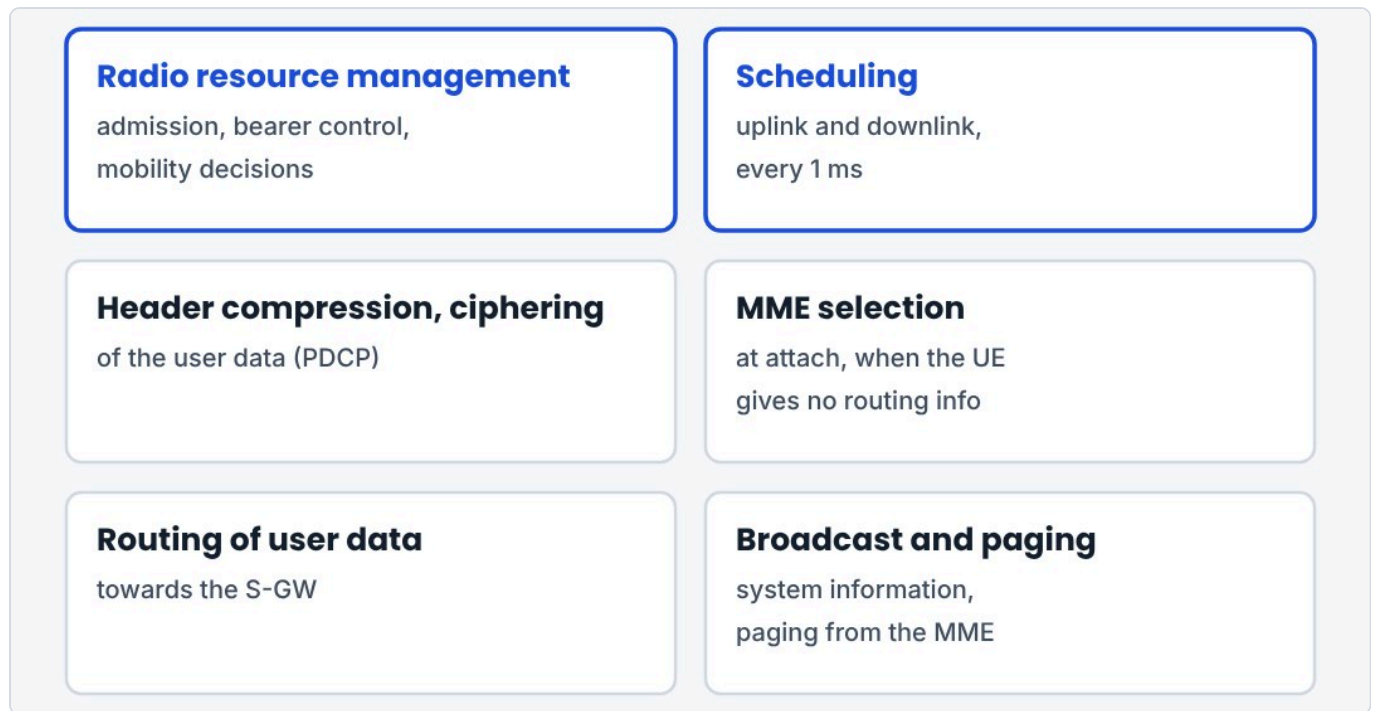


A flat architecture: no RNC between the base station and the core

In 3G the radio network had two levels: the NodeB and the RNC above it. LTE removes the controller. Each eNodeB does the radio control itself and connects straight to the core over S1. Neighbouring eNodeBs talk to each other over X2, mainly for handover and interference coordination.

TS 36.300 §4 · TS 36.401

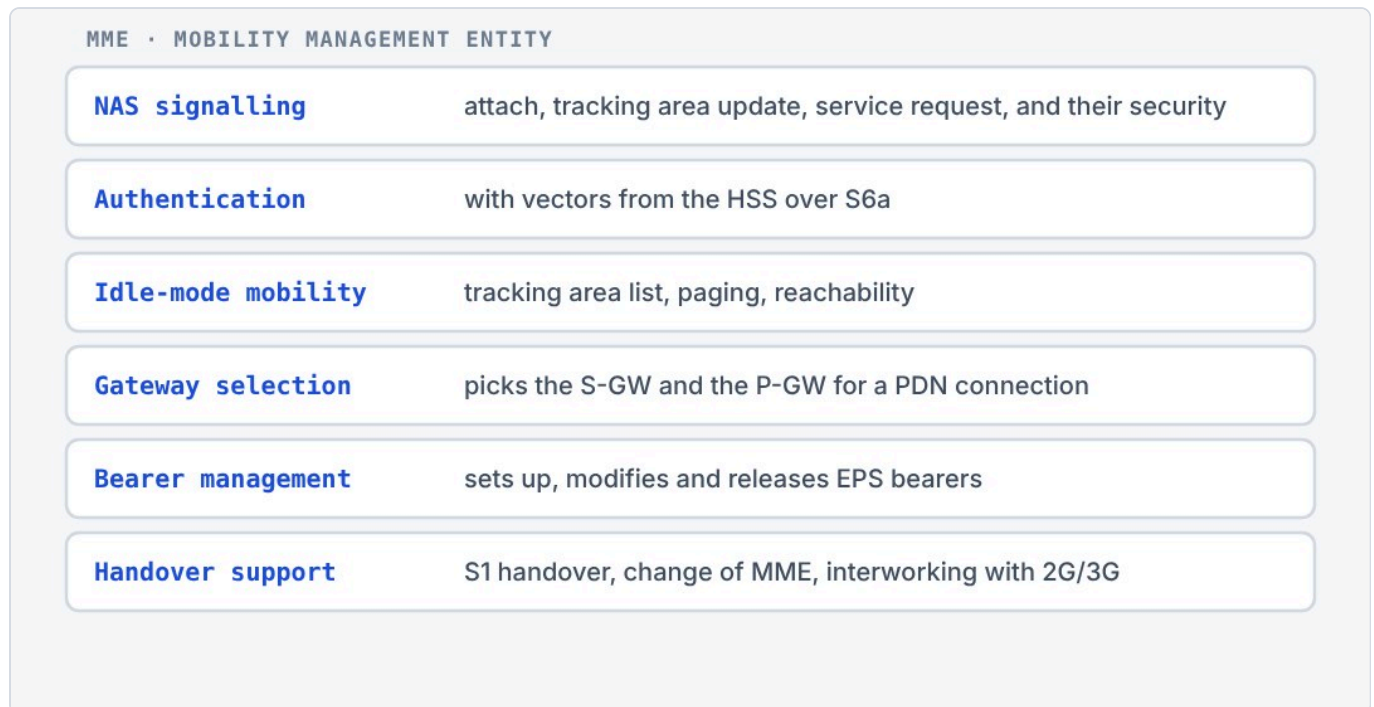
1.3 What the eNodeB does



The eNodeB owns everything on the radio side. It schedules every UE each millisecond, decides handovers, compresses and ciphers the user data, and picks an MME when a UE attaches. It also broadcasts system information and sends paging messages on request of the MME.

TS 36.300 §4.1

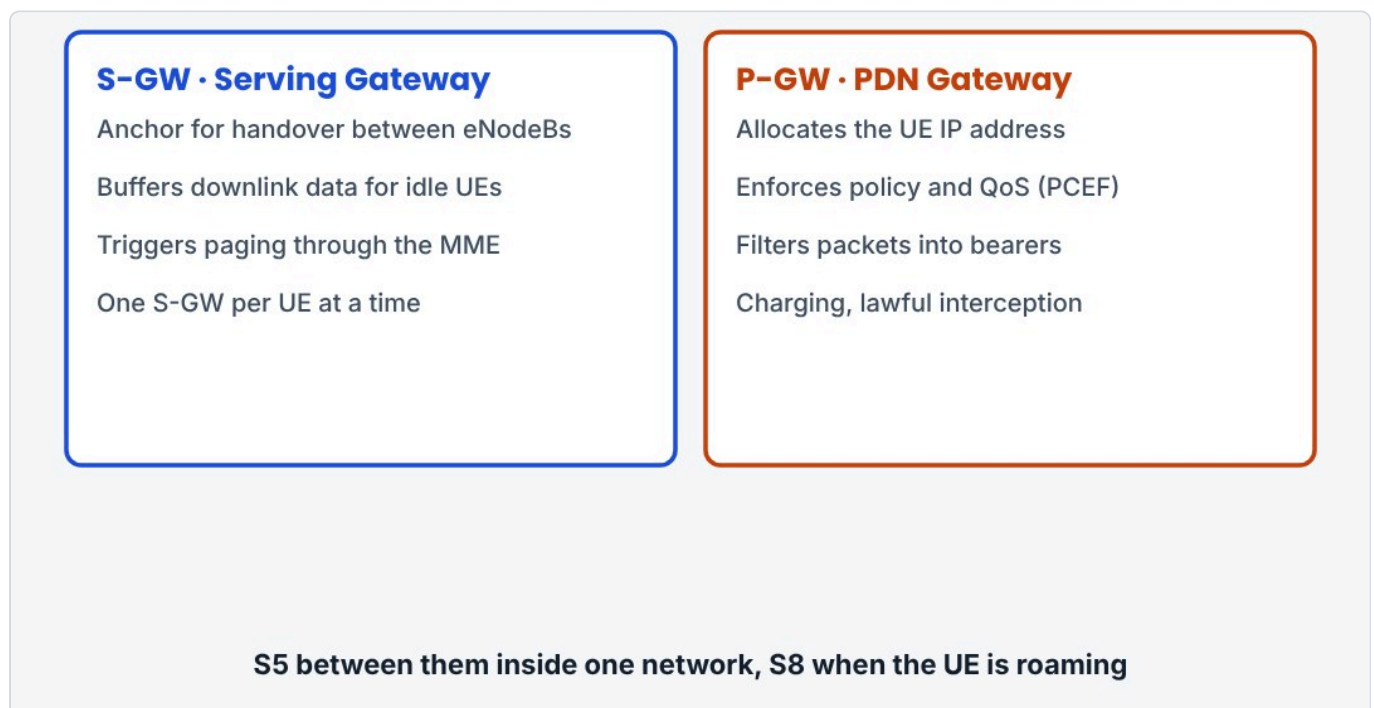
1.4 MME: the control node



The MME is the signalling centre of the EPC. It never touches user data. It authenticates the UE with data from the HSS, runs NAS security, tracks idle UEs at tracking-area level and pages them. It also chooses the S-GW and P-GW and manages the bearers.

TS 23.401 §4.4.2

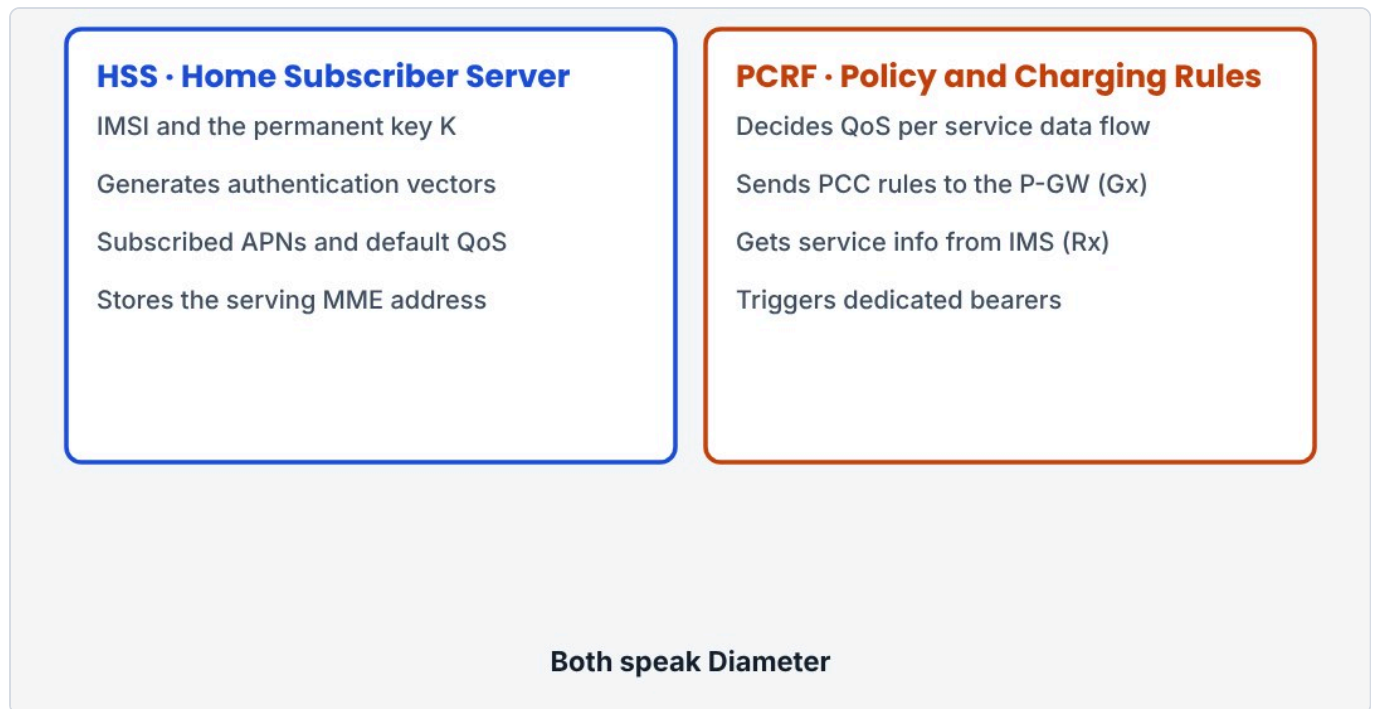
1.5 S-GW and P-GW: the user plane



Two gateways carry the user data. The S-GW is the local anchor: when the UE changes eNodeB the data path is switched here, and it buffers downlink data for idle UEs. The P-GW is the edge to the outside. It gives the UE its IP address and enforces policy and charging.

TS 23.401 §4.4.3

1.6 HSS and PCRF



Two more nodes complete the picture. The HSS is the subscriber database: identities, security keys, subscribed APNs and QoS, and the MME that currently serves the UE. The PCRF decides the policy: which service gets which QoS and how it is charged.

TS 23.401 §4.4 · TS 23.203

1.7 Identities you see in logs

IDENTITY	MADE OF	USED FOR
IMSI	MCC + MNC + MSIN, up to 15 digits	permanent subscriber identity
GUTI	GUMMEI + M-TMSI	temporary identity from the MME
S-TMSI	MME code + M-TMSI	paging, service request
TAI	MCC + MNC + TAC	tracking area
ECGI	PLMN + 28-bit cell identity	one cell, globally
C-RNTI	16 bits	the UE inside one cell
IMEI	15 digits	the device

A handful of identities appear in every LTE trace. The IMSI is permanent and is sent as rarely as possible. The GUTI is the temporary identity given by the MME. Its short form, the S-TMSI, is used for paging. On the radio side the UE is known by its C-RNTI.

TS 23.003 §2, §19

1.8 Three things to remember

1

Flat radio network

Only eNodeBs, linked by X2 and connected to the core by S1.

2

Control and data are split

MME for signalling. S-GW and P-GW for user data.

3

All IP, packet switched only

Voice needs IMS, or a fallback to 2G/3G.

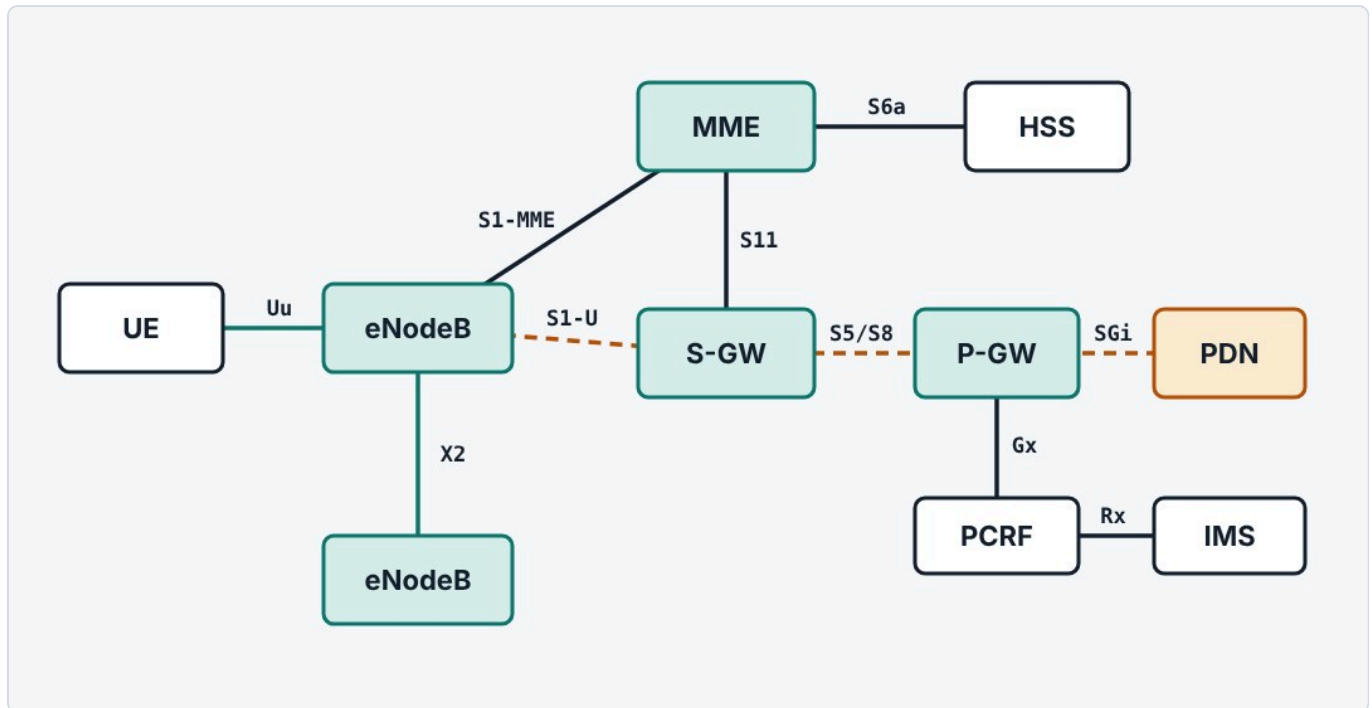
E-UTRAN is flat: the eNodeB does all radio control. The MME handles signalling only, the gateways handle data only. HSS and PCRF supply the subscription and the policy.

TS 23.401 §4

Interfaces and Protocol Stacks

Which protocol runs on which interface, the control and user plane stacks, and the channels of the radio interface.

2.1 The interface map



Every link between two LTE nodes has a name. Uu is the radio. S1 connects the eNodeB to the core, split into S1-MME for signalling and S1-U for data. X2 connects eNodeBs. S11, S5 and S6a are inside the core, and SGi leads to the packet data network.

TS 23.401 §4.2 · TS 36.300 §4.3

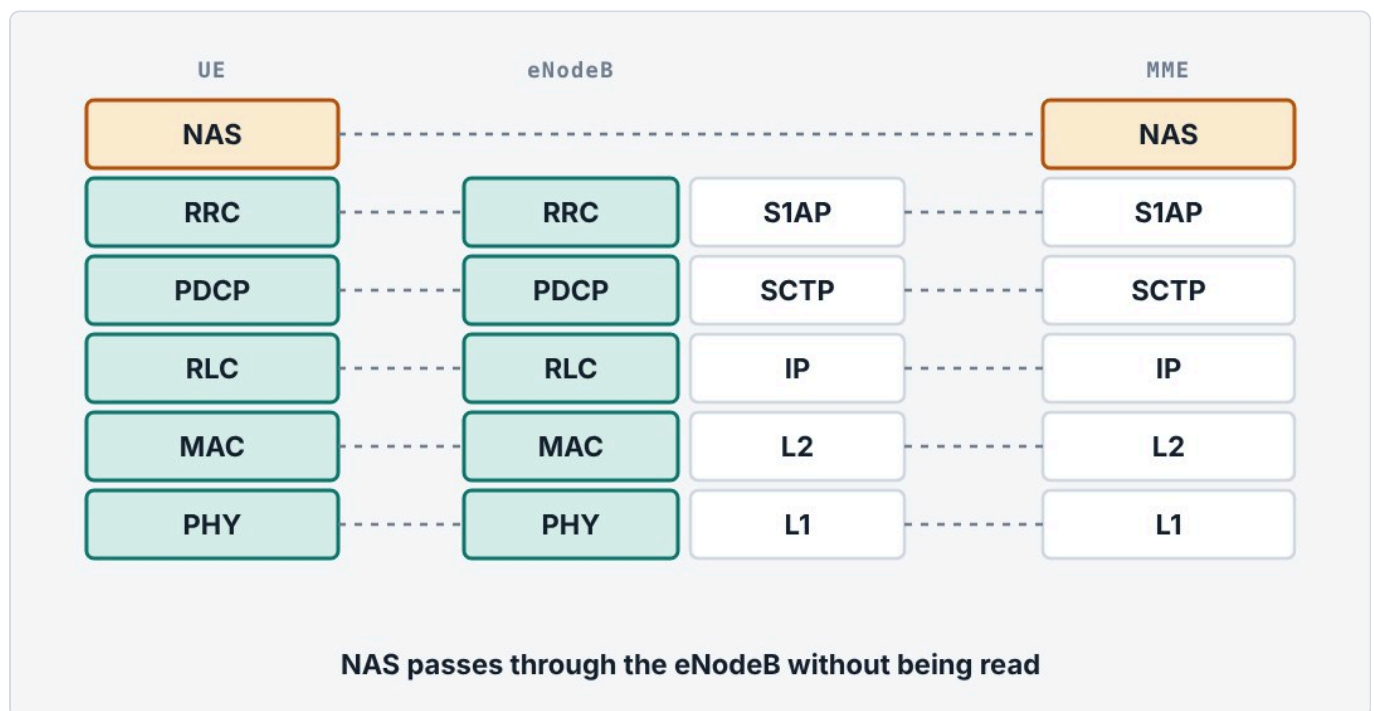
2.2 Interfaces and their protocols

INTERFACE	BETWEEN	PROTOCOL
S1-MME	eNodeB – MME	S1AP over SCTP
S1-U	eNodeB – S-GW	GTP-U over UDP
X2	eNodeB – eNodeB	X2AP over SCTP, GTP-U
S11	MME – S-GW	GTPv2-C over UDP
S5 / S8	S-GW – P-GW	GTPv2-C and GTP-U
S6a	MME – HSS	Diameter over SCTP
Gx	P-GW – PCRF	Diameter
SGi	P-GW – PDN	plain IP

Signalling interfaces towards the eNodeB use application protocols over SCTP: S1AP and X2AP. Inside the core, GTP-C carries the session signalling and Diameter talks to the databases. User data always travels in GTP-U tunnels over UDP.

TS 36.413 · TS 36.423 · TS 29.274 · TS 29.281 · TS 29.272

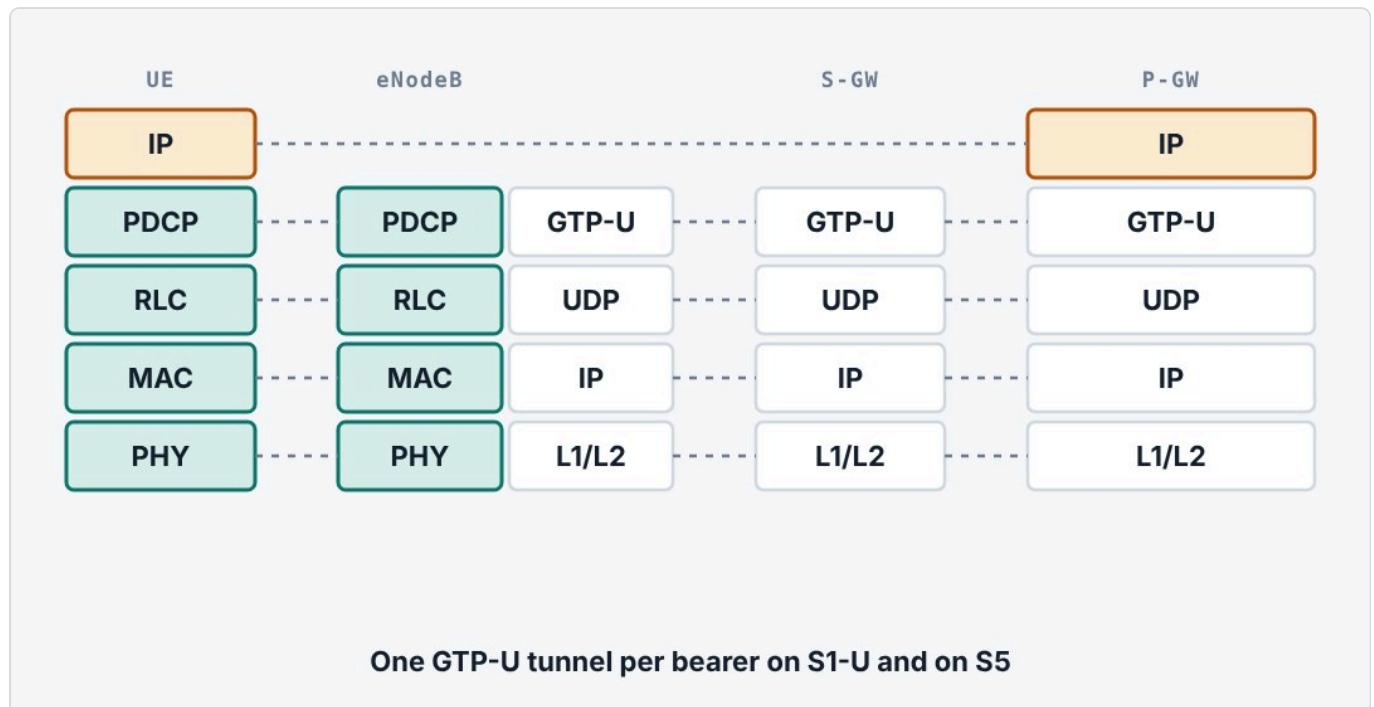
2.3 Control plane stack



NAS messages go from the UE to the MME. The eNodeB does not read them: it carries them inside RRC on the radio side and inside S1AP on the core side. RRC itself ends in the eNodeB. Below RRC the radio uses PDPC, RLC, MAC and the physical layer.

TS 36.300 §4.3.2

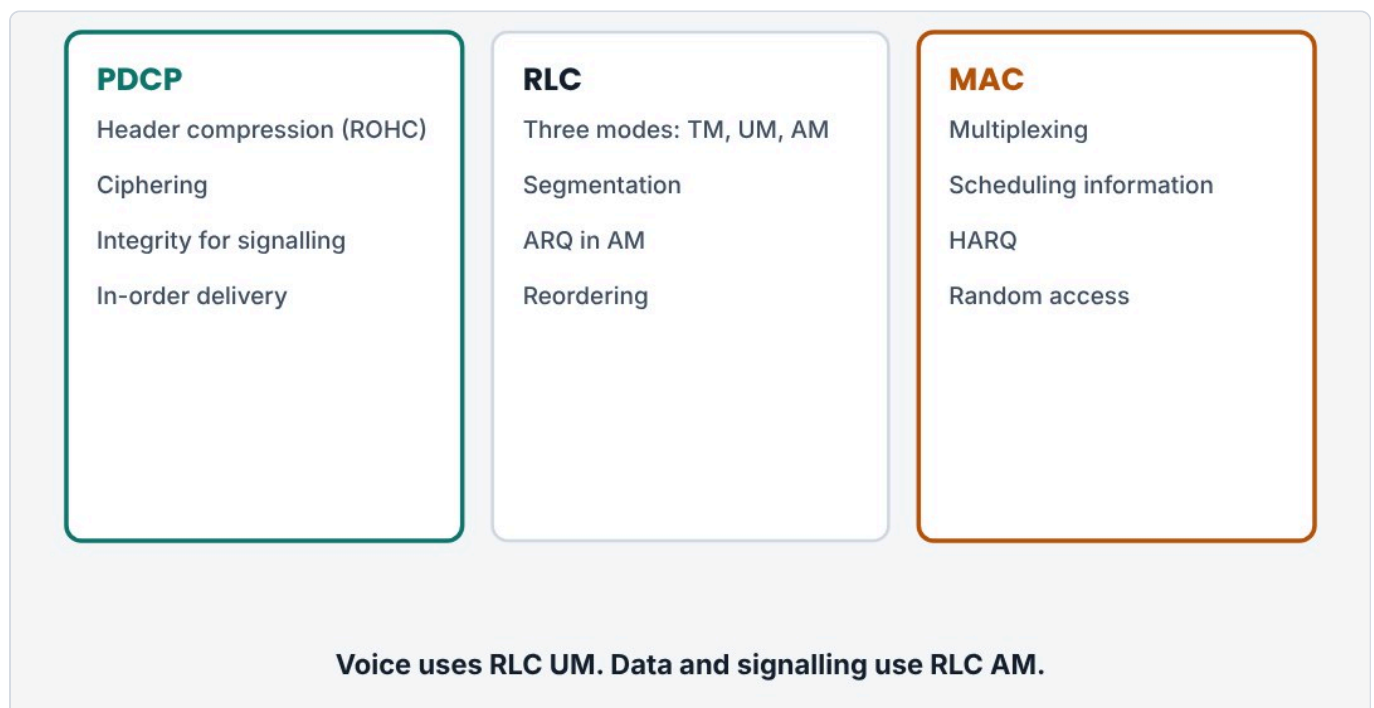
2.4 User plane stack



The IP packet of the user travels from the UE to the P-GW. On the radio it is carried by PDCP, RLC and MAC. From the eNodeB onwards it sits inside a GTP-U tunnel, first to the S-GW and then to the P-GW. The P-GW removes the tunnel and sends the packet to the internet.

TS 36.300 §4.3.1 · TS 29.281

2.5 Layer 2 in three sublayers



PDCP compresses IP headers and ciphers. RLC cuts the data to the size the scheduler grants and, in acknowledged mode, repeats what was lost. MAC multiplexes the logical channels, runs HARQ and handles random access.

TS 36.323 · TS 36.322 · TS 36.321

2.6 Logical, transport and physical channels

LOGICAL	TRANSPORT	PHYSICAL	CONTENT
BCCH	BCH	PBCH	MIB
BCCH	DL-SCH	PDSCH	SIBs
PCCH	PCH	PDSCH	paging
CCCH, DCCH, DTCH	DL-SCH	PDSCH	signalling + data, down
CCCH, DCCH, DTCH	UL-SCH	PUSCH	signalling + data, up
—	RACH	PRACH	random access preamble

RLC offers logical channels that say what is sent. MAC maps them to transport channels that say how it is sent. The physical layer maps those to physical channels. Almost everything ends up on the shared channels PDSCH and PUSCH.

TS 36.300 §5.3, §6.1 · TS 36.321 §4.5

2.7 Radio bearers

BEARER	CHANNEL	CARRIES
SRB0	CCCH, RLC TM	RRC connection request and setup
SRB1	DCCH, RLC AM	RRC messages, NAS until SRB2 exists
SRB2	DCCH, RLC AM	NAS messages, lower priority, after security
DRB	DTCH, RLC AM or UM	user data, one per EPS bearer

Up to 8 data radio bearers per UE in Release 8

A radio bearer is a PDCP and RLC entity pair between UE and eNodeB. Signalling radio bearers carry RRC and NAS. Data radio bearers carry user data, one per EPS bearer. SRB0 needs no setup. SRB1 comes with the RRC connection and SRB2 after security is active.

TS 36.331 §4.2.2

2.8 Three things to remember

1

S1 is split

S1-MME for signalling (S1AP), S1-U for data (GTP-U).

2

NAS is end to end

UE to MME. The eNodeB only transports it.

3

Same Layer 2 for both planes

PDCP, RLC and MAC serve signalling and data bearers.

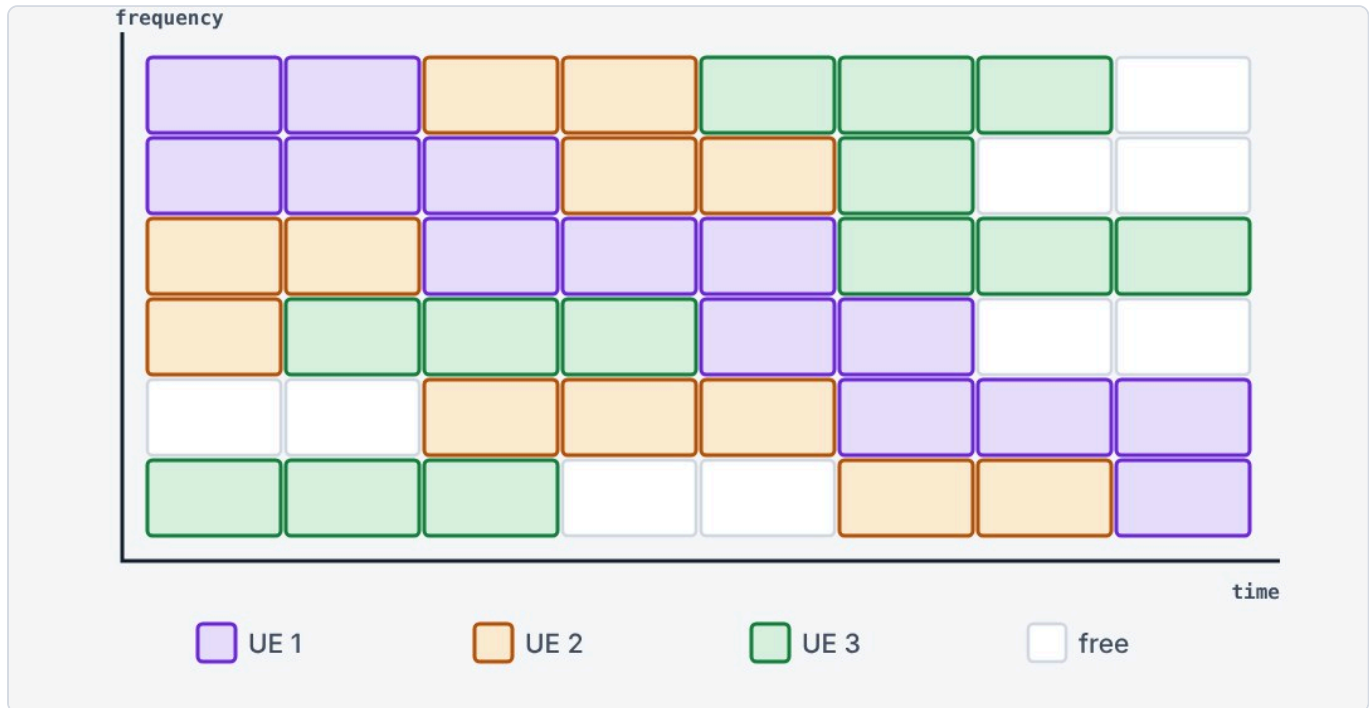
S1AP and X2AP run over SCTP, user data runs in GTP-U. NAS goes end to end between UE and MME. The radio stack is PDCP, RLC, MAC, PHY in both planes.

TS 36.300 §4.3

The LTE Air Interface

OFDMA in the downlink, SC-FDMA in the uplink, the cyclic prefix, modulation and the six channel bandwidths.

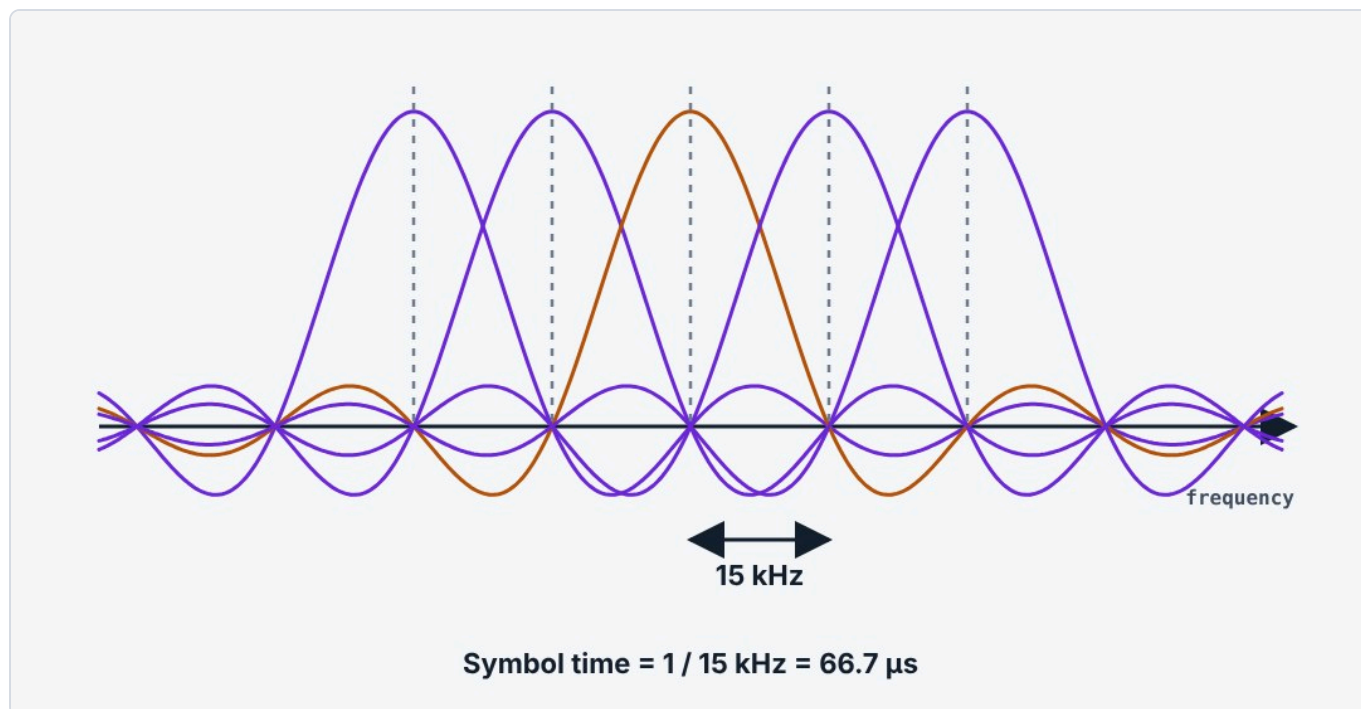
3.1 Downlink: OFDMA



The LTE downlink splits the carrier into subcarriers 15 kHz apart. Different users get different groups of subcarriers, and the allocation changes every millisecond. This is OFDMA: OFDM used as a multiple access scheme.

TS 36.211 §6 · TS 36.300 §5.1

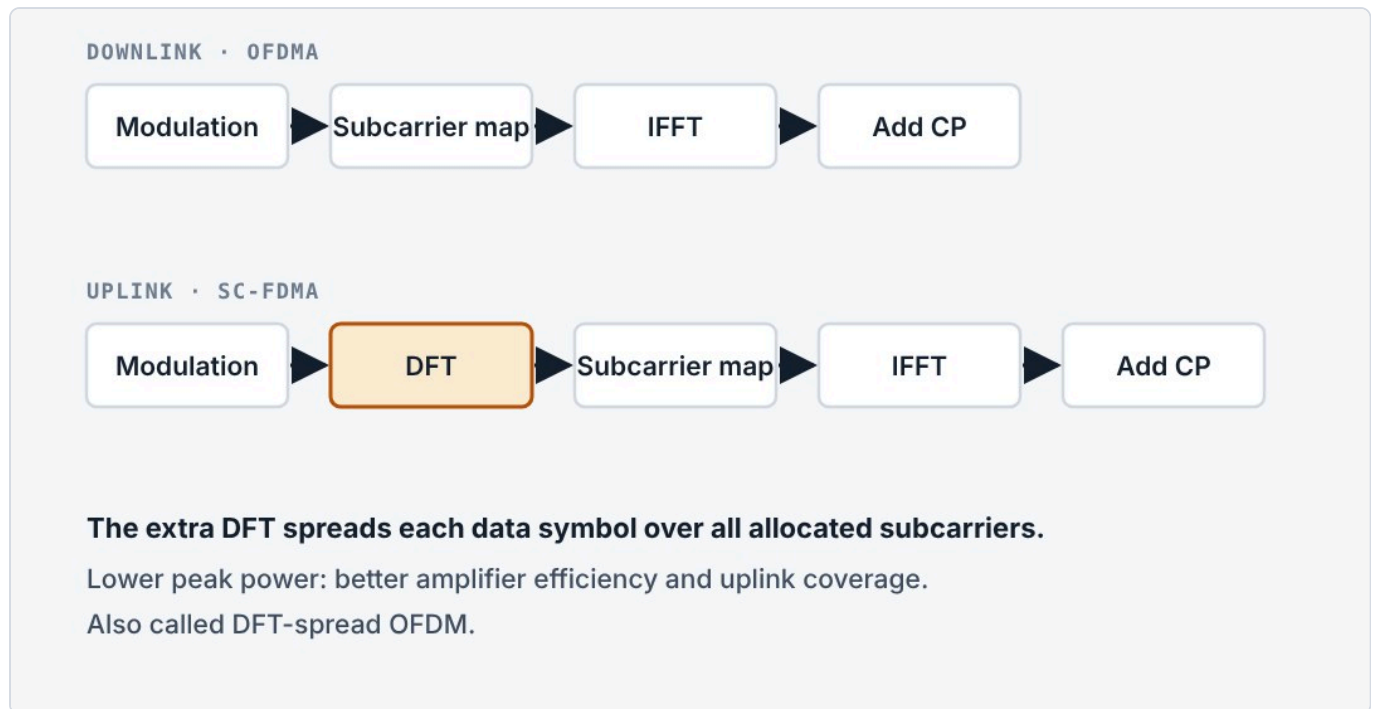
3.2 Why the subcarriers do not disturb each other



Each subcarrier has a sinc-shaped spectrum. With a spacing of exactly one over the symbol time, every subcarrier peaks where all the others are zero. The subcarriers overlap, which saves spectrum, yet the receiver can separate them with an FFT.

TS 36.211 §6.12

3.3 Uplink: SC-FDMA



A plain OFDM signal has high peaks, which a handset amplifier handles badly. The uplink therefore adds a DFT before the subcarrier mapping. The result behaves like a single carrier signal with lower peak to average power. The cost: in Release 8 the allocated resource blocks must be contiguous.

TS 36.211 §5.3 · TS 36.300 §5.2

3.4 The cyclic prefix



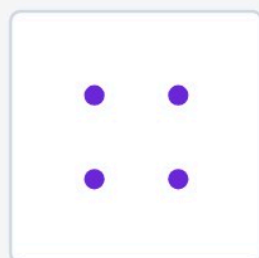
PREFIX	LENGTH	SYMBOLS PER SLOT	ECHO PATH
Normal	4.69 μs (first: 5.21)	7	1.4 km
Extended	16.67 μs	6	5 km

Overhead of the normal prefix: about 6.7 %

Each symbol starts with a copy of its own end. Echoes from the previous symbol fall into this prefix, which the receiver discards. The normal prefix lasts 4.7 microseconds and fits seven symbols into a slot. The extended prefix is for very large cells and broadcast.

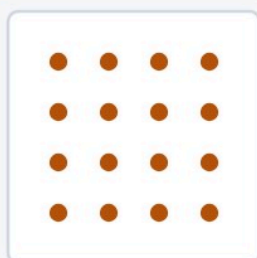
TS 36.211 §5.6, §6.12, Table 6.12-1

3.5 Modulation



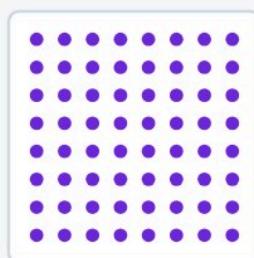
QPSK

2 bits per symbol



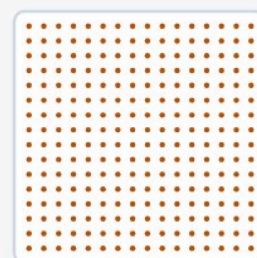
16QAM

4 bits per symbol



64QAM

6 bits per symbol



256QAM

8 bits per symbol

More bits per symbol need a better signal to noise ratio

256QAM: downlink in Release 12, uplink in Release 14

Each resource element carries one modulation symbol. QPSK carries 2 bits and survives poor signal. 16QAM carries 4 bits, 64QAM carries 6. Release 12 added 256QAM in the downlink with 8 bits. The eNodeB picks the modulation from the reported channel quality.

TS 36.211 §7.1 · TS 36.213 §7.1.7

3.6 Six channel bandwidths

CHANNEL	RESOURCE BLOCKS	SUBCARRIERS	FFT SIZE	SAMPLE RATE
1.4 MHz	6	72	128	1.92 MHz
3 MHz	15	180	256	3.84 MHz
5 MHz	25	300	512	7.68 MHz
10 MHz	50	600	1024	15.36 MHz
15 MHz	75	900	1536	23.04 MHz
20 MHz	100	1200	2048	30.72 MHz

LTE defines six channel bandwidths. The subcarrier spacing is always 15 kHz, so only the number of resource blocks changes. A 20 MHz carrier has 100 resource blocks, which is 1200 subcarriers and 18 MHz of occupied spectrum.

TS 36.101 §5.6, Table 5.6-1 · TS 36.104

3.7 Where 150 Mbit/s comes from

20 MHz DOWNLINK

Resource elements 1200 subcarriers × 14 symbols × 1000 per second = 16.8 million / s

× 6 bits (64QAM) 100.8 Mbit/s raw, one layer

– overhead and coding about 75 Mbit/s per layer

× 2 layers (2×2 MIMO) 150 Mbit/s: UE category 4

× 4 layers (4×4 MIMO) 300 Mbit/s: UE category 5

The uplink reaches 50 Mbit/s with 16QAM and one layer

The peak rate follows from the grid. A 20 MHz carrier has 1200 subcarriers and 14 symbols per millisecond. With 64QAM that is about 100 Mbit/s raw for one layer. Control channels, reference signals and coding take roughly a quarter. Two MIMO layers then give 150 Mbit/s.

TS 36.306 Table 4.1-1 · TS 36.213 §7.1.7

3.8 Three things to remember

1

OFDMA down, SC-FDMA up

Same grid. The uplink adds a DFT to lower the peak power.

2

The cyclic prefix absorbs echoes

4.7 μ s normal, 16.7 μ s extended.

3

1.4 to 20 MHz

6 to 100 resource blocks of 180 kHz each.

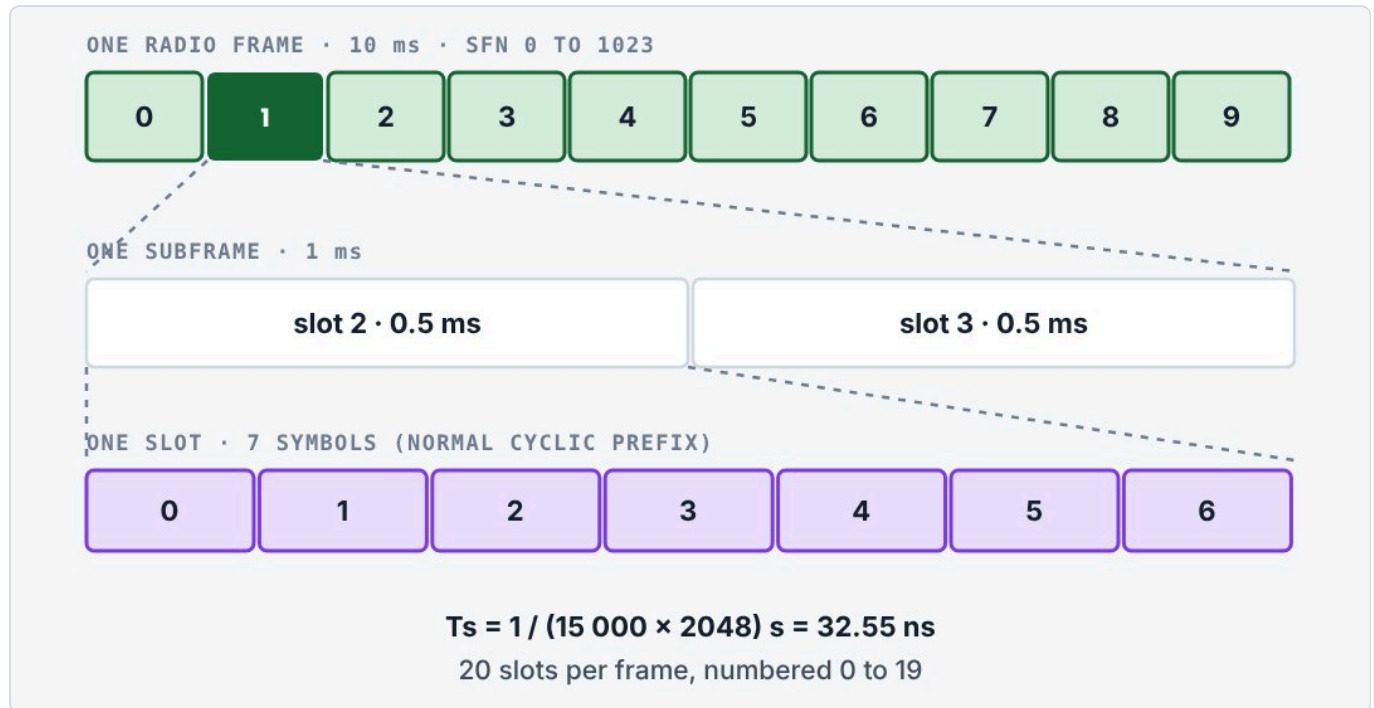
15 kHz subcarriers in both directions. OFDMA down, SC-FDMA up, because of the handset amplifier. From 6 to 100 resource blocks.

TS 36.211 · TS 36.101 §5.6

Frame Structure and Resource Grid

How LTE divides time and frequency: frames, subframes, slots, resource blocks, and the layout of a downlink subframe.

4.1 Frame structure type 1: FDD



A radio frame lasts 10 ms and holds ten subframes of 1 ms. Each subframe has two slots of 0.5 ms. In FDD all ten subframes are available for downlink on one carrier and for uplink on the paired carrier. The basic time unit T_s is $1 / 30.72 \text{ MHz}$.

TS 36.211 §4, §4.1

4.2 Frame structure type 2: TDD

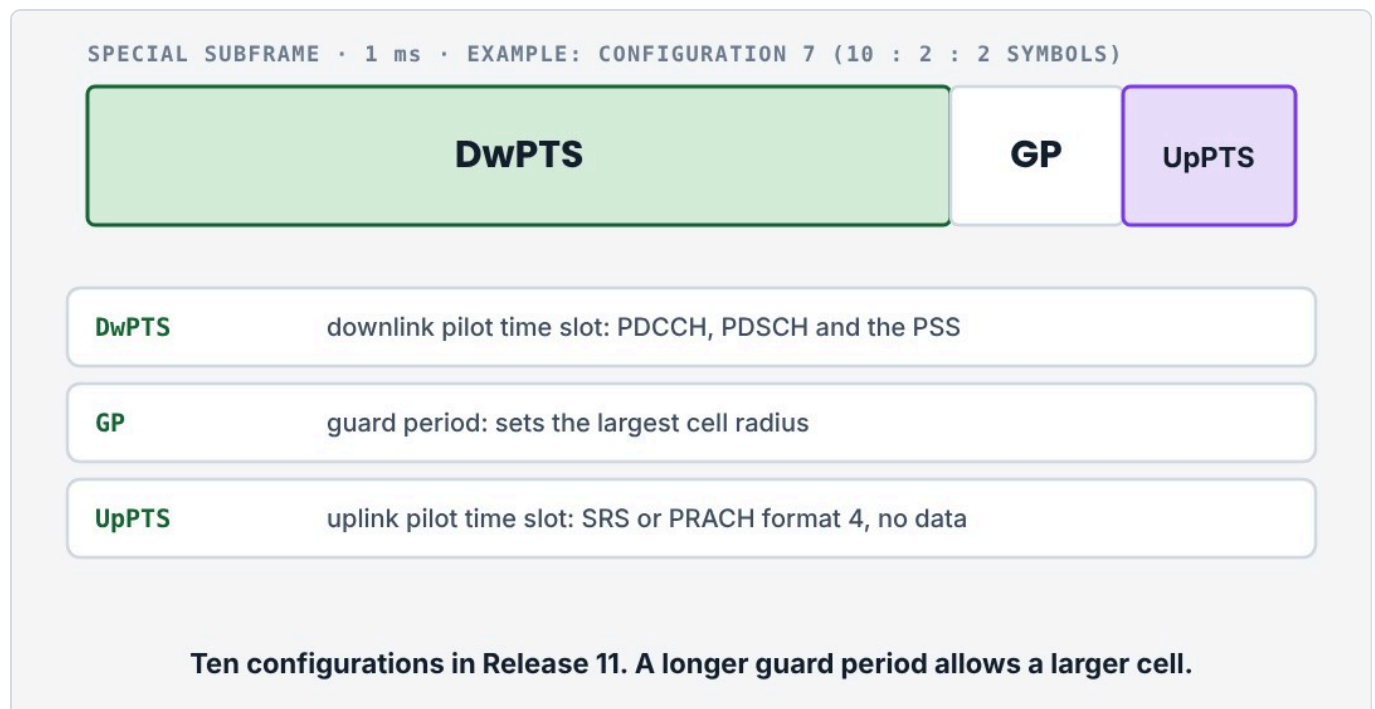
CONFIG	0	1	2	3	4	5	6	7	8	9
0	D	S	U	U	U	D	S	U	U	U
1	D	S	U	U	D	D	S	U	U	D
2	D	S	U	D	D	D	S	U	D	D
3	D	S	U	U	U	D	D	D	D	D
4	D	S	U	U	D	D	D	D	D	D
5	D	S	U	D	D	D	D	D	D	D
6	D	S	U	U	U	D	S	U	U	D

D downlink · U uplink · S special subframe
Configs 0, 1, 2 and 6 repeat every 5 ms. Configs 3, 4 and 5 every 10 ms.

In TDD one carrier serves both directions. The frame keeps its ten subframes, but each one is downlink, uplink or special. Seven configurations exist, from uplink-heavy to downlink-heavy. Subframes 0 and 5 are always downlink, and subframe 2 is always uplink.

TS 36.211 §4.2, Table 4.2-2

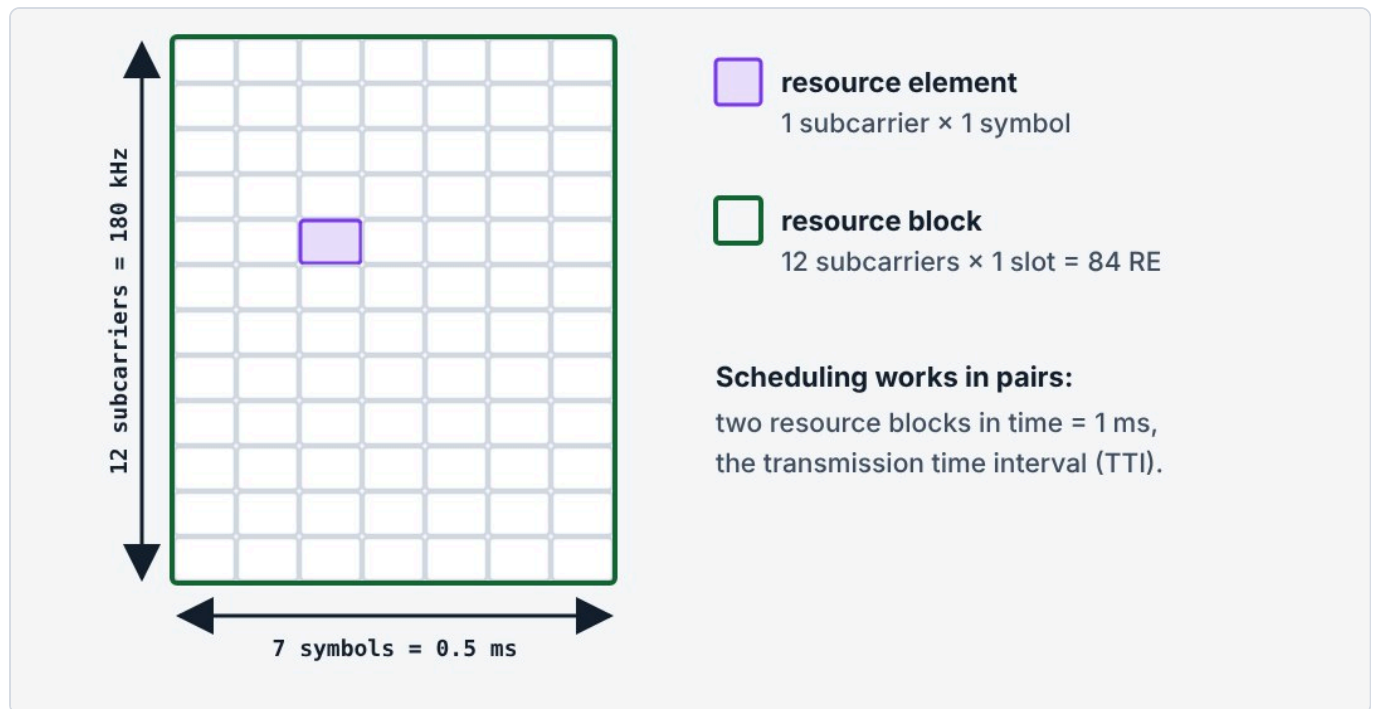
4.3 The special subframe



The special subframe is where TDD turns from downlink to uplink. It has three parts. DwPTS is a shortened downlink subframe. The guard period is silent, so that the downlink of far cells has died out and the UE can advance its uplink. UpPTS is a short uplink part for sounding or a short random access preamble.

TS 36.211 §4.2, Table 4.2-1

4.4 The resource block



A resource block is 12 subcarriers wide and one slot long. With the normal cyclic prefix that is $12 \times 7 = 84$ resource elements, covering 180 kHz and 0.5 ms. A resource element is one subcarrier during one symbol and carries one modulation symbol.

TS 36.211 §5.2, §6.2

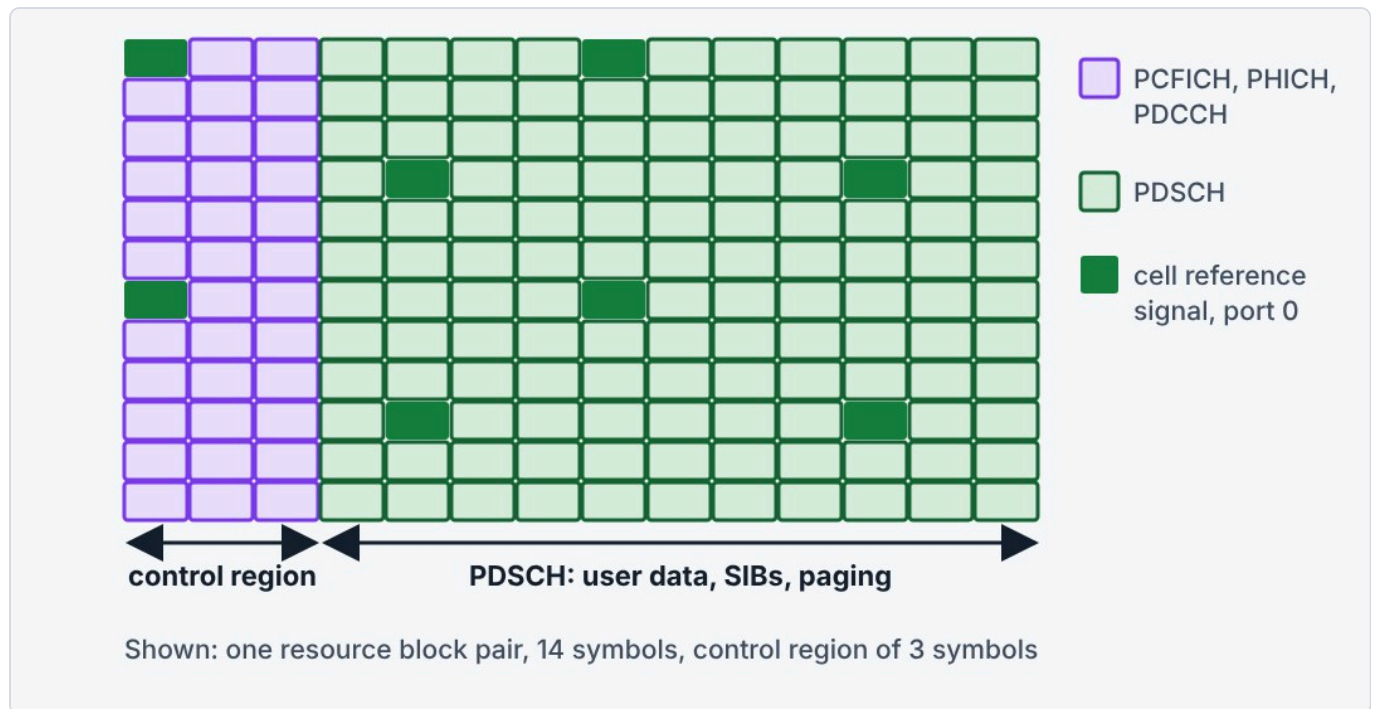
4.5 Units built from resource elements

UNIT	SIZE	USED FOR
RE	1 subcarrier $\times$ 1 symbol	one modulation symbol
RB	12 subcarriers $\times$ 1 slot	the unit of the grid
RB pair	12 subcarriers $\times$ 1 ms	scheduling of PDSCH and PUSCH
RBG	1 to 4 RBs, by bandwidth	resource allocation type 0
REG	4 resource elements	PCFICH, PHICH, PDCCH mapping
CCE	9 REGs = 36 RE	PDCCH: 1, 2, 4 or 8 CCEs

Several units are built on the resource element. Data is scheduled in resource block pairs. Control channels use smaller units: four resource elements form a resource element group, and nine groups form a control channel element, the building block of the PDCCH.

TS 36.211 §6.2.4 · TS 36.213 §7.1.6

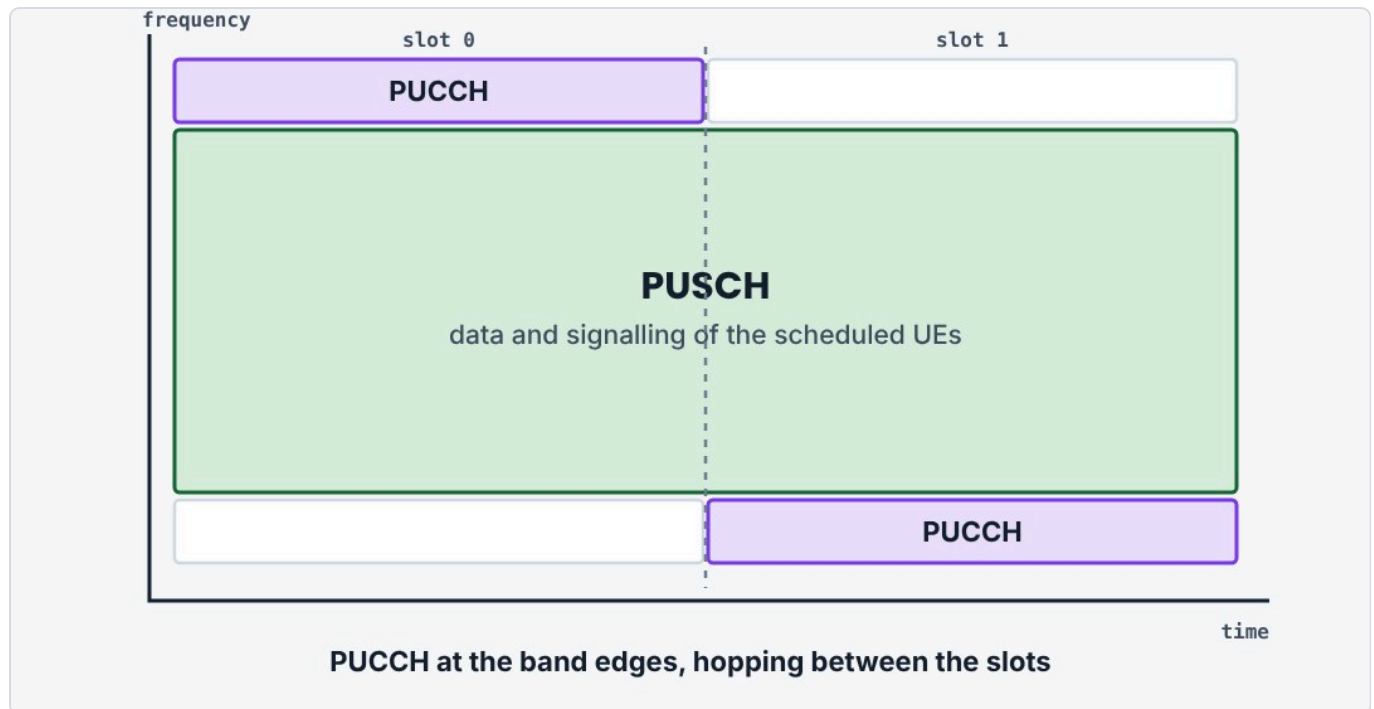
4.6 Layout of a downlink subframe



Every downlink subframe starts with a control region of one to three symbols. It holds the PCFICH, which tells how long the region is, the PHICH with the uplink HARQ feedback, and the PDCCH with the scheduling. The rest of the subframe carries the PDSCH. Reference signals are spread over the whole grid.

TS 36.211 §6.7 to §6.10 · TS 36.213 §9.1

4.7 Uplink subframe layout



The uplink keeps the single-carrier property by placing the control channel at the two edges of the band. The PUCCH hops from one edge to the other between the two slots. The middle is the PUSCH. One symbol per slot carries the demodulation reference signal.

TS 36.211 §5.4, §5.5

4.8 Three things to remember

1

1 ms is the scheduling unit

One subframe, two slots, 14 symbols.

2

Resource block = 12×7

84 resource elements, 180 kHz wide, 0.5 ms long.

3

Control region first

1 to 3 symbols of PDCCH, then the PDSCH.

10 ms frame, 1 ms subframe, 0.5 ms slot. A resource block is 180 kHz by one slot. Control first, then data.

TS 36.211 §4, §5.2, §6.2

Physical Channels and Signals

From cell search to random access: synchronisation signals, broadcast, control, data and reference signals.

5.1 Downlink channels at a glance

CHANNEL	PURPOSE
PSS, SSS	synchronisation and physical cell identity
PBCH	master information block (MIB)
PCFICH	size of the control region: 1, 2 or 3 symbols
PHICH	ACK or NACK for uplink data
PDCCH	downlink control information: scheduling, power control
PDSCH	user data, system information, paging, RRC
CRS	cell reference signal: channel estimation, RSRP

The downlink has three synchronisation and broadcast signals that a UE can read without any prior knowledge, three control channels at the start of every subframe, and one shared data channel. Reference signals run through all of them.

TS 36.211 §6 · TS 36.300 §5.1

5.2 Cell search: PSS and SSS

1 • PSS

Zadoff-Chu sequence

Gives 5 ms timing

NID2 = 0, 1 or 2

2 • SSS

Two interleaved M-sequences

Gives the frame start

NID1 = 0 ... 167

3 • PCI

$PCI = 3 \times NID1 + NID2$

504 values, 0 ... 503

Sets the CRS position

FDD: PSS in the last symbol of slots 0 and 10, SSS one symbol before

TDD: SSS in subframes 0 and 5, PSS in subframes 1 and 6

Centre 6 resource blocks, every 5 ms

The UE first finds the primary synchronisation signal. It gives the 5 ms timing and one of three identities. The secondary signal then gives the frame start and one of 168 group identities. Together they form the physical cell identity. Both sit in the central 62 subcarriers, so no bandwidth knowledge is needed.

TS 36.211 §6.11 · TS 36.213 §4.1

5.3 PBCH and the MIB

MASTER INFORMATION BLOCK

dl-Bandwidth	6, 15, 25, 50, 75 or 100 resource blocks
phich-Config	PHICH duration and resource: needed to find the PDCCH
systemFrameNumber	the 8 most significant bits of the 10-bit SFN
Antenna ports	1, 2 or 4: found from the CRC mask of the PBCH
Timing	subframe 0, first 4 symbols of slot 1, new content every 40 ms

After synchronisation the UE reads the PBCH. It carries the master information block: the downlink bandwidth, the PHICH configuration and the system frame number. The MIB is repeated every 10 ms and renewed every 40 ms. It also sits in the centre 6 resource blocks.

TS 36.331 §6.2.2 MasterInformationBlock · TS 36.211 §6.6

5.4 System information blocks

BLOCK	CONTENT
SIB1	PLMN, tracking area code, cell identity, scheduling of other SIBs
SIB2	common radio configuration: RACH, PUCCH, PUSCH, timers, access barring
SIB3	cell reselection, common part
SIB4, 5	intra-frequency and inter-frequency neighbours
SIB6, 7, 8	reselection to UMTS, GSM and CDMA2000
SIB10, 11, 12	earthquake and public warning messages (ETWS, CMAS)

SIB1: every 80 ms, in subframe 5 of even frames

Everything else is in the system information blocks on the PDSCH. SIB1 has a fixed schedule and tells where the others are. SIB1 and SIB2 are the minimum a UE needs before it can access the cell.

TS 36.331 §5.2, §6.3.1

5.5 PDCCH and DCI

DCI FORMAT	USE
0	uplink grant for PUSCH
1	downlink, one codeword
1A	compact downlink, also random access order
1C	very compact: SIBs, paging, RAR
2, 2A	downlink MIMO: closed loop, open loop
3, 3A	power control commands for a group of UEs

RNTIs: C-RNTI (one UE), SI-RNTI, P-RNTI, RA-RNTI, SPS C-RNTI, TPC-RNTI

Search spaces: common (16 CCEs) and UE-specific

The PDCCH carries downlink control information. A DCI tells one UE where its data is and how it is coded. The CRC is scrambled with an RNTI, so the UE finds its own DCI by blind decoding a limited search space. A PDCCH uses 1, 2, 4 or 8 control channel elements depending on the radio conditions.

TS 36.212 §5.3.3 · TS 36.213 §9.1.1

5.6 Reference signals

SIGNAL	DIRECTION	PURPOSE
CRS	downlink	channel estimation, RSRP and RSRQ, ports 0 to 3
DM-RS	downlink	demodulation with beamforming, ports 5, 7 to 14
CSI-RS	downlink	channel state for up to 8 ports (Release 10)
PRS	downlink	positioning (Release 9)
DM-RS	uplink	demodulation of PUSCH and PUCCH
SRS	uplink	sounding: uplink channel quality and timing

Reference signals are known symbols that let the receiver estimate the channel. The cell reference signal is always on and is what RSRP measures. UE-specific signals follow the precoding of one UE. Release 10 added CSI-RS so that up to eight antenna ports can be measured with little overhead.

TS 36.211 §6.10, §5.5

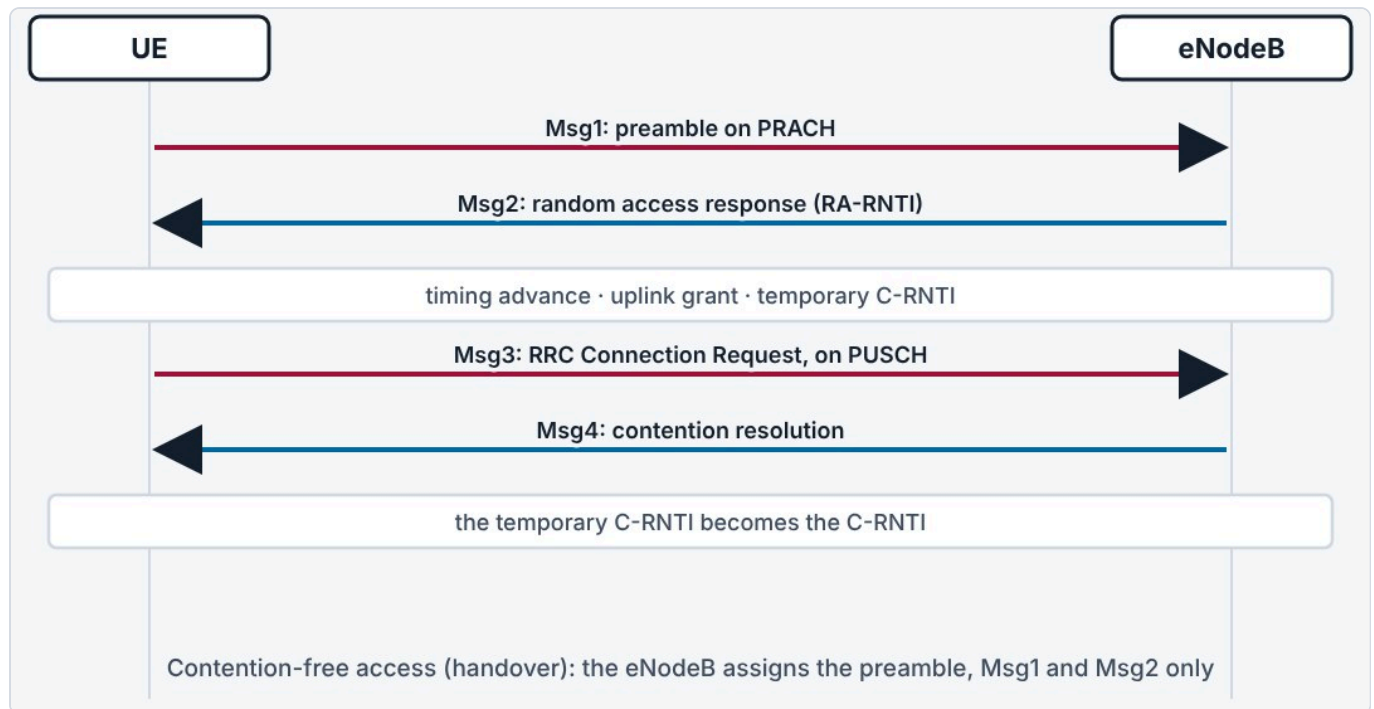
5.7 Uplink channels

CHANNEL	CARRIES
PUSCH	user data, RRC, and control when multiplexed
PUCCH format 1	scheduling request
PUCCH format 1a, 1b	HARQ ACK or NACK, 1 or 2 bits
PUCCH format 2, 2a, 2b	CQI, PMI, RI reports
PUCCH format 3	many ACK bits for carrier aggregation (Release 10)
PRACH	preamble: Zadoff-Chu, length 839, 6 resource blocks

The uplink has three channels. The PUSCH carries data and, when scheduled, the control information too. The PUCCH carries control when there is no PUSCH: scheduling requests, HARQ feedback and channel quality reports. The PRACH carries the random access preamble.

TS 36.211 §5.3, §5.4, §5.7

5.8 Random access



Random access gives the UE uplink timing and a first grant. The UE sends one of 64 preambles. The eNodeB answers with a timing advance, a temporary identity and a grant. The UE sends its identity in Msg3, and Msg4 resolves the case where two UEs chose the same preamble.

TS 36.321 §5.1 · TS 36.213 §6

5.9 Three things to remember

1

Search, then read

PSS → SSS → PBCH (MIB) → SIB1 → SIB2 → random access.

2

PDCCH is the key

No DCI, no data. The RNTI says who the DCI is for.

3

CRS is always on

It is the reference for RSRP, RSRQ and channel estimation.

PSS and SSS give timing and cell identity. MIB, then SIB1, then SIB2. The PDCCH tells each UE where its data is.

TS 36.211 · TS 36.213 · TS 36.331 §5.2

MIMO and Transmission Modes

What several antennas are used for, how the UE reports the channel, and the ten transmission modes.

6.1 Three uses of multiple antennas

Transmit diversity

- Same data on all antennas
- Protects against fading
- Works at the cell edge
- No gain in peak rate

Spatial multiplexing

- Different data per layer
- Multiplies the data rate
- Needs a good signal and a rich channel

Beamforming

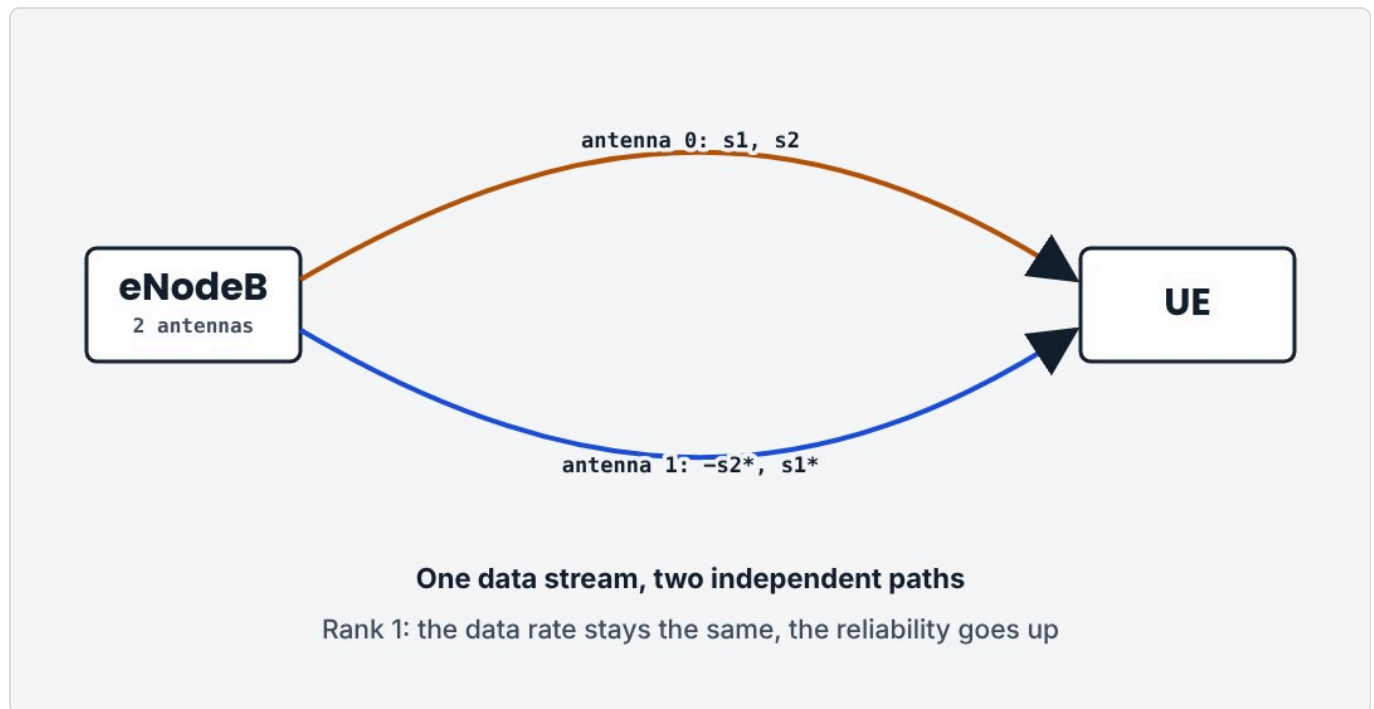
- Antennas form one beam
- Raises the signal level
- Extends the range
- Needs channel knowledge

An $M \times N$ system has M transmit and N receive antennas

Several antennas can be used in three ways. Diversity sends the same data over different paths, for robustness. Spatial multiplexing sends different data streams at once, for speed. Beamforming points the energy at one UE, for range. Which one the eNodeB uses depends on the signal quality and the channel.

TS 36.300 §5.1.5 · TS 36.211 §6.3

6.2 Transmit diversity



With two antennas LTE uses space-frequency block coding. Each pair of symbols is sent on two neighbouring subcarriers from both antennas, the second time swapped and conjugated. If one path fades, the other still delivers. The control channels and the PBCH always use this scheme when the cell has more than one antenna port.

TS 36.211 §6.3.3.3, §6.3.4.3

6.3 Spatial multiplexing



Layers $\leq \min(\text{transmit antennas, receive antennas, channel rank})$

Release 8: up to 4 layers. Release 10: up to 8 layers in the downlink.

With good signal and enough scattering, the antennas can carry separate data streams, called layers, on the same resource blocks. A 2×2 system carries two layers and doubles the peak rate. The number of layers can never exceed the smaller antenna count, nor the rank of the channel.

TS 36.211 §6.3.3.2, §6.3.4.2

6.4 What the UE reports: CQI, PMI, RI

CHANNEL STATE INFORMATION

RI · rank indicator

number of useful layers: 1 to 4 (8 in Release 10)

PMI · precoding matrix

index into the codebook of antenna weights

CQI · channel quality

0 to 15: highest modulation and code rate at 10 % BLER

Periodic reports

on PUCCH, small, at a configured period

Aperiodic reports

on PUSCH, detailed, requested by an uplink grant

The eNodeB cannot see the downlink channel, so the UE reports it. The rank indicator says how many layers the channel supports. The precoding matrix indicator proposes the antenna weights. The channel quality indicator says which modulation and code rate would give about 10 percent block errors.

TS 36.213 §7.2

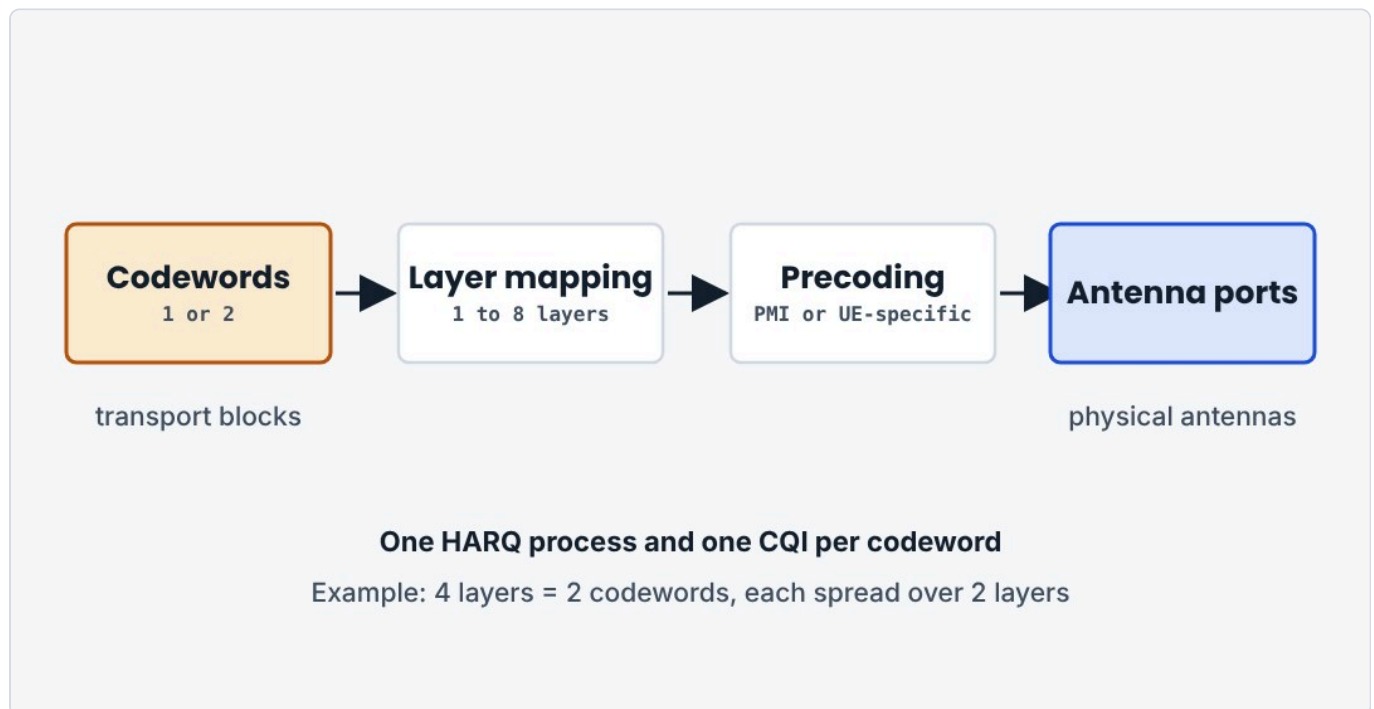
6.5 The ten transmission modes

MODE	SCHEME	RELEASE
TM1	single antenna, port 0	8
TM2	transmit diversity	8
TM3	open-loop spatial multiplexing	8
TM4	closed-loop spatial multiplexing	8
TM5	multi-user MIMO	8
TM6	closed-loop, one layer	8
TM7	beamforming, one layer, port 5	8
TM8	beamforming, two layers	9
TM9	up to 8 layers, CSI-RS based	10
TM10	coordinated multipoint (CoMP)	11

RRC configures one transmission mode per UE. It fixes which MIMO scheme the PDSCH uses and which DCI formats the UE must look for. Modes 1 to 7 are Release 8. The later modes rely on UE-specific reference signals and allow more layers.

TS 36.213 §7.1, Table 7.1-5

6.6 Codewords, layers, antenna ports



The path from data to antennas has three steps. Up to two codewords, each with its own HARQ process and modulation, are mapped onto the layers. Precoding then maps the layers onto the antenna ports. So even eight layers use only two codewords and two ACK bits.

TS 36.211 §6.3

6.7 UE categories

CATEGORY	DOWNLINK	UPLINK	TYPICAL SETUP
Cat 1	10 Mbit/s	5 Mbit/s	1 layer
Cat 3	100 Mbit/s	50 Mbit/s	20 MHz, 2 layers
Cat 4	150 Mbit/s	50 Mbit/s	20 MHz, 2 layers
Cat 6	300 Mbit/s	50 Mbit/s	2 carriers, 2 layers
Cat 9	450 Mbit/s	50 Mbit/s	3 carriers, 2 layers
Cat 12	600 Mbit/s	100 Mbit/s	3 carriers, 256QAM

The UE category sums up what a device can do: peak rate, layers and modulation. Categories 1 to 5 belong to Release 8. Higher categories need carrier aggregation. From Release 12 the downlink and uplink categories are signalled separately.

TS 36.306 §4.1, Tables 4.1-1, 4.1-2

6.8 Three things to remember

1

Layers multiply the rate

But only with good signal and a channel of sufficient rank.

2

The UE measures, the eNodeB decides

RI, PMI and CQI are proposals, not commands.

3

TM3 and TM4 are the workhorses

Open-loop and closed-loop spatial multiplexing.

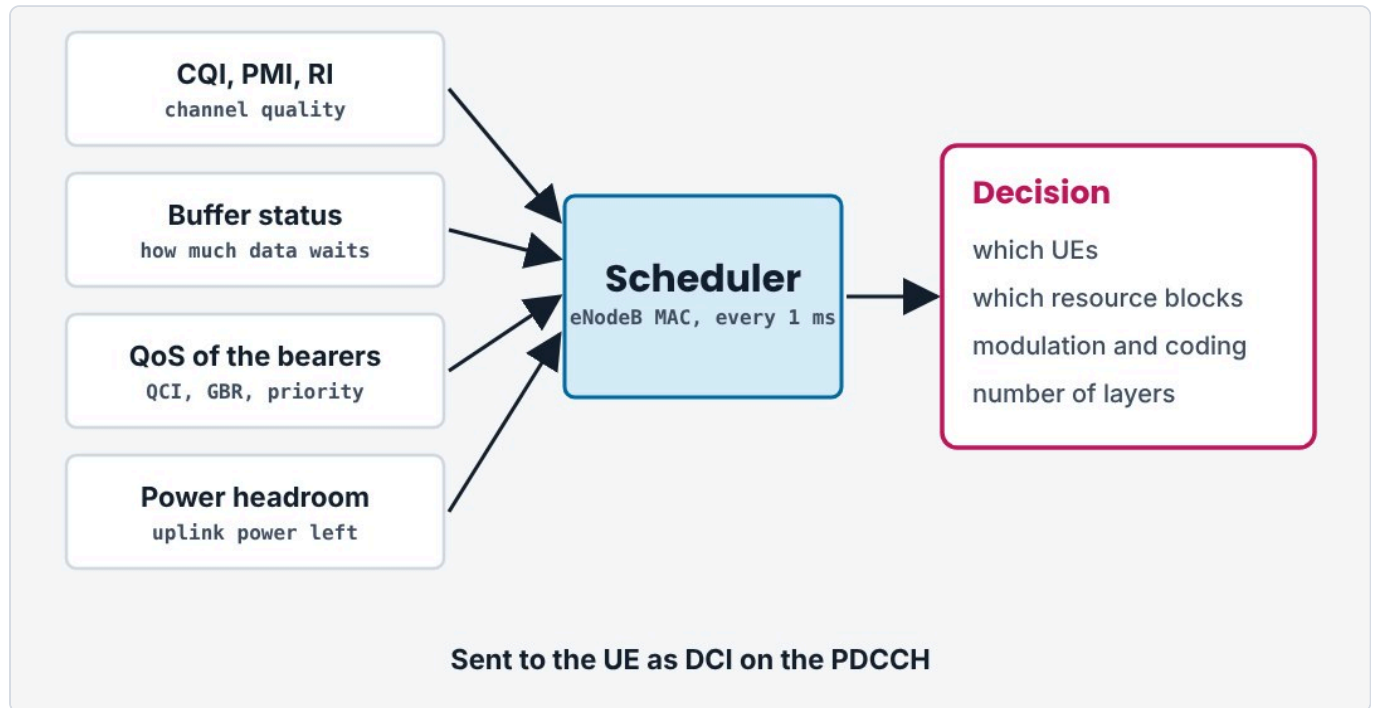
Diversity for robustness, multiplexing for speed, beamforming for range. The UE reports RI, PMI and CQI. The transmission mode sets the scheme.

TS 36.213 §7.1, §7.2

Scheduling, Link Adaptation and HARQ

How the eNodeB shares the radio every millisecond, picks the modulation, and repairs errors.

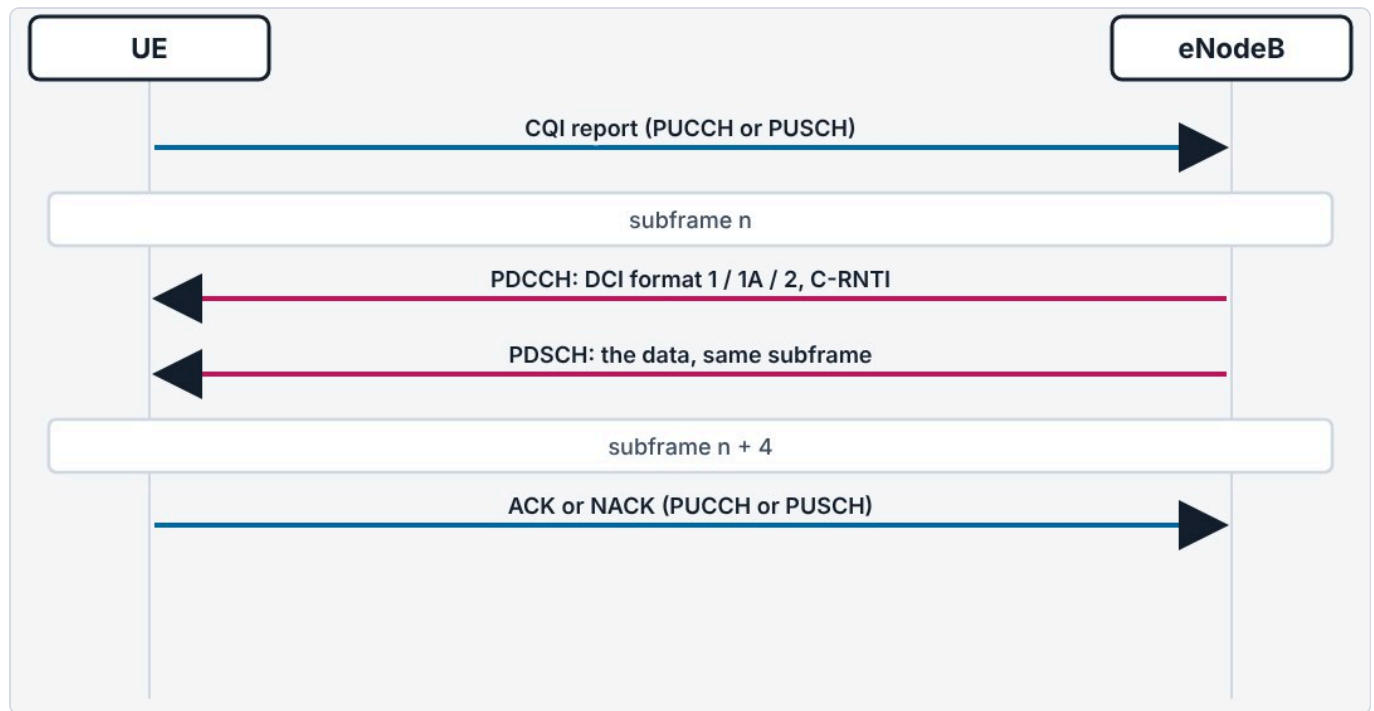
7.1 The scheduler



The scheduler sits in the MAC layer of the eNodeB and runs every millisecond. It decides which UEs transmit, on which resource blocks, with which modulation and coding. It works for both directions. The algorithm itself is not standardised, only the signalling around it.

TS 36.300 §11 · TS 36.321 §5.3, §5.4

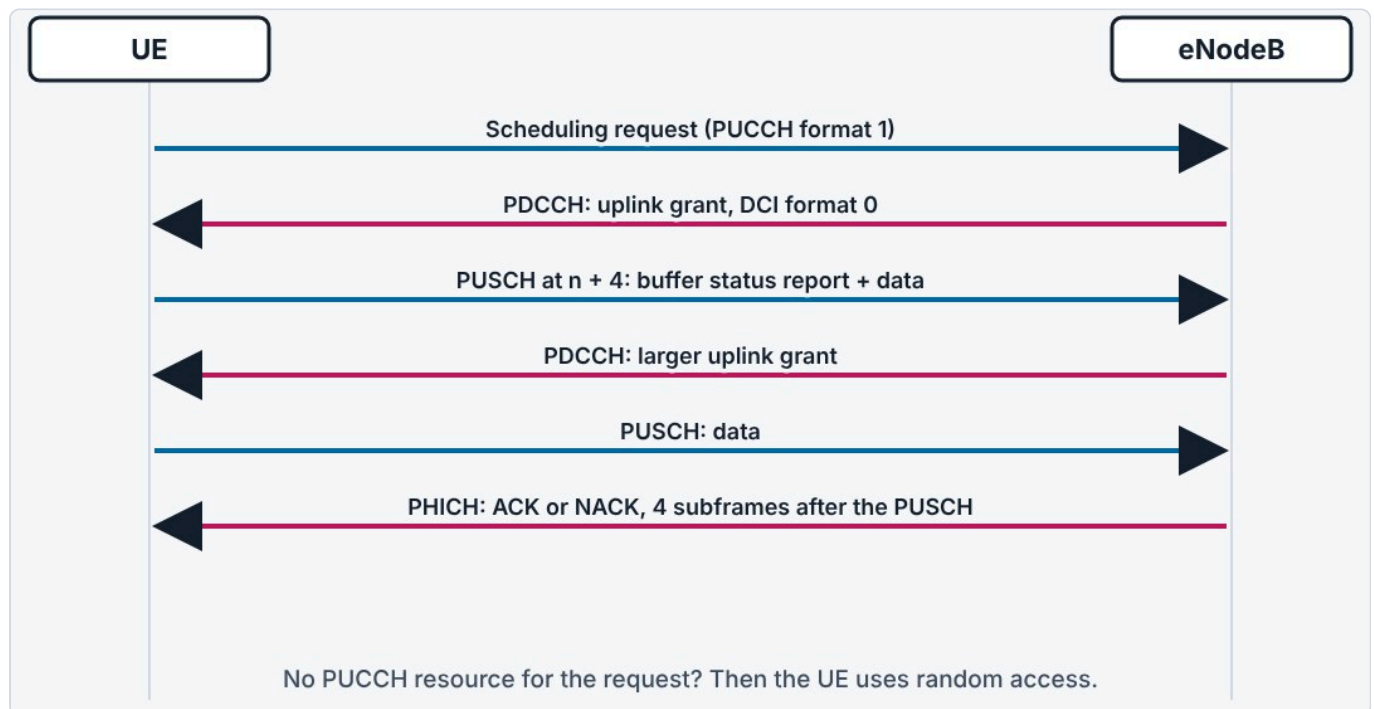
7.2 Downlink scheduling



In the downlink the grant and the data come in the same subframe. The UE finds a DCI with its C-RNTI on the PDCCH, reads the PDSCH resource blocks it points to, and answers four subframes later with an ACK or NACK.

TS 36.213 §7.1, §10.1

7.3 Uplink scheduling



In the uplink the UE must first ask. It sends a scheduling request on the PUCCH. The eNodeB gives a small grant, and the UE uses it to report how much data it has. Larger grants follow. Each grant is valid four subframes after it was received.

TS 36.321 §5.4 · TS 36.213 §8

7.4 Link adaptation

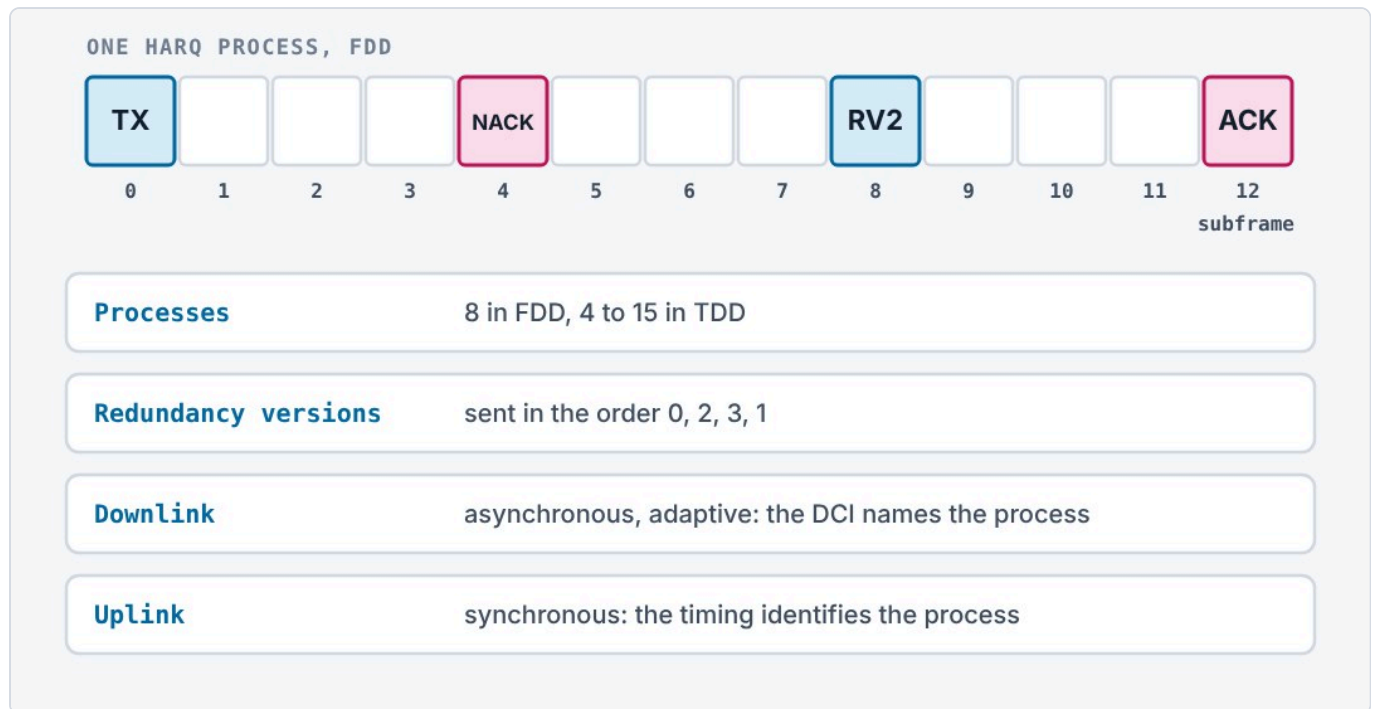
CQI	MODULATION	CODE RATE	BITS PER SYMBOL
1	QPSK	0.08	0.15
4	QPSK	0.30	0.60
7	16QAM	0.37	1.48
10	64QAM	0.46	2.73
13	64QAM	0.75	4.52
15	64QAM	0.93	5.55

MCS 0 to 28 select modulation and transport block size.
MCS 29 to 31 are for retransmissions.

The eNodeB matches the modulation and code rate to the channel. The CQI from the UE is the input. The result is an MCS index from 0 to 28, which together with the number of resource blocks gives the transport block size. The target is about 10 percent errors on the first transmission.

TS 36.213 §7.1.7, Tables 7.2.3-1, 7.1.7.1-1

7.5 HARQ



HARQ repairs errors within milliseconds. The receiver keeps a failed block and combines it with the retransmission, which carries different redundancy bits. FDD uses eight parallel stop-and-wait processes, so the data flow never stops. A retransmission comes eight subframes after the first attempt.

TS 36.321 §5.3.2, §5.4.2 · TS 36.213 §10

7.6 HARQ and ARQ work together

HARQ · MAC

Feedback after 4 ms
Soft combining of attempts
Usually up to 4 transmissions
Residual error about 0.1 to 1 %

ARQ · RLC AM

Status reports with sequence numbers
Retransmits or re-segments PDUs
Slower: tens of milliseconds
Residual error near 0.0001 %

RLC UM (voice): HARQ only

Two layers repair errors. HARQ in MAC is fast but gives up after a few attempts, and its ACK can be misread. RLC acknowledged mode sits above and catches what is left with status reports and retransmissions. Voice skips the RLC part, because a late packet is useless.

TS 36.322 §5.2 · TS 36.321 §5.3.2

7.7 Semi-persistent scheduling



One activation, then no PDCCH for the following packets

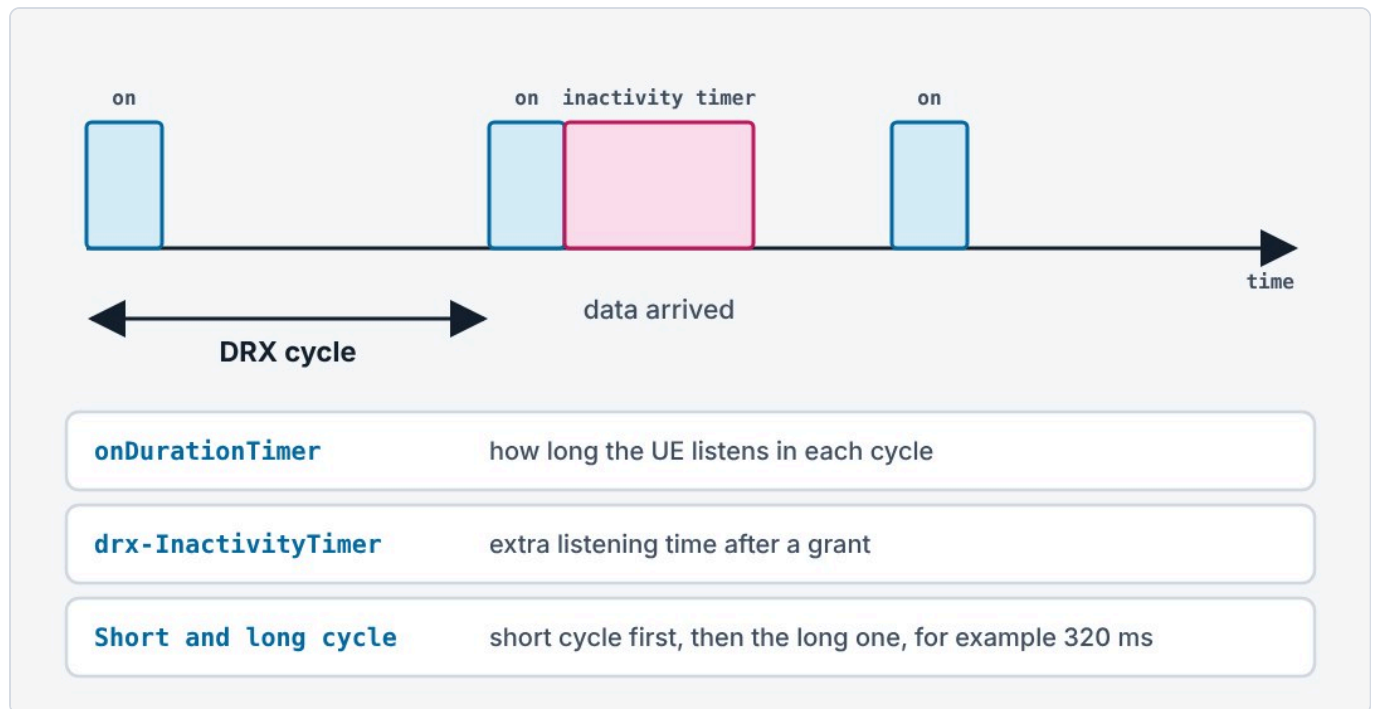
RRC: semiPersistSchedIntervalDL / UL = 20 ms

Retransmissions are still scheduled dynamically

A voice call sends a small packet every 20 ms. Granting each one over the PDCCH would use up the control channel. With semi-persistent scheduling RRC sets the period, and one PDCCH with the SPS C-RNTI activates a grant that then repeats by itself until it is released.

TS 36.321 §5.10 · TS 36.331 SPS-Config

7.8 DRX in connected mode



A connected UE does not need to watch the PDCCH all the time. With DRX it wakes for a short on-duration in each cycle. If it is scheduled, an inactivity timer keeps it awake. If not, it sleeps until the next cycle. This saves battery at the price of some delay.

TS 36.321 §5.7 · TS 36.331 DRX-Config

7.9 Three things to remember

- 1 n + 4 is the LTE rhythm**
Uplink grant to PUSCH, and data to ACK: four subframes.
- 2 Target: 10 % first-attempt errors**
HARQ and RLC ARQ clean up the rest.
- 3 SPS and DRX**
Fewer grants for voice, less listening for the UE.

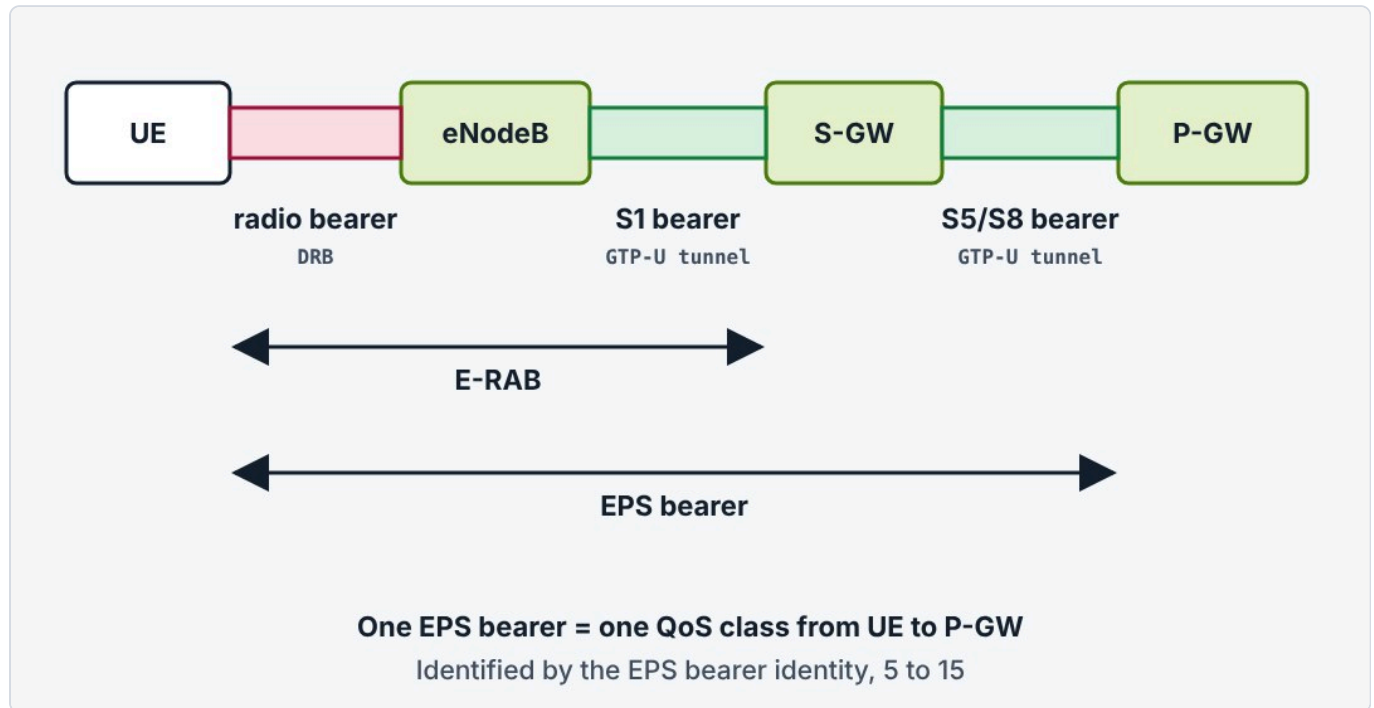
The eNodeB schedules both directions every millisecond. CQI drives the MCS. HARQ answers after four subframes.

TS 36.321 · TS 36.213

EPS Bearers and QoS

How LTE gives different traffic different treatment: bearers, QCI, traffic flow templates and dedicated bearer setup.

8.1 The EPS bearer



An EPS bearer is a pipe with one QoS from the UE to the P-GW. It is made of three pieces joined one to one: a radio bearer, an S1 bearer and an S5 bearer. The last two are GTP-U tunnels. All packets in one bearer get the same treatment.

TS 23.401 §4.7.2 · TS 36.300 §13

8.2 Default and dedicated bearers

Default bearer

- Created with the PDN connection
- Gives the UE its IP address
- Always non-GBR
- QoS from the subscription (HSS)
- Stays for the whole connection

Dedicated bearer

- Added when a service needs it
- Same IP address, no new one
- GBR or non-GBR
- QoS from the PCRF
- Has a traffic flow template

Up to 11 bearers per UE in Release 8 (8 data radio bearers)

Each PDN connection starts with one default bearer. It comes with the IP address, is always non-GBR and stays as long as the connection. Dedicated bearers are added on top when a service needs different QoS. They share the same IP address and are selected by packet filters.

TS 23.401 §4.7.2.1

8.3 QoS parameters

BEARER LEVEL QoS

QCI	class identifier: resource type, priority, delay, loss
ARP	allocation and retention priority, 1 to 15, with pre-emption flags
GBR	guaranteed bit rate, GBR bearers only
MBR	maximum bit rate, GBR bearers only
APN-AMBR	limit for all non-GBR bearers of one APN, enforced in the P-GW
UE-AMBR	limit for all non-GBR bearers of the UE, enforced in the eNodeB

Every bearer has a QCI and an ARP. The QCI points to a standard set of characteristics. The ARP decides who is admitted or dropped under congestion. GBR bearers add a guaranteed and a maximum bit rate. Non-GBR bearers share two aggregate limits.

TS 23.401 §4.7.3 · TS 23.203 §6.1.7

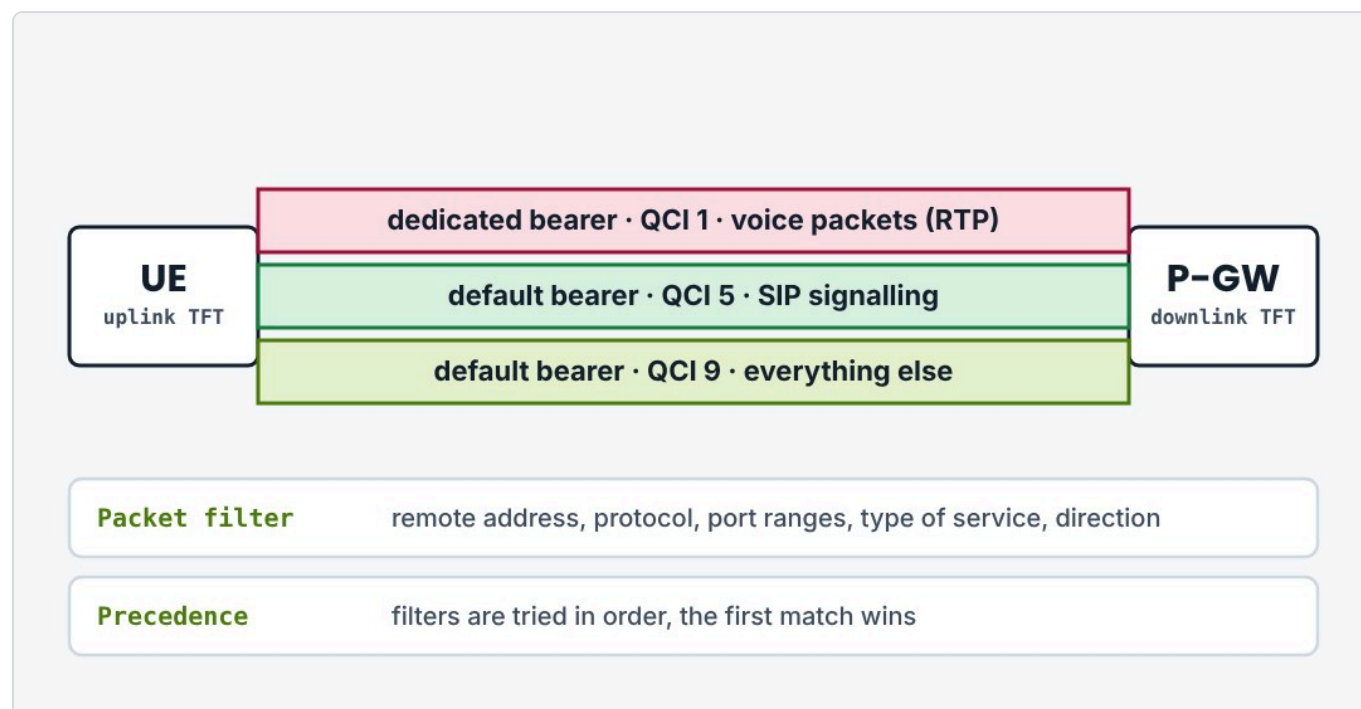
8.4 The standardised QCI values

QCI	TYPE	PRIORITY	DELAY	LOSS	EXAMPLE
1	GBR	2	100 ms	10^{-2}	voice
2	GBR	4	150 ms	10^{-3}	video call
3	GBR	3	50 ms	10^{-3}	real-time gaming
4	GBR	5	300 ms	10^{-6}	buffered video
5	non-GBR	1	100 ms	10^{-6}	IMS signalling
6	non-GBR	6	300 ms	10^{-6}	video, TCP
7	non-GBR	7	100 ms	10^{-3}	voice, live video
8	non-GBR	8	300 ms	10^{-6}	video, TCP
9	non-GBR	9	300 ms	10^{-6}	default data

Nine QCI values come from Release 8. Each one fixes the resource type, a priority, a packet delay budget and a packet error loss rate. The eNodeB turns these into scheduler weights and RLC and PDCP settings.

TS 23.203 Table 6.1.7

8.5 Traffic flow templates



A traffic flow template is a set of packet filters. It decides which IP packets go into which bearer. The UE applies the uplink template, the P-GW the downlink one. Packets that match no filter go into the default bearer.

TS 23.401 §4.7.2.2 · TS 24.008 §10.5.6.12

8.6 A VoLTE phone has three bearers

BEARER	QCI	RLC MODE	CARRIES
Default, internet APN	9	AM	web, apps
Default, IMS APN	5	AM	SIP signalling
Dedicated, IMS APN	1	UM	voice (RTP), GBR

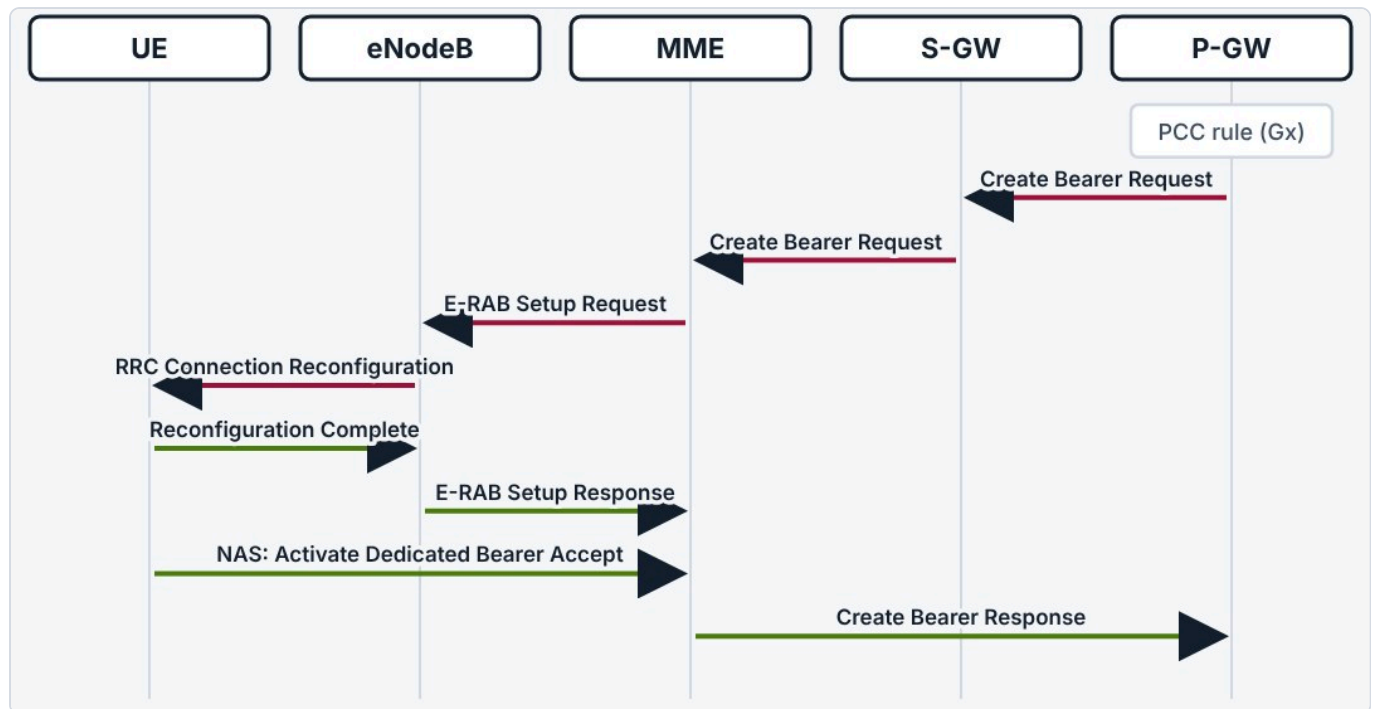
The dedicated bearer exists only during the call.

A video call adds a second dedicated bearer, QCI 2.

A typical VoLTE phone shows the model well. It has two PDN connections: one to the internet APN and one to the IMS APN. Each has a default bearer. During a call the network adds a dedicated bearer for the voice packets.

TS 23.401 §4.7 · GSMA IR.92

8.7 Dedicated bearer activation



The network starts a dedicated bearer. The PCRF sends a rule to the P-GW, which asks for a new bearer. The MME packs a NAS request into the E-RAB setup, and the eNodeB delivers it in an RRC reconfiguration that adds the data radio bearer.

TS 23.401 §5.4.1

8.8 Three things to remember

1

Bearer = QCI + ARP

Plus bit rates for GBR bearers.

2

QCI 1, 5 and 9

Voice, IMS signalling and default data.

3

Filters choose the bearer

Uplink TFT in the UE, downlink TFT in the P-GW.

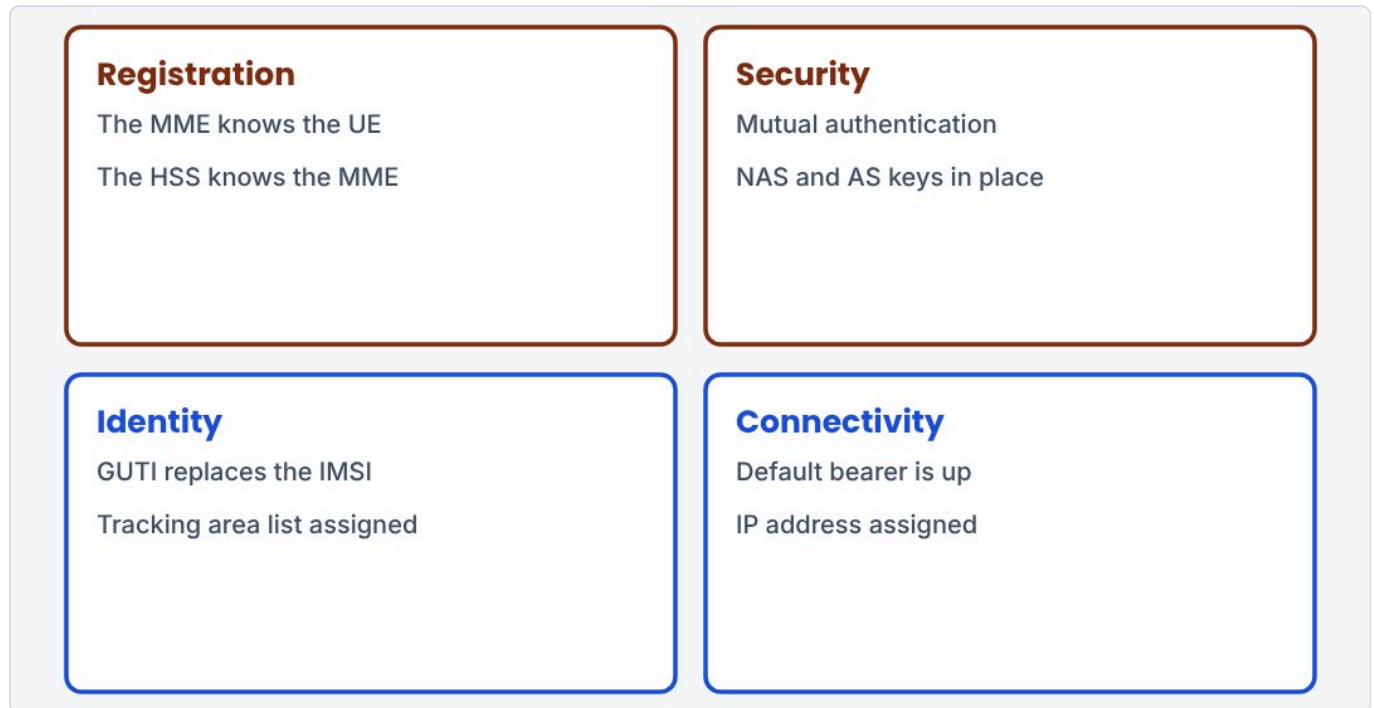
A bearer is a QoS pipe from UE to P-GW. The default bearer comes with the IP address. Dedicated bearers are added by the network when a service needs them.

TS 23.401 §4.7 · TS 23.203

Attach, Authentication and Security

What happens between switching the phone on and having an IP address: the attach procedure, EPS-AKA and the two security layers.

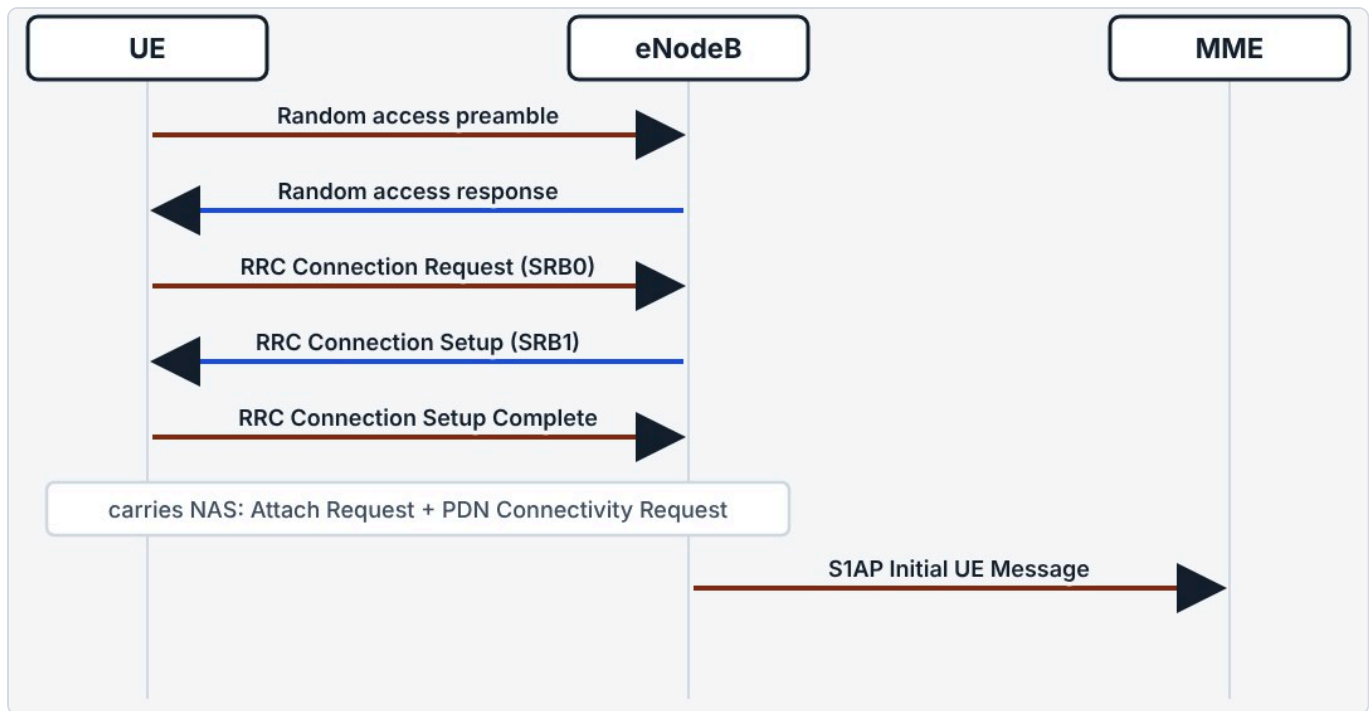
9.1 What attach achieves



Attach registers the UE in the network. In LTE it always sets up a default bearer too, so the UE is online as soon as attach is done. At the end the UE has a temporary identity, a tracking area list, security keys and an IP address.

TS 23.401 §5.3.2 · TS 24.301 §5.5.1

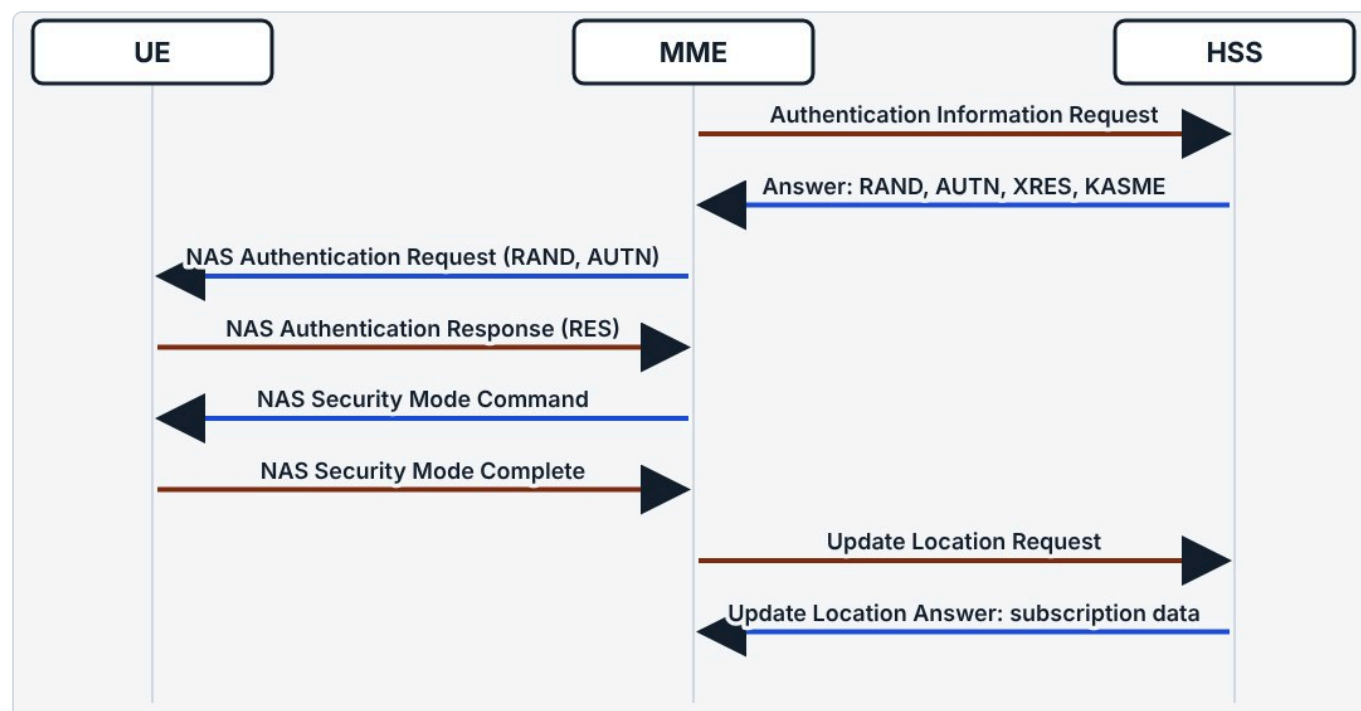
9.2 Step 1: RRC connection



The UE first needs a signalling connection to the eNodeB. After random access it sends the RRC Connection Request and receives the setup of SRB1. Its first NAS message, the Attach Request with a PDN Connectivity Request inside, rides in the RRC Connection Setup Complete.

TS 36.331 §5.3.3

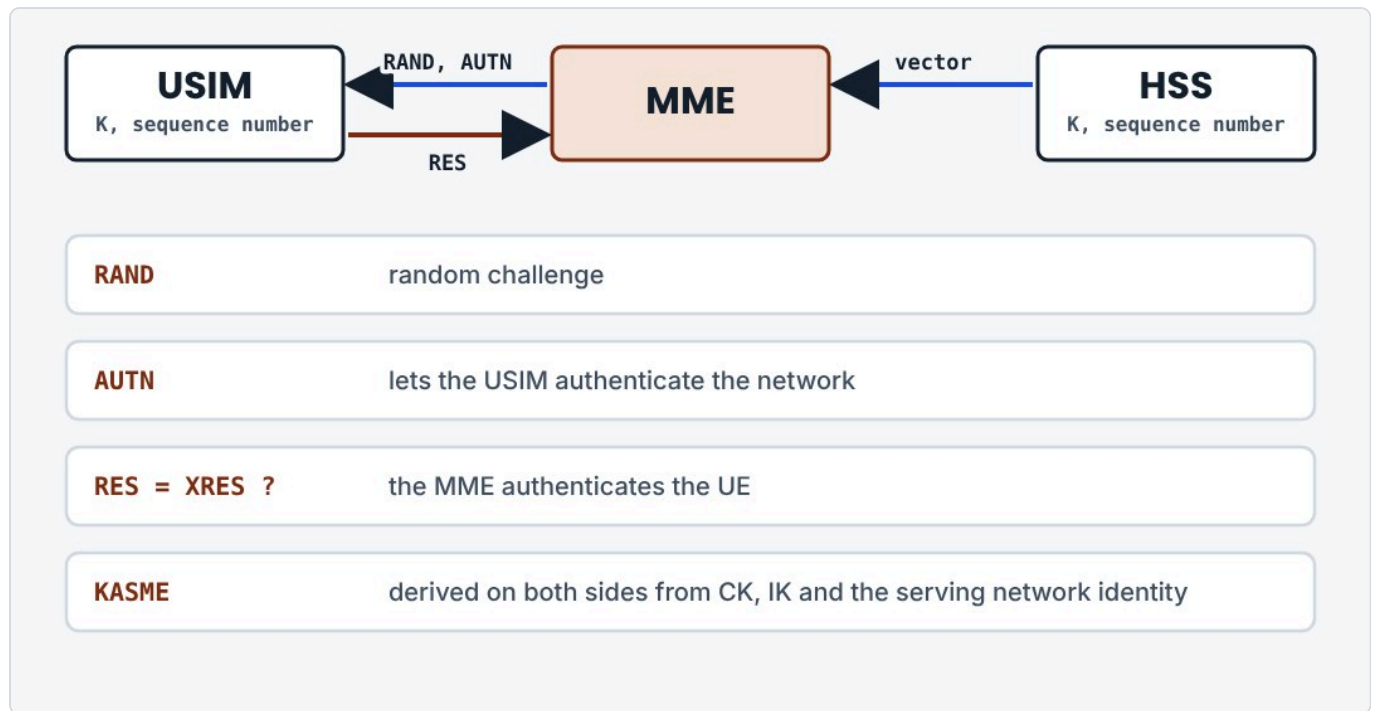
9.3 Step 2: authentication and NAS security



The MME fetches authentication vectors from the HSS and challenges the UE. When the answer matches, it starts NAS security, so all later NAS messages are integrity protected and ciphered. Then it registers itself in the HSS and receives the subscription data.

TS 23.401 §5.3.2.1 steps 5 to 11 · TS 33.401 §6.1

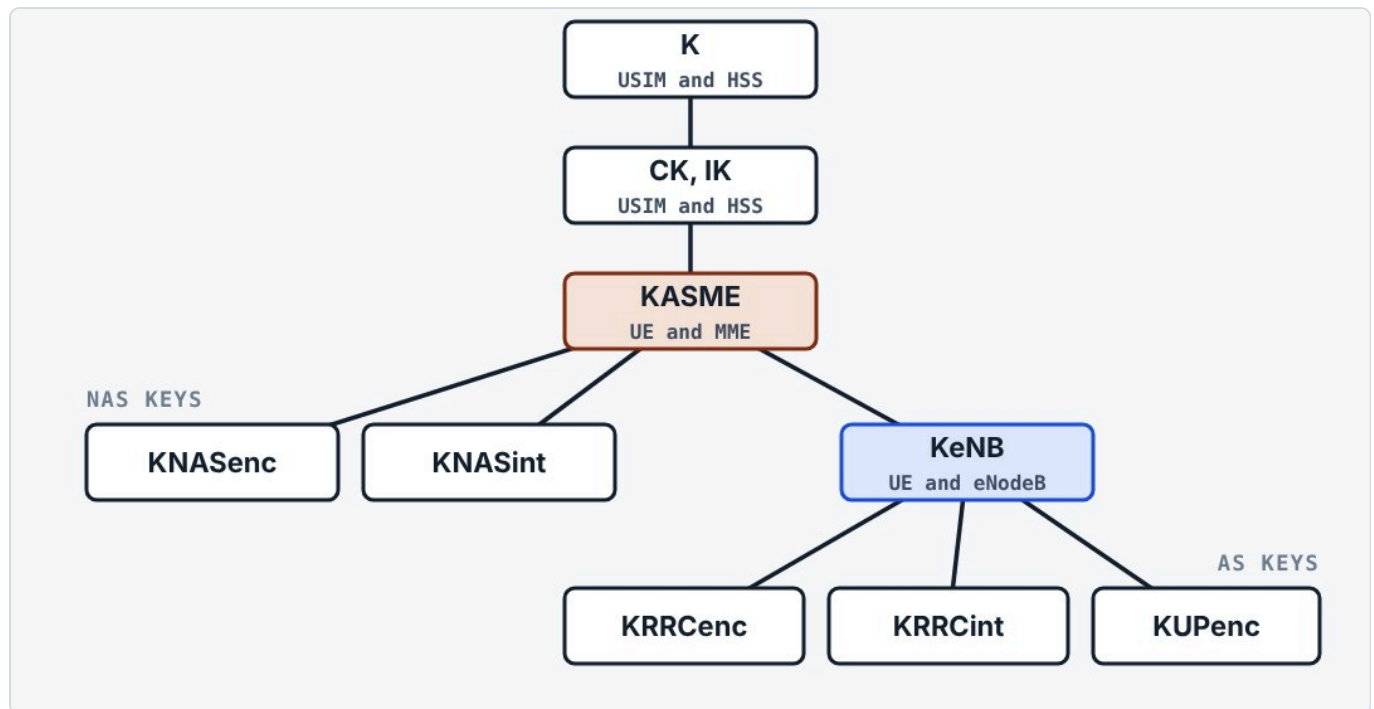
9.4 EPS-AKA: mutual authentication



Authentication works both ways. The USIM checks AUTN to verify that the network is genuine. The MME compares RES with the expected value to verify the UE. Both sides derive the same key KASME without ever sending it over the radio. The permanent key K never leaves the USIM and the HSS.

TS 33.401 §6.1 · TS 33.102 §6.3

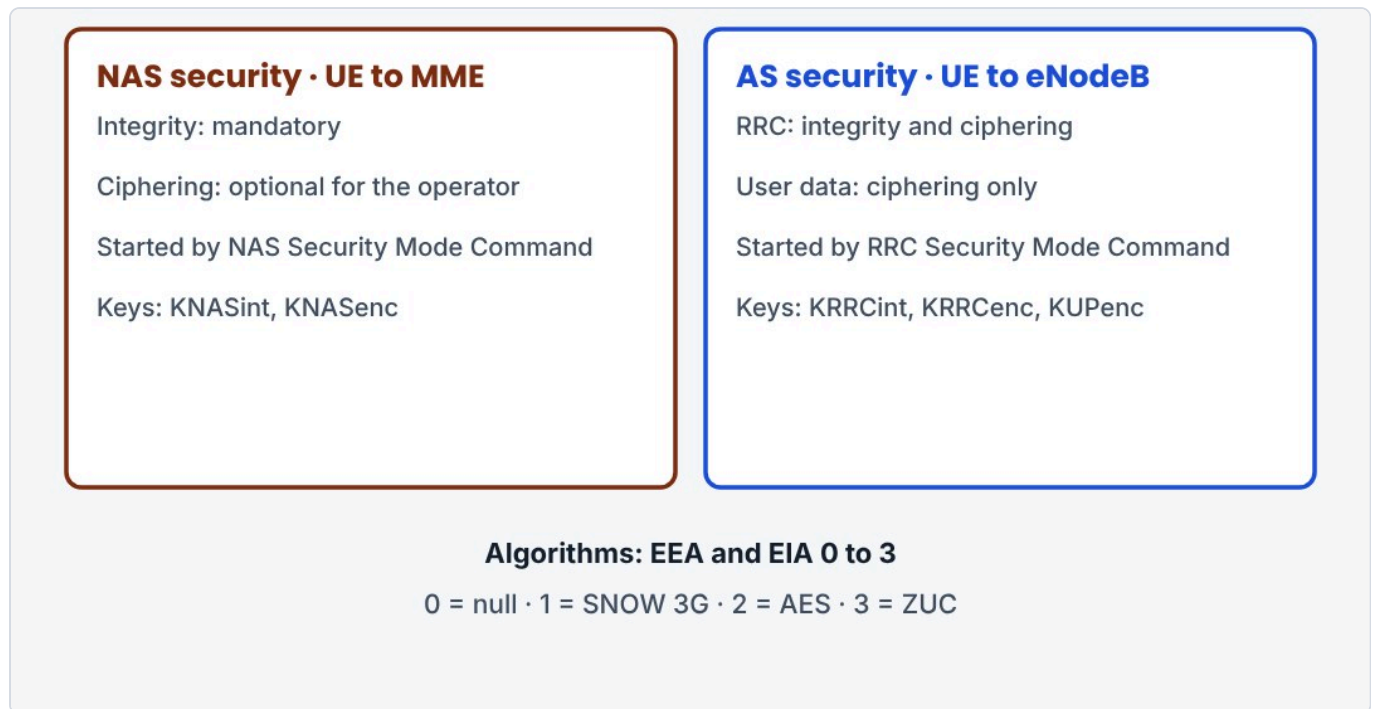
9.5 The key hierarchy



All keys grow from the permanent key **K**. **KASME** is the root for one MME session. From it come the two NAS keys and **KeNB**, which the MME hands to the eNodeB. The eNodeB derives the RRC and user plane keys. A stolen eNodeB key therefore reveals nothing about NAS.

TS 33.401 §6.2

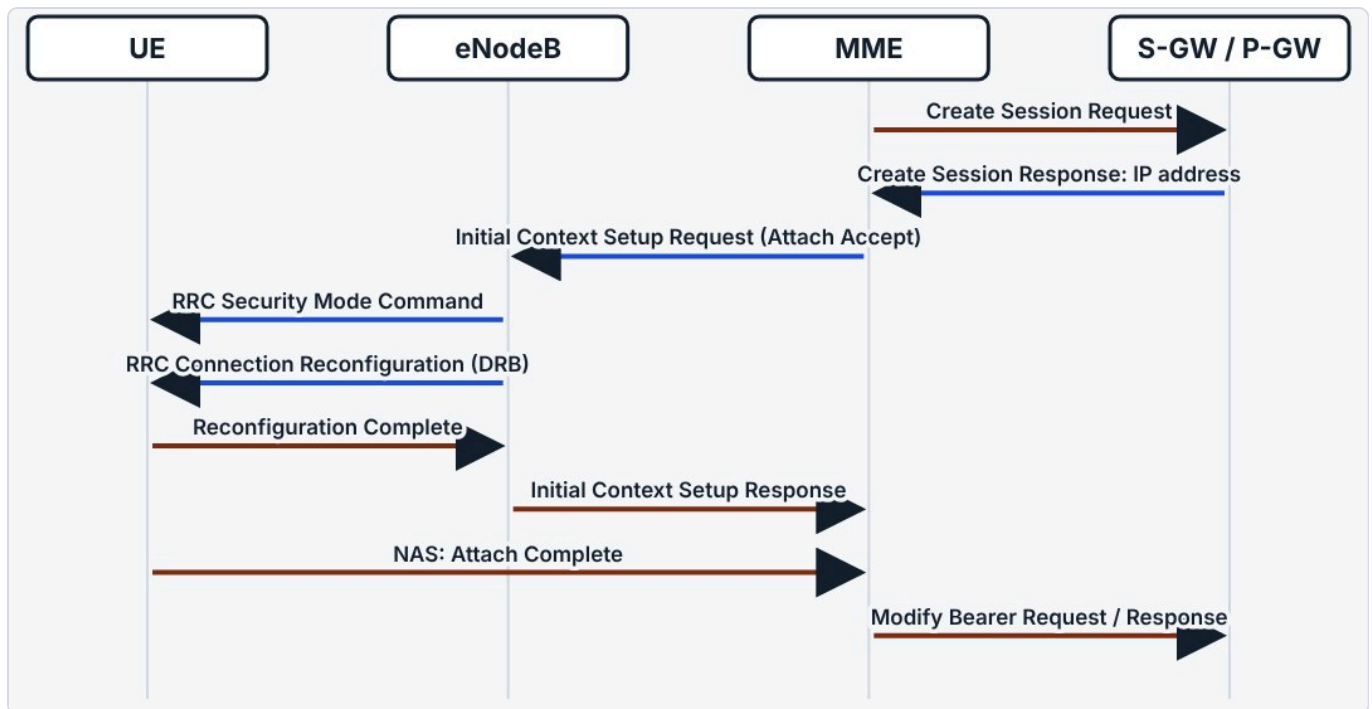
9.6 Two security layers



LTE protects signalling twice. NAS security runs between UE and MME. AS security runs between UE and eNodeB, in PDCP, and covers RRC and the user data. Integrity protection is mandatory for signalling. User data is ciphered but not integrity protected.

TS 33.401 §5.1, §7 · TS 24.301 §4.4

9.7 Step 3: default bearer and attach accept



The MME creates the session through the S-GW at the P-GW, which allocates the IP address. It then sends the Attach Accept inside the Initial Context Setup Request. The eNodeB activates AS security and adds the data radio bearer. The UE confirms, and the downlink path is switched on.

TS 23.401 §5.3.2.1 steps 12 to 24

9.8 Three things to remember

1 Always-on from the start
Attach ends with an IP address and a default bearer.

2 K never travels
Only RAND, AUTN and RES cross the radio.

3 Two security mode commands
One from the MME for NAS, one from the eNodeB for AS.

Attach is registration plus default bearer. Authentication is mutual and leaves both sides with KASME. NAS and AS are protected separately.

TS 23.401 §5.3.2 · TS 33.401

Idle Mode: Tracking, Paging, Service Request

What the UE and the network do when no data flows: states, cell reselection, tracking area update, paging and the way back to connected.

10.1 Three pairs of states

STATE	MEANING	NETWORK KNOWS
EMM-DEREGISTERED	not attached	nothing
EMM-REGISTERED	attached, has a GUTI	at least the tracking area list
ECM-IDLE	no S1 connection	tracking area list
ECM-CONNECTED	S1 connection exists	the serving eNodeB
RRC_IDLE	no radio connection	–
RRC_CONNECTED	radio connection exists	the cell

ECM-IDLE goes with RRC_IDLE, ECM-CONNECTED with RRC_CONNECTED

Three state machines describe a UE. EMM says whether it is registered. ECM says whether a signalling connection to the MME exists. RRC says whether a radio connection exists. In idle, the network knows the UE only by its tracking area list. In connected, it knows the cell.

TS 23.401 §4.6 · TS 36.331 §4.2.1

10.2 Cell selection and reselection

IDLE MODE MOBILITY · PARAMETERS FROM SIB1 AND SIB3

Selection: S criterion

$S_{rxlev} > 0$ and $S_{qual} > 0$

S_{rxlev}

measured RSRP – $q_{RxLevMin}$ – offsets – power compensation

Priorities

0 to 7 per frequency or RAT, from SIB3 to SIB8 or dedicated

Ranking: R criterion

R_s = serving + hysteresis · R_n = neighbour – offset

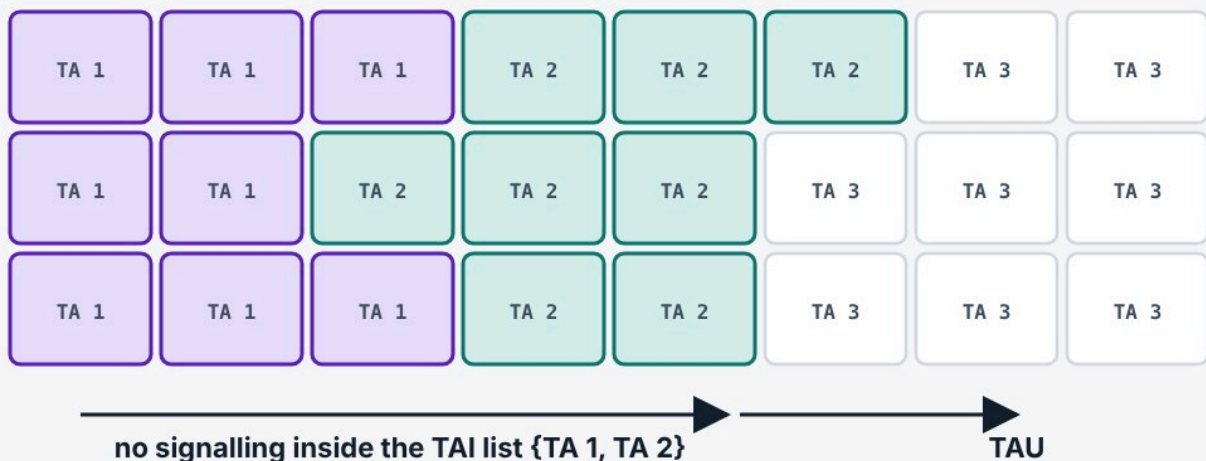
Treselection

the better cell must stay better for this time, 0 to 7 s

In idle the UE chooses its cell by itself. A cell is suitable when its level and quality are above the minimum broadcast in SIB1. The UE then keeps comparing: higher-priority frequencies first, and within one priority the best ranked cell after a hysteresis and a waiting time.

TS 36.304 §5.2.3, §5.2.4

10.3 Tracking areas and the TAI list



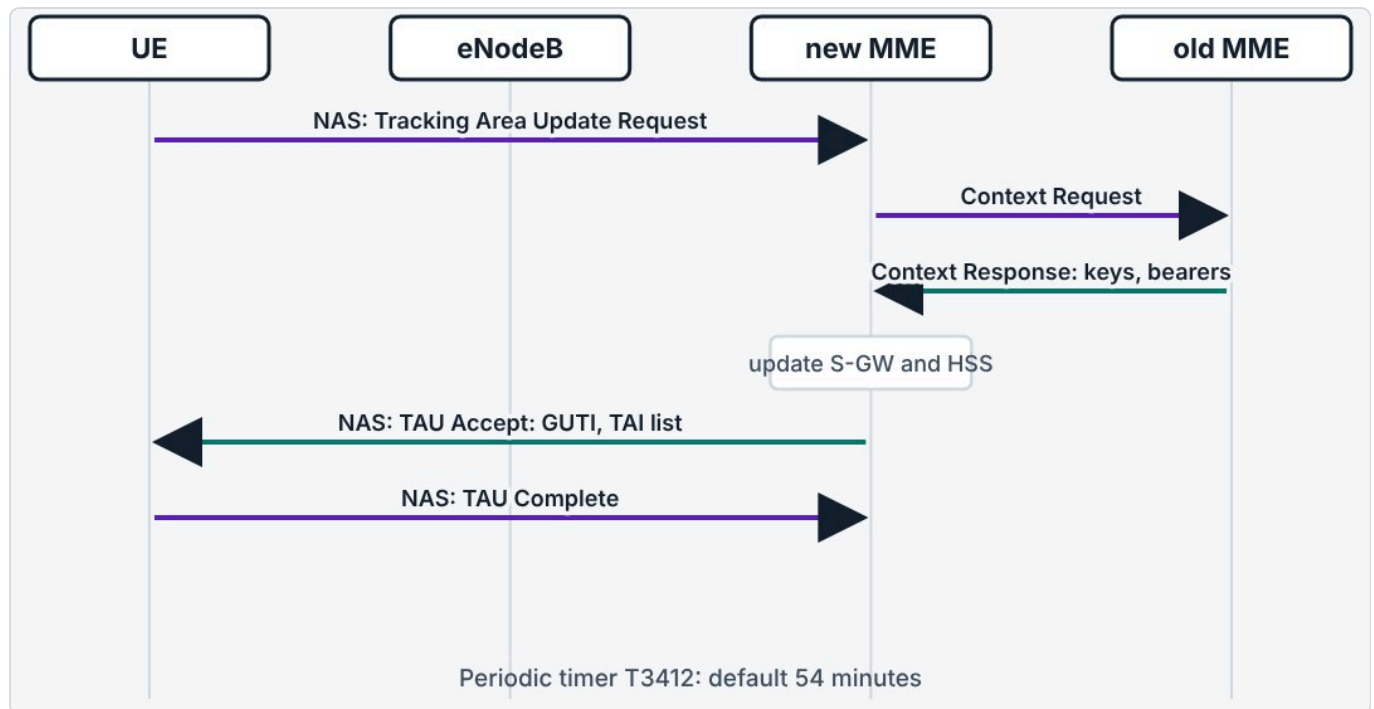
A list instead of one area avoids ping-pong updates at the border

Paging goes to all cells of the list

Cells are grouped into tracking areas. At attach the MME gives the UE a list of tracking areas. Inside the list the UE moves without any signalling. Only when it enters a cell whose tracking area is not in the list does it tell the network.

TS 23.401 §4.3.5.3 · TS 24.301 §5.3.4

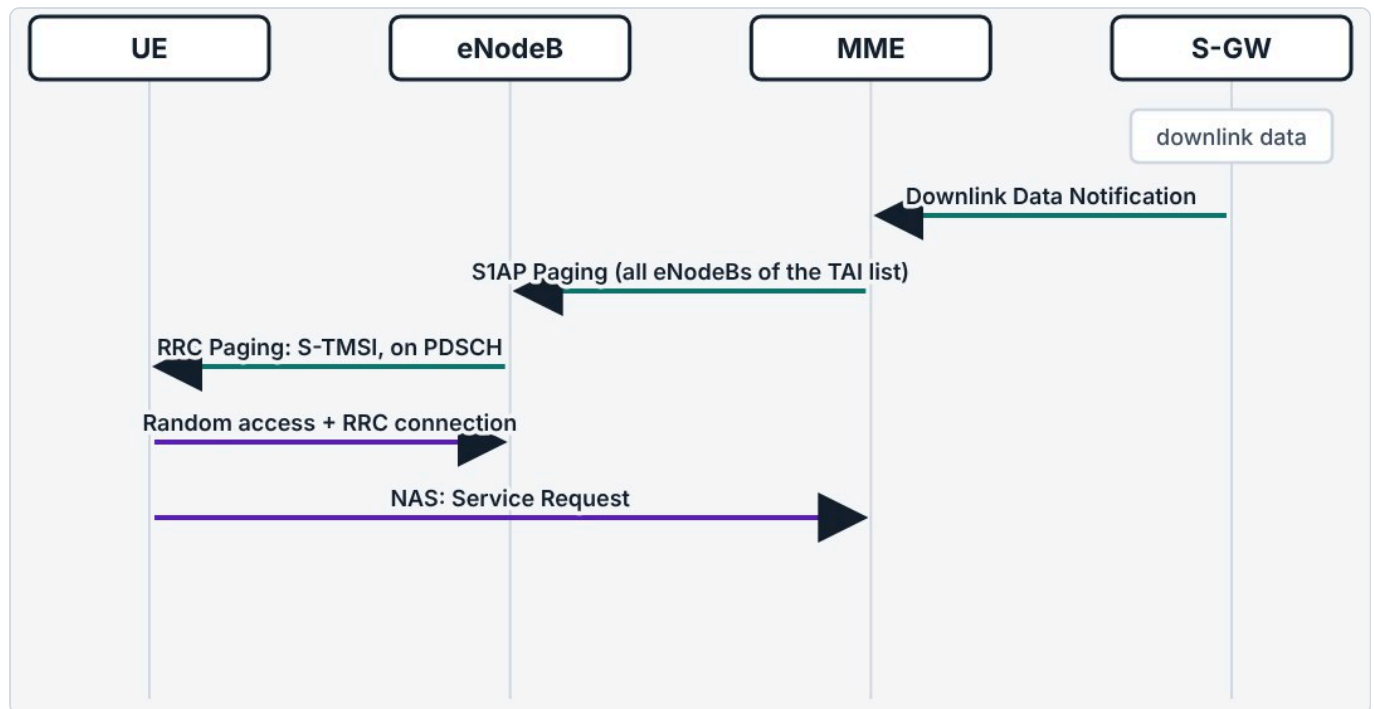
10.4 Tracking area update



The UE sends a tracking area update when it leaves its list, when the periodic timer T3412 expires, or after some changes such as a return from 2G or 3G. The MME answers with a new list and often a new GUTI. If the MME has changed, the new one fetches the context from the old one.

TS 23.401 §5.3.3 · TS 24.301 §5.5.3

10.5 Paging



When downlink data arrives for an idle UE, the S-GW buffers it and notifies the MME. The MME sends a paging message to every eNodeB in the tracking area list. Each cell broadcasts the S-TMSI at the paging occasion of the UE. The UE answers with a service request.

TS 23.401 §5.3.4.3 · TS 36.331 §5.3.2

10.6 When the UE listens for paging

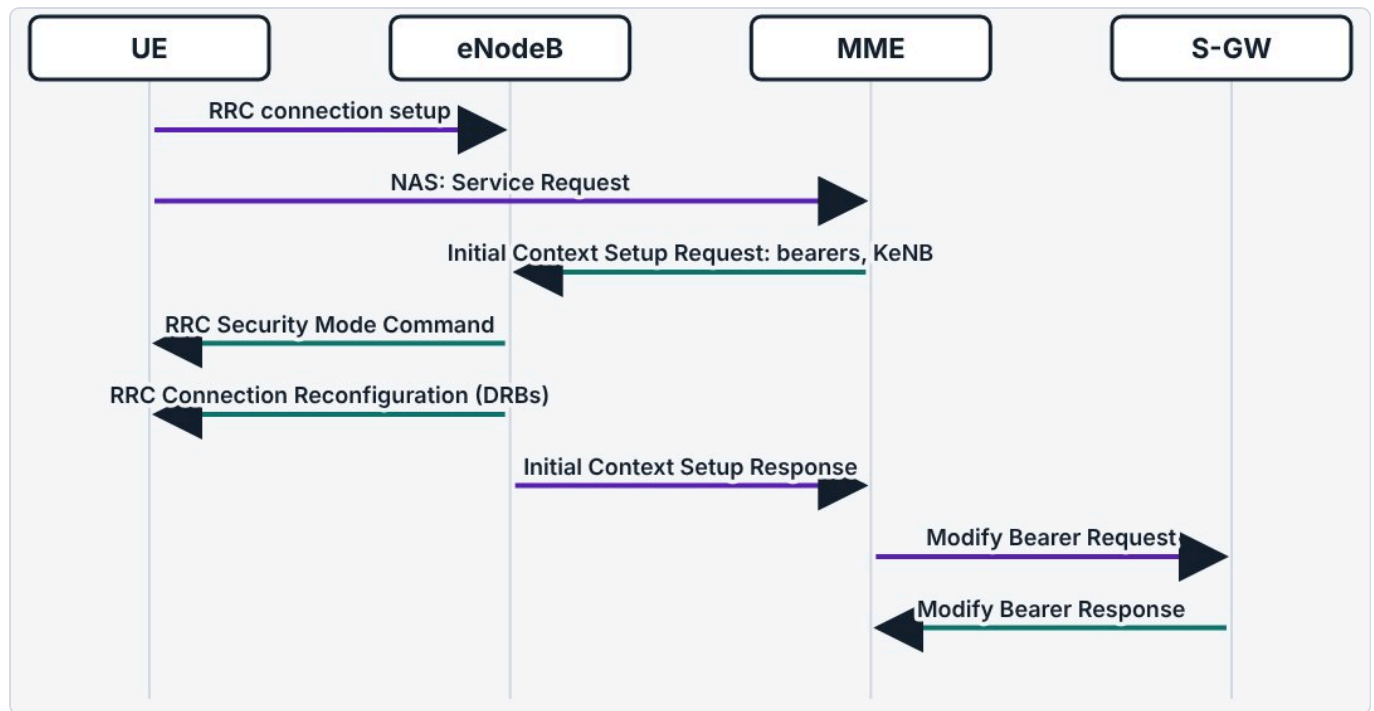
$$\text{SFN mod } T = (T / N) \times (\text{UE_ID mod } N)$$

T	DRX cycle: 32, 64, 128 or 256 frames = 0.32 to 2.56 s
UE_ID	IMSI mod 1024
N	min(T, nB): number of paging frames per cycle
nB	from SIB2: 4T, 2T, T, T/2 ... T/32
Paging occasion	subframe 9 when nB ≤ T (FDD); also 0, 4, 5 for larger nB

An idle UE sleeps almost all the time. It wakes once per DRX cycle, in one subframe of one frame, and checks the PDCCH for the P-RNTI. The frame and subframe follow from the IMSI, so the eNodeB knows when each UE is listening.

TS 36.304 §7

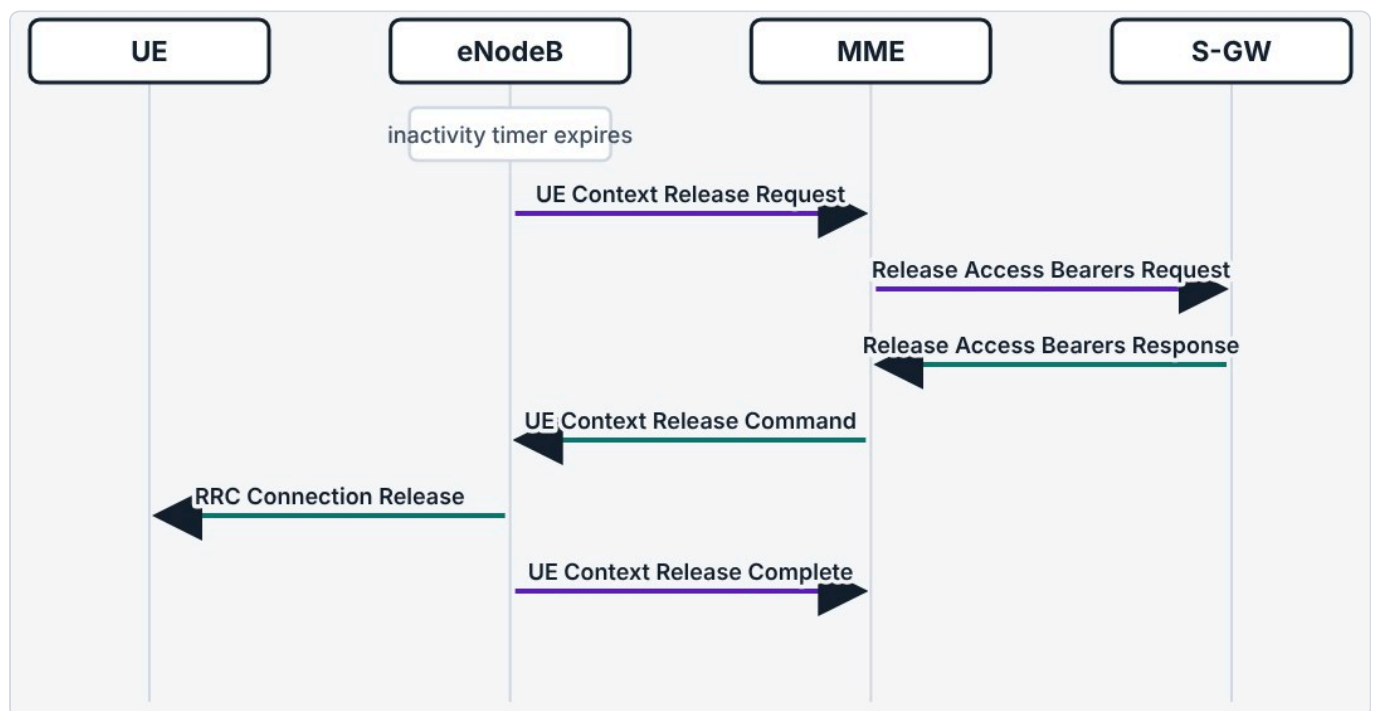
10.7 Service request



The service request brings an idle UE back to connected. It is short, because the MME already holds the context. The eNodeB gets the bearers and the key, activates security and sets up the data radio bearers. From request to first data typically takes well under 100 ms.

TS 23.401 §5.3.4.1

10.8 S1 release: back to idle



After some seconds without data the eNodeB asks for release. The MME tells the S-GW to drop the eNodeB address and releases the UE context in the eNodeB. The RRC connection is released. The bearers stay in the core, so the next service request is quick.

TS 23.401 §5.3.5

10.9 Three things to remember

1

Idle: the UE moves by itself

Cell reselection with priorities and ranking.

2

TAU when leaving the list

Or when T3412 expires.

3

Paging → service request

The core keeps the bearers, only the radio part is rebuilt.

Idle UEs are known per tracking area list. They are reached by paging and come back with a service request.

TS 23.401 §5.3 · TS 36.304

Mobility in Connected Mode

Measurements, the events that trigger a report, and the X2 and S1 handover procedures.

11.1 What the UE measures

UE MEASUREMENTS

RSRP

average power of the resource elements that carry the CRS

RSRP range

-140 to -44 dBm, reported in 1 dB steps (0 to 97)

RSSI

total received power in the measured symbols, all sources

RSRQ $N \times \text{RSRP} / \text{RSSI}$, with N = number of resource blocks**RSRQ range**

-19.5 to -3 dB, reported in 0.5 dB steps (0 to 34)

Handover decisions rest on two measurements of the cell reference signal. RSRP is the signal strength of one reference signal resource element. RSRQ relates it to the total received power, so it also reflects load and interference.

TS 36.214 §5.1 · TS 36.133 §9.1

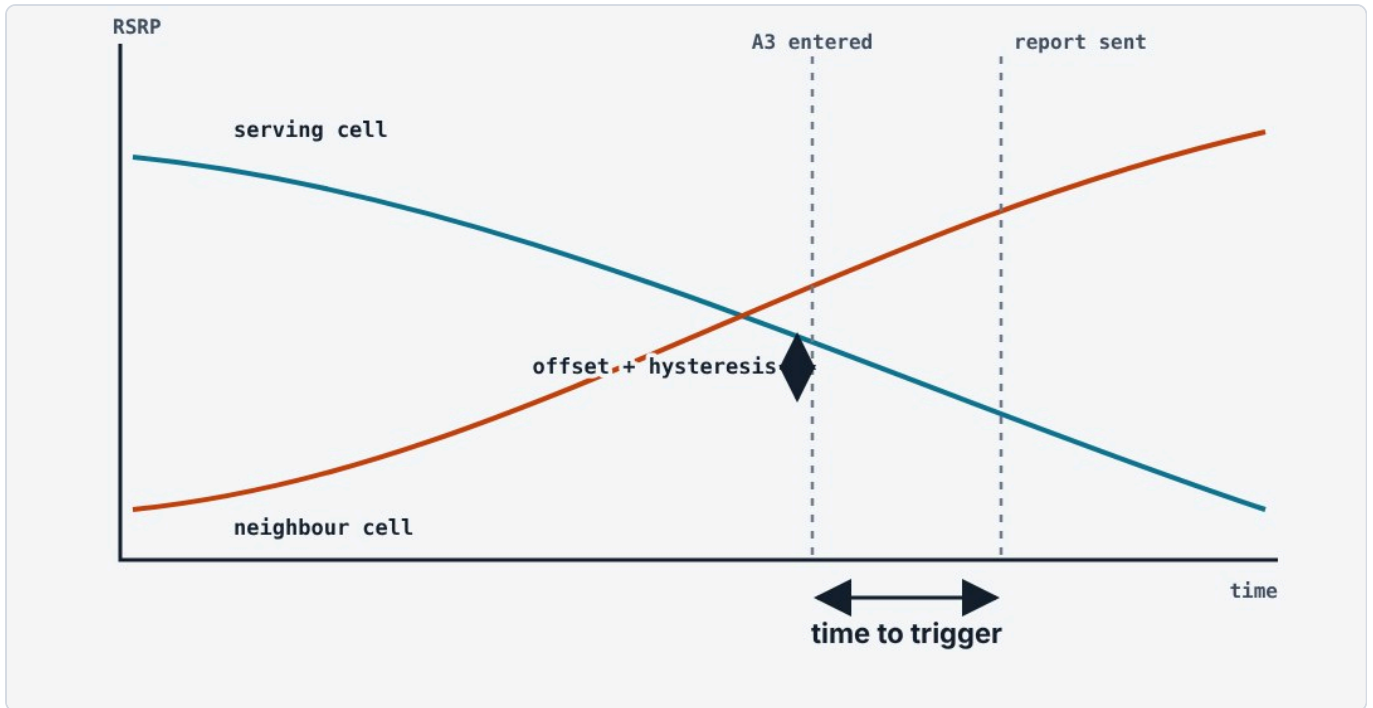
11.2 Measurement events

EVENT	CONDITION
A1	serving cell becomes better than a threshold
A2	serving cell becomes worse than a threshold
A3	neighbour becomes an offset better than the serving cell
A4	neighbour becomes better than a threshold
A5	serving worse than threshold 1 and neighbour better than threshold 2
A6	neighbour becomes an offset better than an SCell (Release 10)
B1	inter-RAT neighbour becomes better than a threshold
B2	serving worse than threshold 1 and inter-RAT better than threshold 2

The eNodeB does not want a constant stream of reports. It configures events, and the UE reports only when one is fulfilled for a set time. A3 is the usual handover trigger. A2 often starts inter-frequency measurements, and B1 or B2 handle other radio technologies.

TS 36.331 §5.5.4

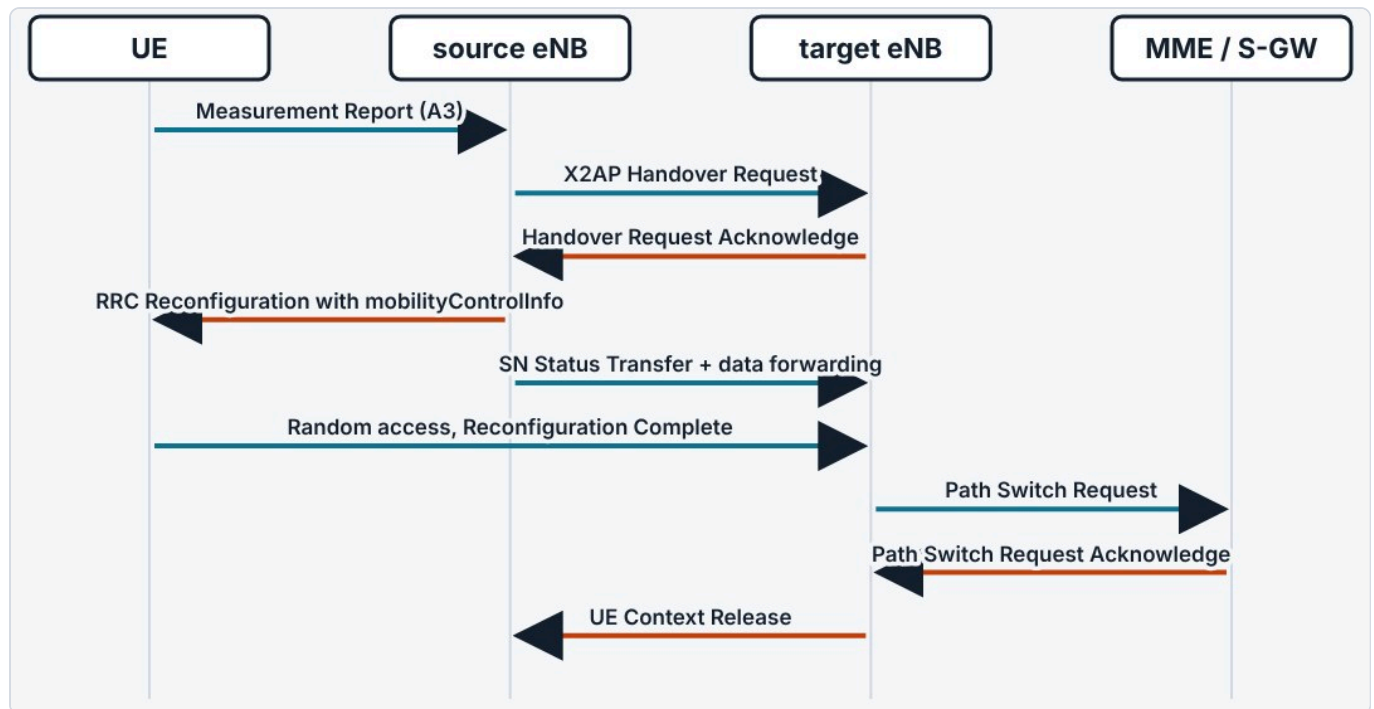
11.3 Event A3 in a picture



As the UE moves, the serving cell fades and the neighbour grows. The event starts when the neighbour exceeds the serving cell by the offset plus the hysteresis. It must stay so for the time to trigger. Only then is the report sent. These three values trade handover speed against ping-pong.

TS 36.331 §5.5.4.4

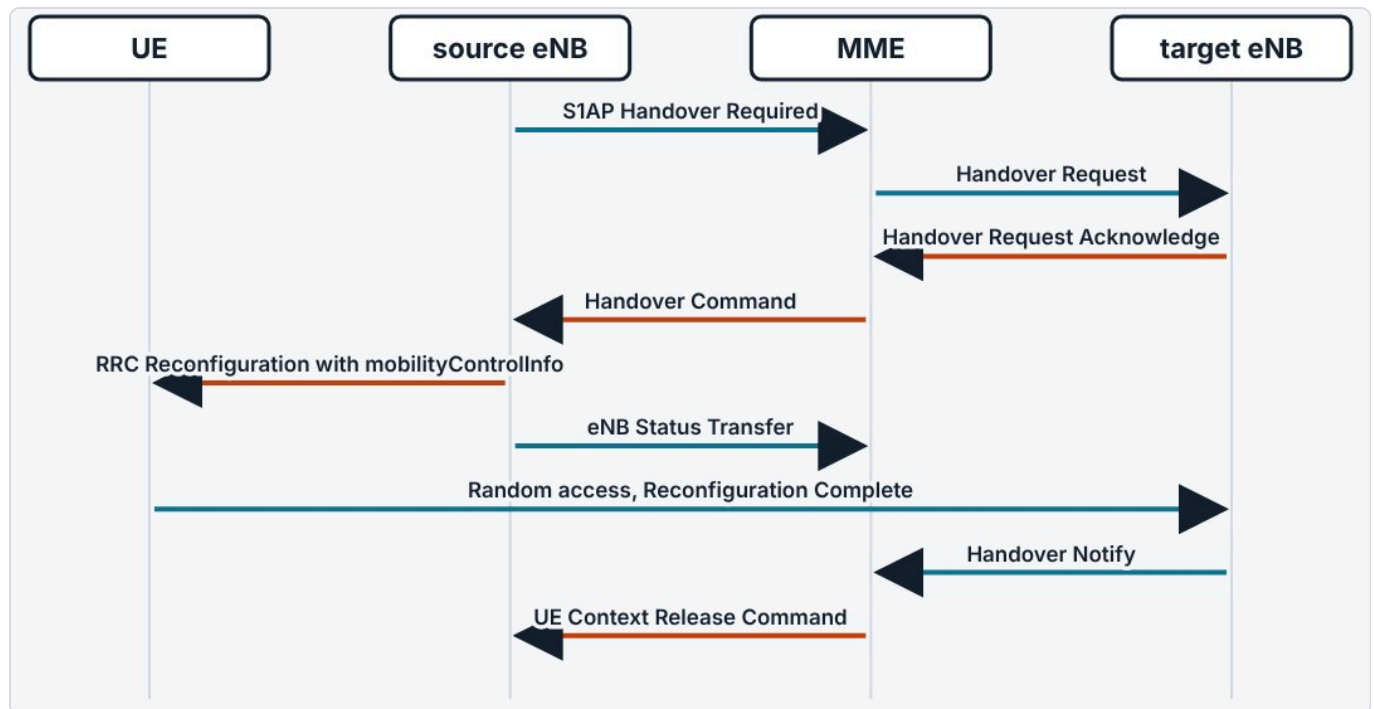
11.4 X2 handover



Most handovers run directly between two eNodeBs. The source asks the target, which reserves resources and returns the command the UE will receive. The UE moves and synchronises with the target by random access. Only then is the core told to switch the downlink path.

TS 36.300 §10.1.2.1 · TS 23.401 §5.5.1.1

11.5 S1 handover



When there is no X2 link, or the MME or S-GW must change, the handover goes through the core. The source sends Handover Required to the MME, which prepares the target. The command returns the same way. The procedure is slower, but the UE sees the same RRC message.

TS 23.401 §5.5.1.2

11.6 No data is lost

SN Status Transfer

Uplink and downlink PDCP
sequence numbers for
RLC AM bearers

Data forwarding

Undelivered downlink packets
go from source to target
over X2 (or via the S-GW)

End marker

The S-GW sends it on the
old path after the switch.
Target then uses new data

Interruption on the radio: a few tens of milliseconds

The UE re-sends unacknowledged PDCP packets after the handover

LTE handover is a hard handover: the UE leaves the source before it reaches the target. Three things keep the data safe. The source tells the target the PDCP sequence numbers, forwards the packets not yet delivered, and the S-GW marks the end of the old path.

TS 36.300 §10.1.2.3

11.7 Leaving LTE

METHOD	HOW IT WORKS
PS handover	target prepared in advance, RRC MobilityFromEUTRACommand
Redirection	RRC Connection Release with redirectedCarrierInfo
Cell change order	to GSM, with optional system information (NACC)
Cell reselection	idle mode, by priorities and thresholds from SIB6 and SIB7
SRVCC	hands a voice call over to the circuit-switched domain

When LTE coverage ends, the network can move the UE to another technology. A packet handover prepares the target first and is the fastest. Redirection simply releases the connection with a target frequency. In idle mode the UE reselects by itself, using the priorities from the SIBs.

TS 36.331 §5.4, §5.3.8 · TS 23.401 §5.5.2

11.8 Three things to remember

1 A3 + time to trigger
The standard handover trigger and its ping-pong brake.

2 Prepare, execute, complete
Target reserves, UE moves, path is switched.

3 Forwarding avoids loss
SN status, forwarded packets and the end marker.

The UE measures, the eNodeB decides. X2 handover is the normal case. The core only switches the path at the end.

TS 36.300 §10.1 · TS 36.331 §5.5

Voice in LTE: CSFB, VoLTE and SRVCC

LTE has no circuit-switched domain. Three mechanisms make phone calls work anyway.

12.1 The problem and three answers

CSFB

- Call runs on 2G or 3G
- UE leaves LTE for the call
- Needs the SGs interface
- No IMS required

VoLTE

- Call runs on LTE as IP
- SIP signalling through IMS
- Dedicated bearer, QCI 1
- Data stays on LTE

SRVCC

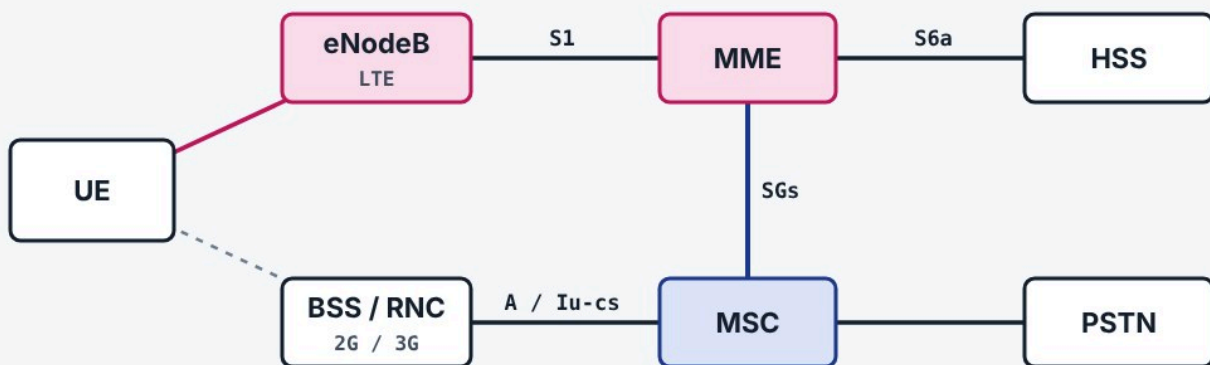
- Handover of a VoLTE call from LTE to 2G or 3G
- Needs the Sv interface
- Call continues in CS

SMS can run over SGs or over IMS in the same way

LTE carries only packets, so a classic phone call has no place in it. The first networks sent the UE back to 2G or 3G for each call: circuit-switched fallback. VoLTE then moved voice onto IP with IMS. SRVCC keeps a VoLTE call alive when the UE leaves LTE coverage.

TS 23.272 · TS 23.228 · TS 23.216

12.2 CSFB: the SGs interface

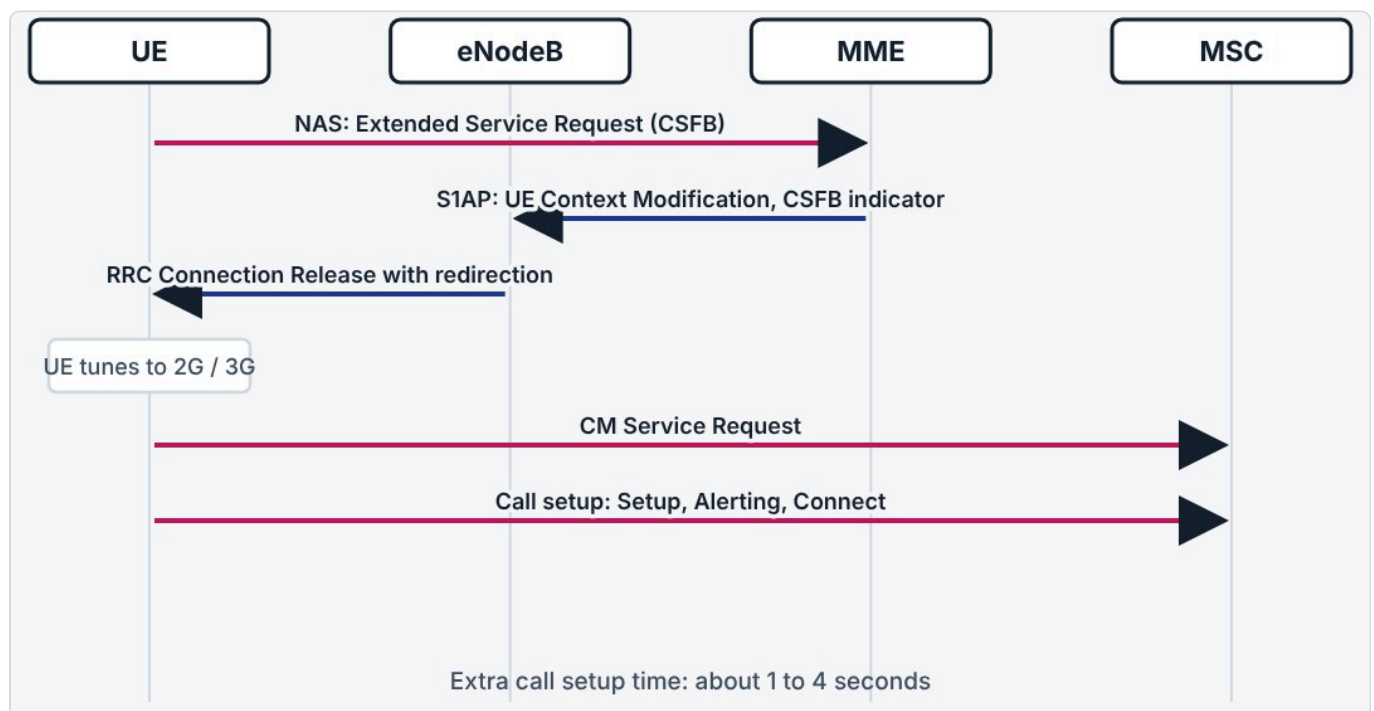


Combined EPS/IMSI attach: registered in the MME and in the MSC
SGs carries paging and SMS between MSC and MME

For fallback the UE does a combined attach. The MME registers it in the MSC over the SGs interface, so the circuit-switched network believes the UE is reachable there. The UE stays on LTE and no 2G or 3G radio is used until a call comes.

TS 23.272 §4.2, §5.2

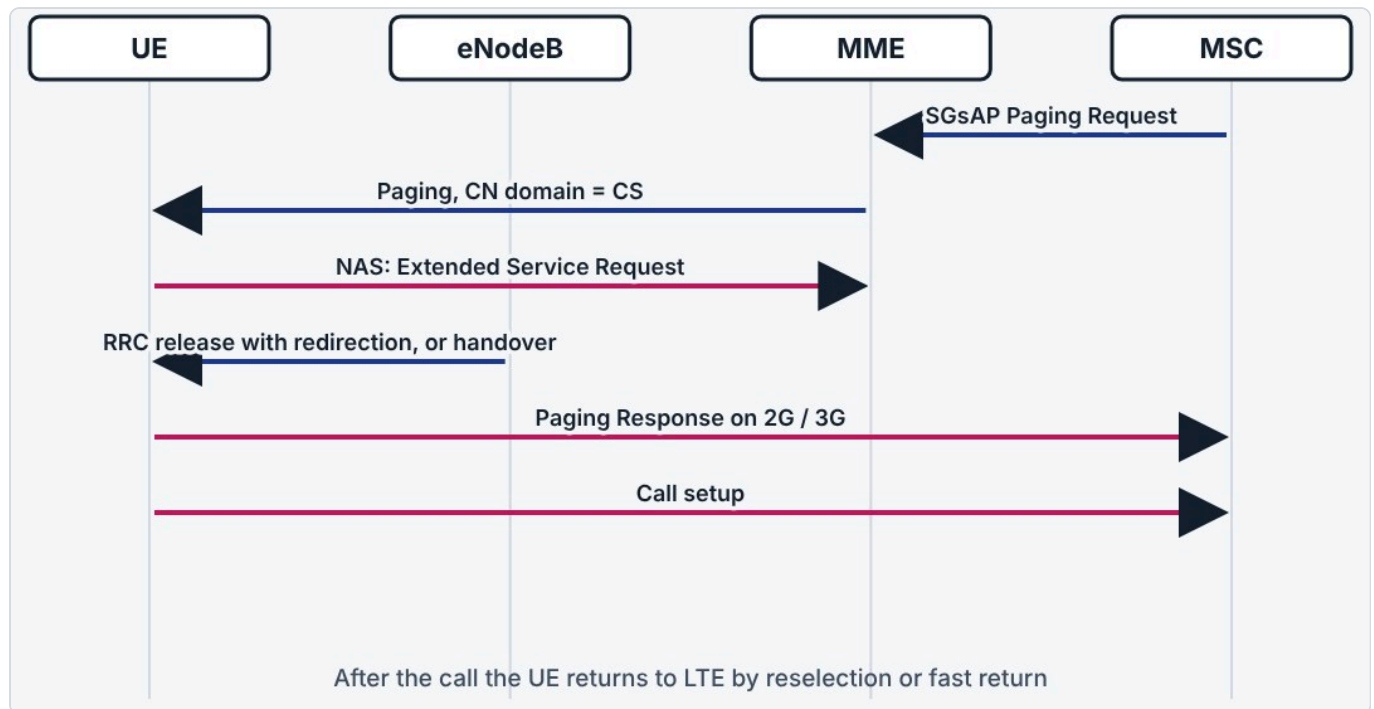
12.3 CSFB: an outgoing call



The user dials. The UE sends an Extended Service Request to the MME, which tells the eNodeB that a fallback is needed. The eNodeB moves the UE to 2G or 3G by redirection or handover. There the UE sets up the call in the usual circuit-switched way.

TS 23.272 §6.2, §6.3

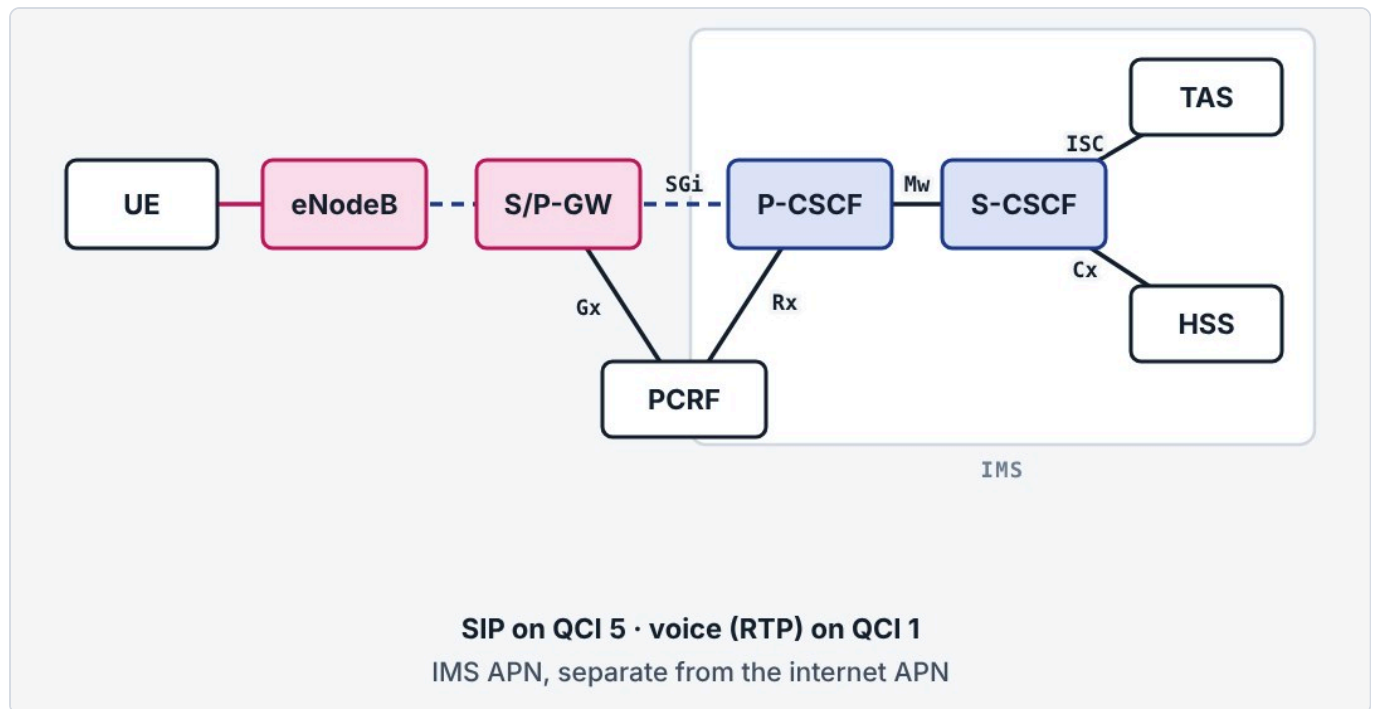
12.4 CSFB: an incoming call and the cost



An incoming call reaches the MSC, which pages over SGs. The MME pages the UE on LTE with the domain set to CS, and the same fallback follows. The price of CSFB is a longer call setup and, on 2G, a break in the data session for the length of the call.

TS 23.272 §7

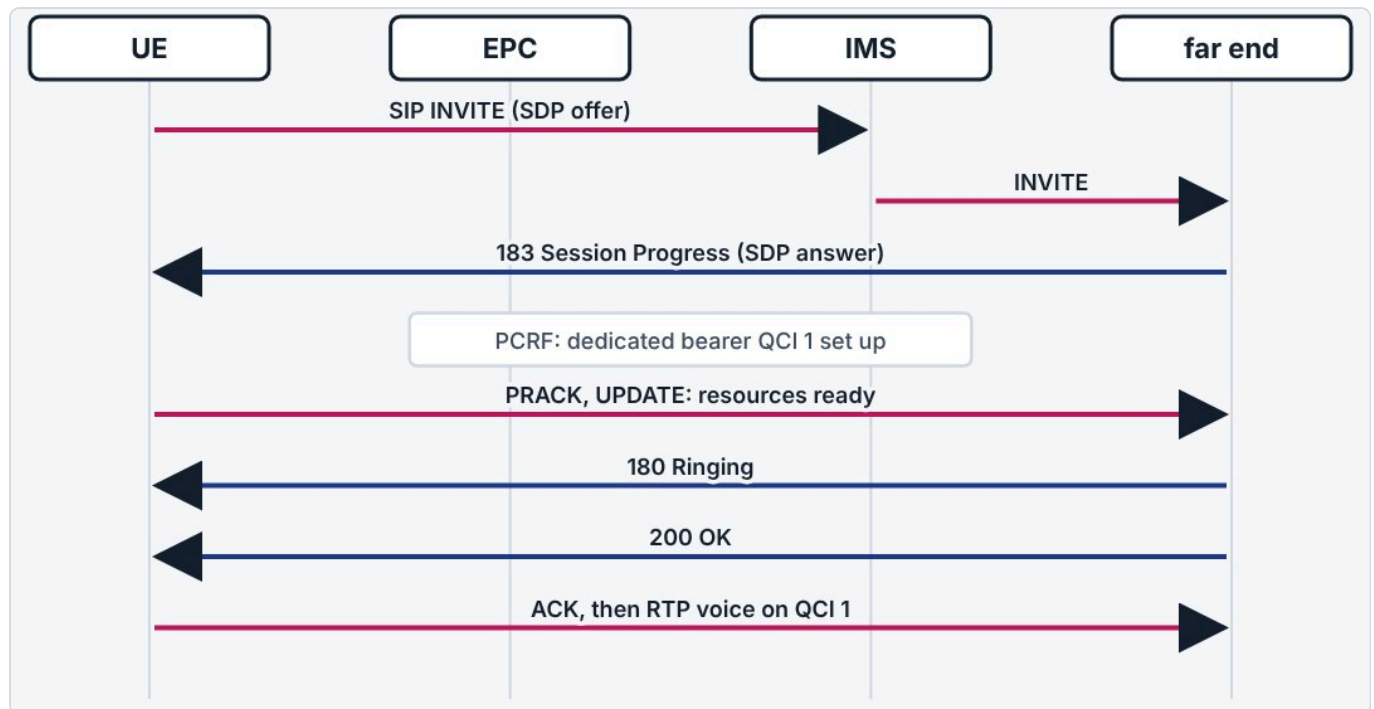
12.5 VoLTE: the architecture



VoLTE treats voice as an IP service. The UE registers in the IMS over a default bearer on the IMS APN. The P-CSCF is its first contact, the S-CSCF controls the session, and the telephony application server provides the services. The PCRF links IMS and EPC, so that the EPC can create the voice bearer.

TS 23.228 §4 · GSMA IR.92

12.6 VoLTE: a call setup



The call is set up with SIP on the QCI 5 bearer. When the codecs are agreed, the P-CSCF tells the PCRF, and the EPC creates a dedicated QCI 1 bearer. The phone rings only after the bearer is ready. Then voice flows as RTP packets.

TS 24.229 · TS 23.228 §5 · TS 23.203

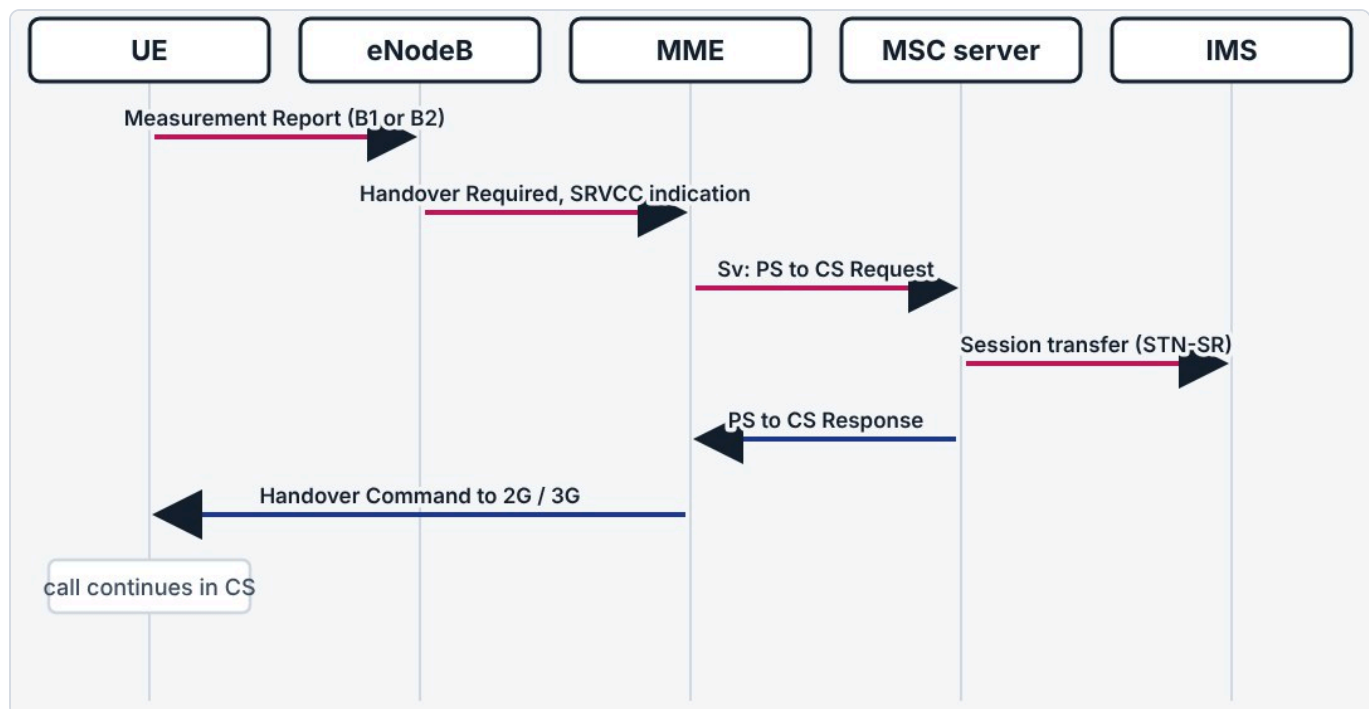
12.7 What the radio does for voice

ROHC	header compression: IP/UDP/RTP 40 bytes down to about 3
RLC UM	no ARQ, a late voice frame is useless
SPS	one grant, repeated every 20 ms
TTI bundling	same packet in 4 consecutive subframes, for the cell edge
DRX	sleep between voice frames
Codec	AMR-WB 12.65 kbit/s (HD voice), EVS in newer devices

A voice bearer is small but delay sensitive. Several radio features serve it. Header compression shrinks the 40 to 60 byte IP headers to a few bytes. RLC unacknowledged mode avoids late retransmissions. Semi-persistent scheduling and TTI bundling save control overhead and extend uplink coverage.

TS 36.323 §5.5 · TS 36.321 §5.10, §5.4.2.1 · GSMA IR.92

12.8 SRVCC: leaving LTE during a call



When a VoLTE call reaches the edge of LTE coverage, the eNodeB starts a handover to 2G or 3G. The MME splits off the voice bearer and asks the MSC server over Sv to prepare a circuit-switched channel. In the IMS the call is re-anchored, so the other party notices only a short gap.

TS 23.216 §6.2 · TS 23.237

12.9 Three things to remember

1 CSFB: SGs, Extended Service Request
Simple, but slower call setup.

2 VoLTE: SIP on QCI 5, voice on QCI 1
Fast setup, HD voice, data stays on LTE.

3 SRVCC: Sv, PS to CS
One-way handover of an ongoing VoLTE call.

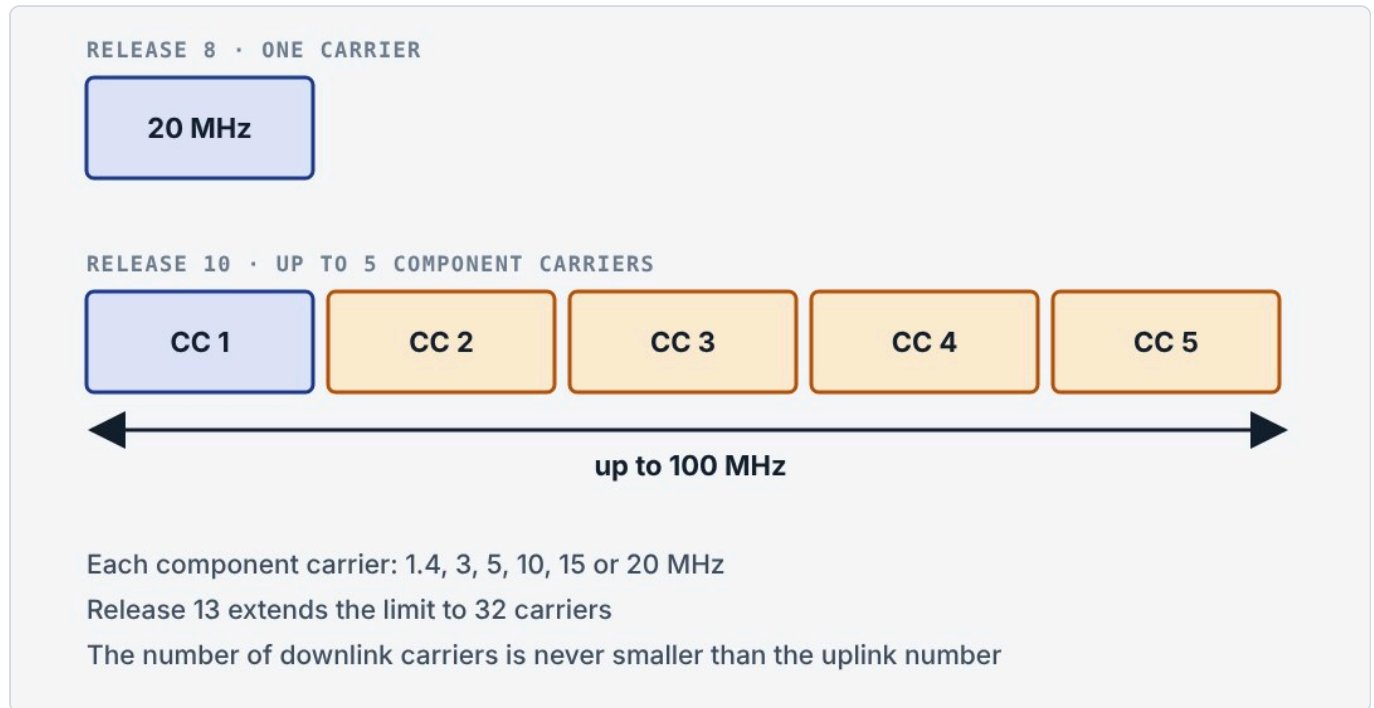
CSFB moves the UE, VoLTE moves the voice. VoLTE needs IMS and a QCI 1 bearer. SRVCC is the safety net at the LTE border.

TS 23.272 · TS 23.228 · TS 23.216

LTE-Advanced: Carrier Aggregation

How Release 10 combines several LTE carriers for one UE: component carriers, PCell and SCell, activation and scheduling.

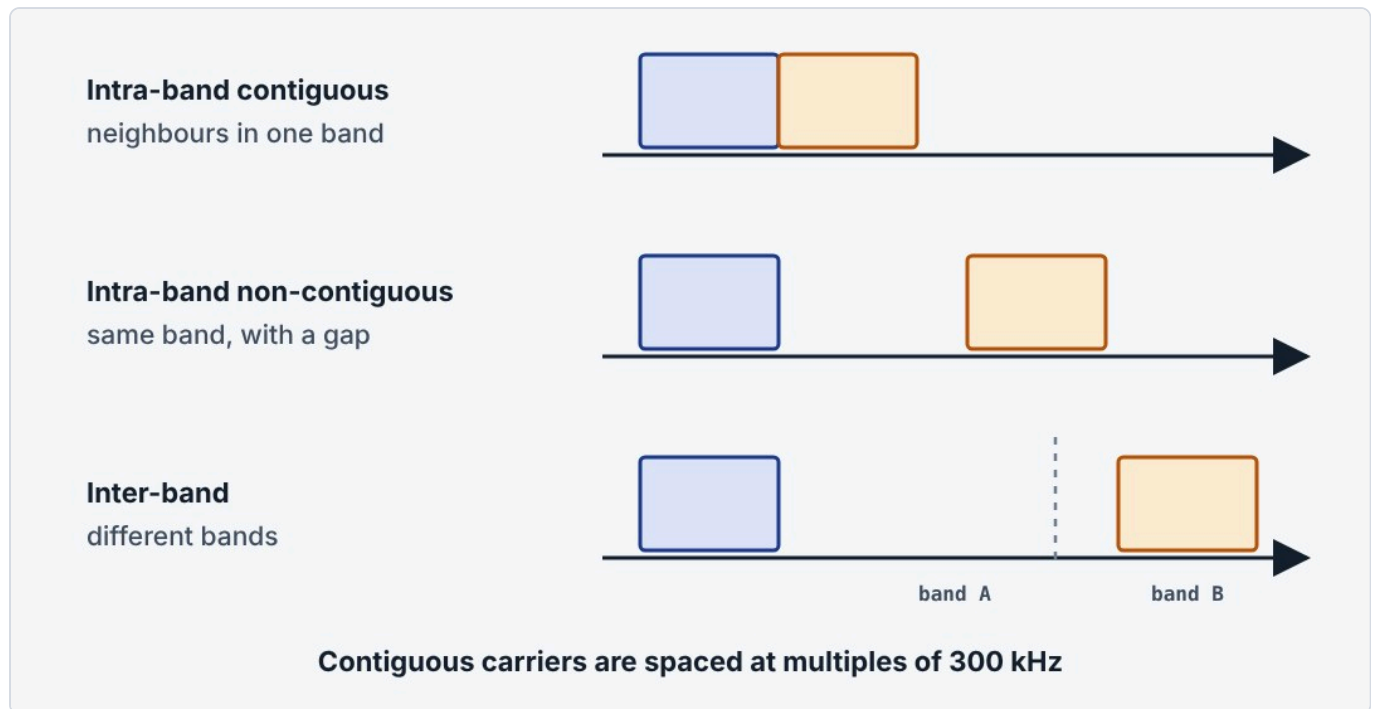
13.1 Why carrier aggregation



One LTE carrier is at most 20 MHz wide. The IMT-Advanced target of 1 Gbit/s needs more. Release 10 lets a UE use up to five carriers at once, 100 MHz in total. Each one is a normal Release 8 carrier, so old UEs can still use any of them alone.

TS 36.300 §5.5 • TR 36.913

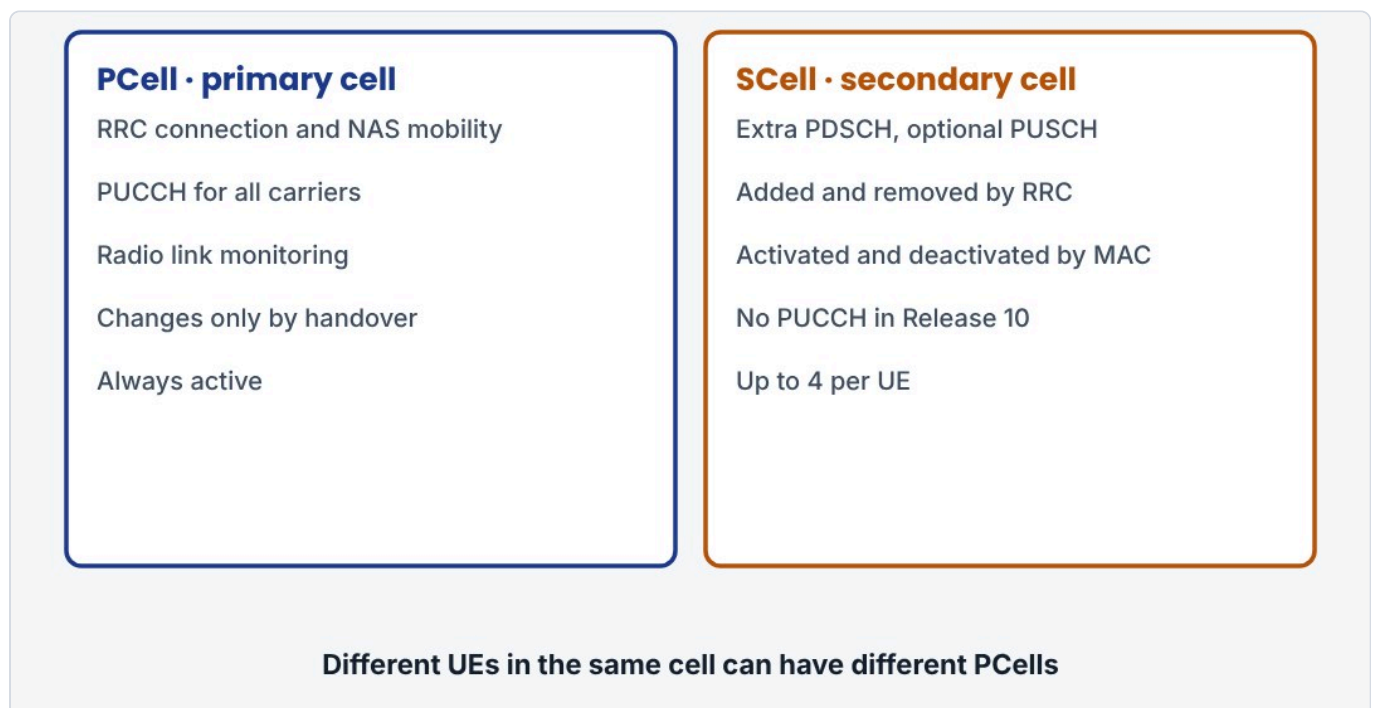
13.2 Three ways to combine carriers



Operators rarely own 100 MHz in one piece. Carrier aggregation therefore works in three ways. Carriers can sit side by side in one band, apart in the same band, or in different bands. The last is the most common in practice and the hardest for the radio hardware.

TS 36.101 §5.5A, §5.6A

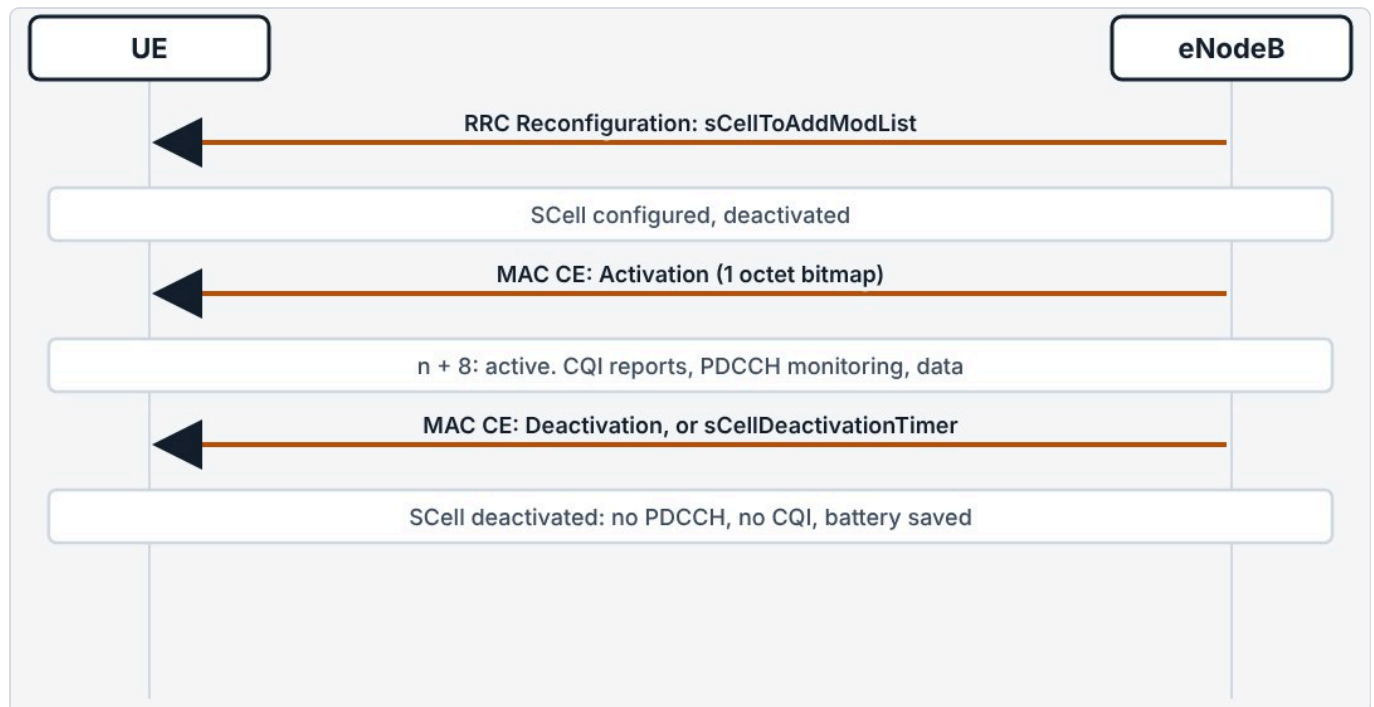
13.3 PCell and SCells



One carrier is the primary cell. It carries the RRC connection, the NAS information, the security input and the PUCCH. Radio link failure is judged on it alone. The other carriers are secondary cells: extra capacity that the network adds and removes as needed.

TS 36.300 §7.5 · TS 36.331 §5.3.10.3b

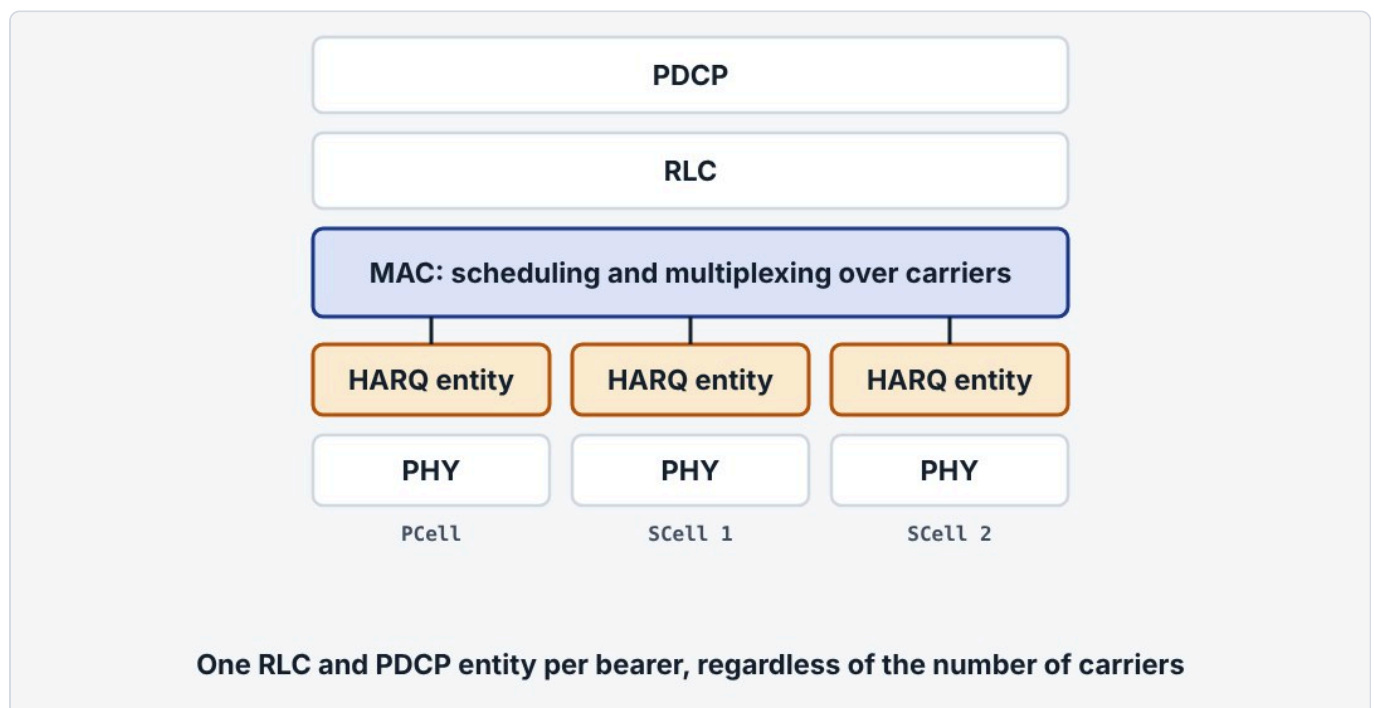
13.4 The life of an SCell



An SCell goes through two steps. RRC adds it with a reconfiguration, and it starts deactivated. A MAC control element then activates it, and eight subframes later the UE can be scheduled on it. A timer or another MAC control element deactivates it when traffic stops.

TS 36.321 §5.13, §6.1.3.8 · TS 36.213 §4.3

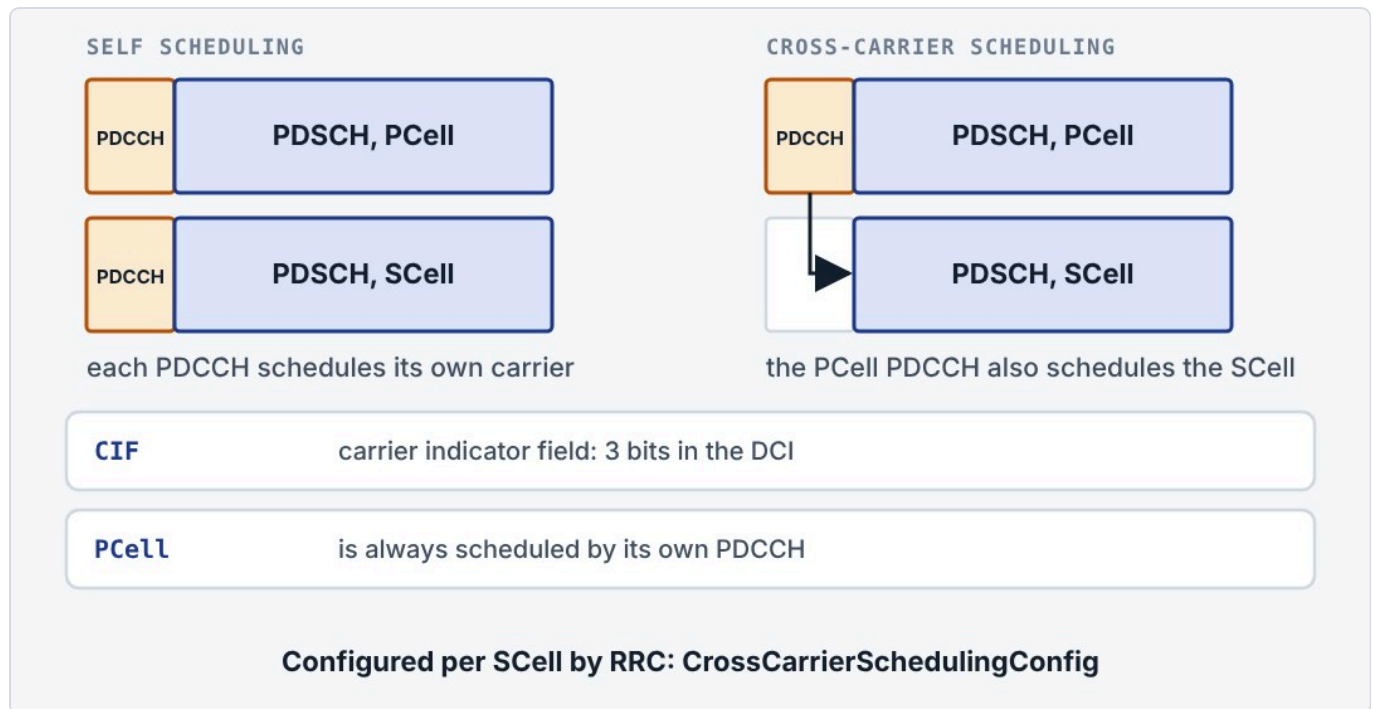
13.5 What changes in the protocol stack



Carrier aggregation is a MAC feature. PDCP and RLC see one data stream and do not know how many carriers exist. MAC splits the data over the carriers. Each carrier has its own HARQ entity, its own transport blocks and its own physical layer.

TS 36.300 §6.4, §7.5

13.6 Self scheduling and cross-carrier scheduling



By default every carrier has its own PDCCH for its own PDSCH. With cross-carrier scheduling one carrier carries the grants for another, marked by a 3-bit carrier indicator field. This helps when the control region of one carrier suffers from interference, as in mixed macro and small cell layouts.

TS 36.213 §9.1.1 · TS 36.212 §5.3.3.1

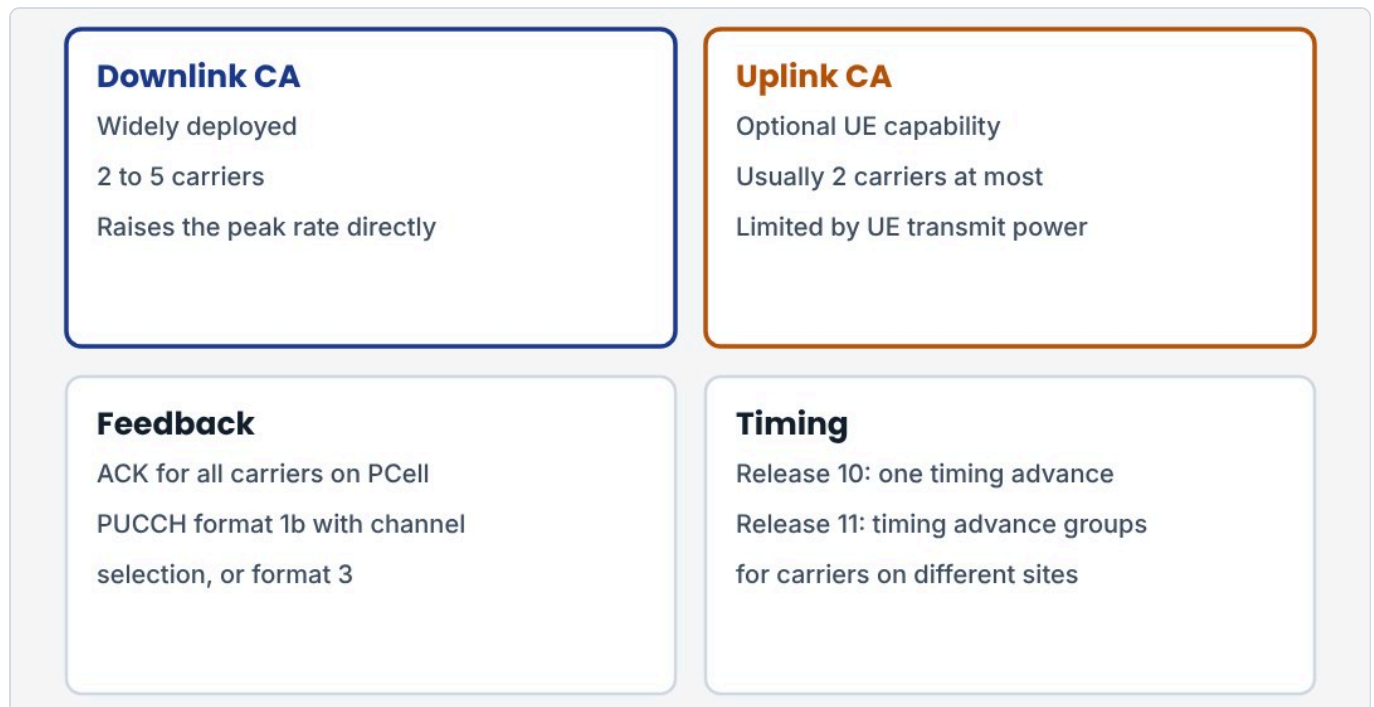
13.7 How combinations are written

NOTATION	MEANING	
CA_3A-7A	inter-band: one carrier in band 3, one in band 7	
CA_3C	intra-band contiguous: two carriers in band 3	
CA_3A-3A	intra-band non-contiguous: two separate carriers in band 3	
CA_1A-3A-7A	three carriers in three bands	
CLASS	AGGREGATED RBs	CARRIERS
A	up to 100	1
B	25 to 100	2
C	100 to 200	2

A band combination is written as CA, then the bands with a bandwidth class letter each. Class A is a single carrier. Class C is two contiguous carriers. The UE lists the combinations it supports in its capabilities, and the network may only configure one of those.

TS 36.101 Table 5.6A-1 · TS 36.331 UE-EUTRA-Capability

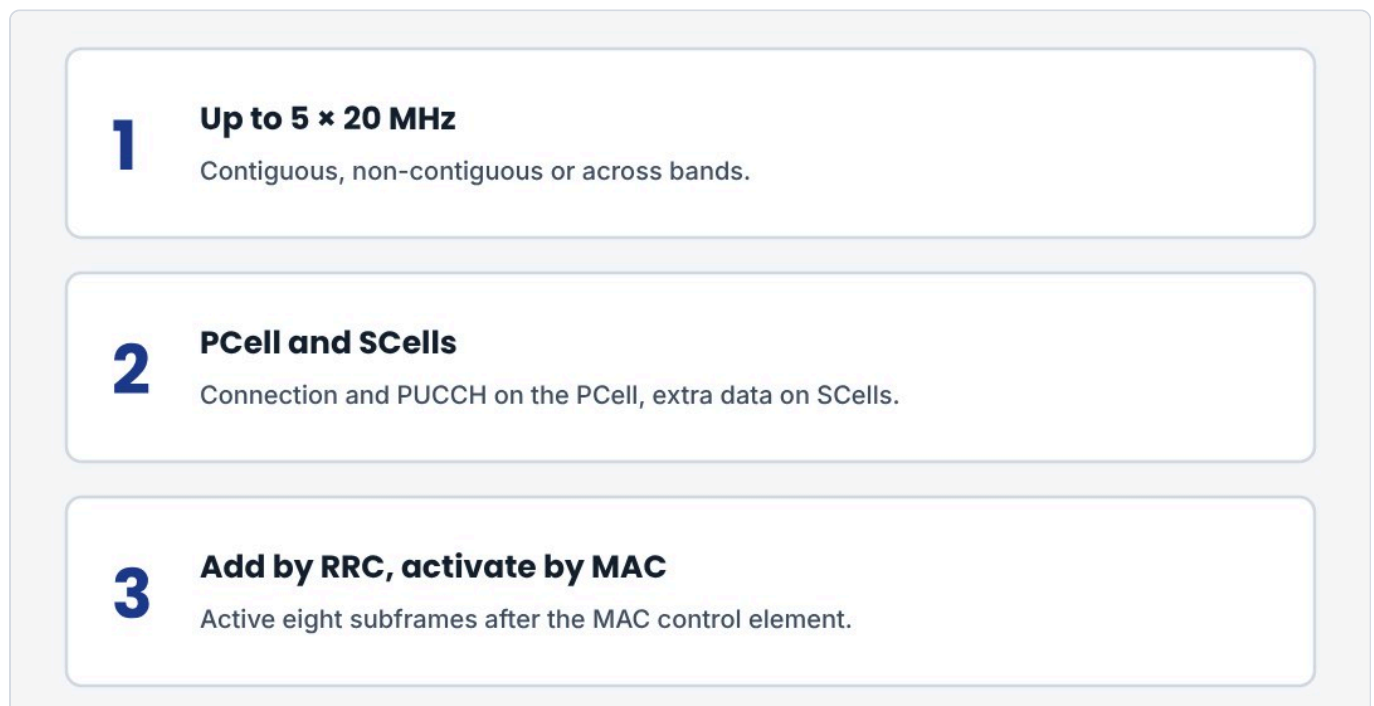
13.8 Uplink aggregation and its limits



Uplink carrier aggregation is optional and less common. The handset has limited power, and two uplink carriers raise the peak power. In Release 10 all feedback goes on the PUCCH of the PCell and all carriers share one timing advance. Release 11 added several timing advance groups.

TS 36.300 §5.5 · TS 36.213 §10.1 · TS 36.321 §5.2

13.9 Three things to remember



Carrier aggregation adds up Release 8 carriers. The PCell holds the connection, SCells add capacity. RRC configures them, MAC switches them on and off.

TS 36.300 §5.5, §7.5 · TS 36.321 §5.13

LTE-Advanced: The Other Features

Beyond carrier aggregation: the release timeline, enhanced MIMO, small cells, CoMP, relays, dual connectivity and LTE-Advanced Pro.

14.1 From LTE to LTE-Advanced Pro

RELEASE	NAME	MAIN ADDITIONS
8	LTE	OFDMA, SC-FDMA, 4×4 MIMO, flat architecture
9	LTE	eMBMS, positioning, dual-layer beamforming
10	LTE-Advanced	carrier aggregation, 8×8 MIMO, eICIC, relays
11	LTE-Advanced	CoMP, ePDCCH, multiple timing advances
12	LTE-Advanced	256QAM downlink, dual connectivity, D2D
13	LTE-Advanced Pro	32 carriers, LAA, NB-IoT, LTE-M, FD-MIMO
14	LTE-Advanced Pro	V2X, 256QAM uplink

LTE is a family of releases. Release 8 is the first LTE. Release 10 is LTE-Advanced, the version that meets the IMT-Advanced requirements for 4G. Release 13 and later carry the name LTE-Advanced Pro and prepare the ground for 5G.

3GPP release descriptions · TR 36.912

14.2 Enhanced MIMO

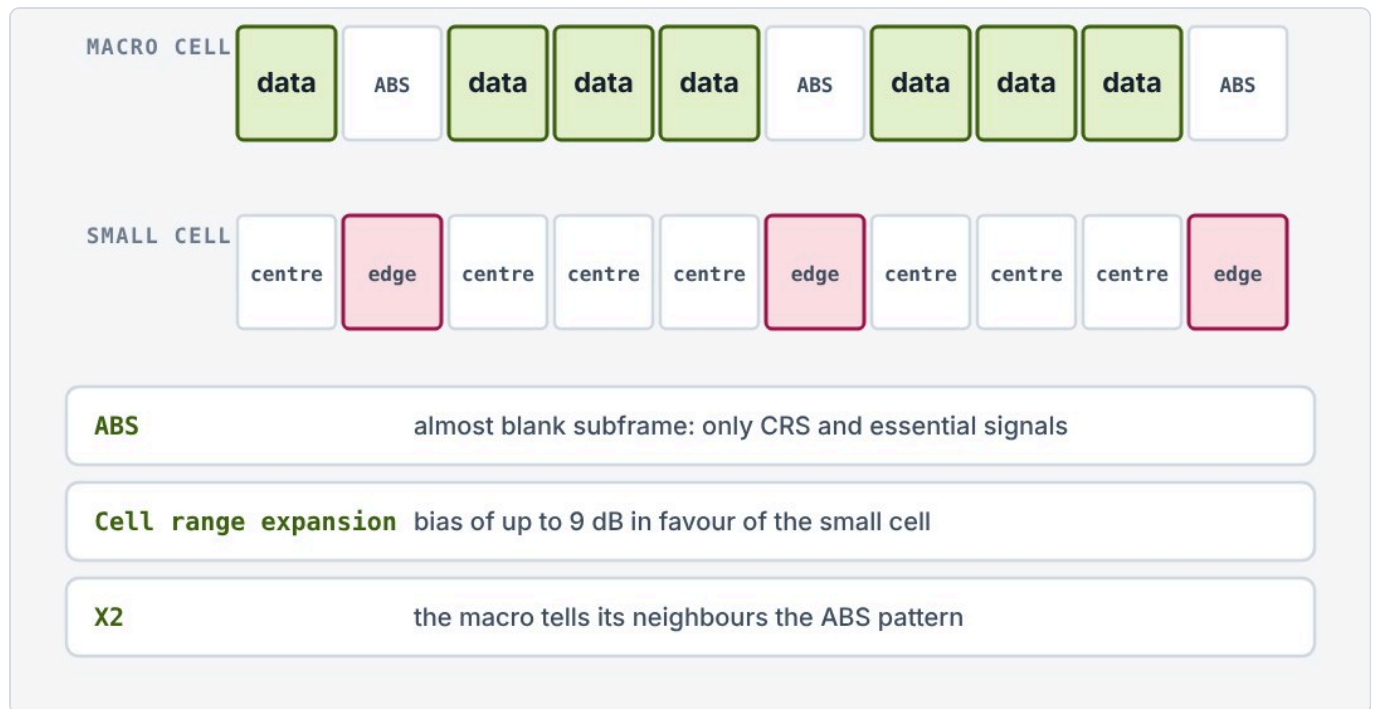
MIMO IN RELEASE 10

Downlink	up to 8 layers, transmission mode 9
Uplink	up to 4 layers, single-user MIMO, new in Release 10
CSI-RS	measurement only, every 5 to 80 ms, up to 8 ports
DM-RS	ports 7 to 14, precoded like the data
Codebook	dual-stage for 8 antennas: wideband W1, subband W2
Multi-user MIMO	improved: dynamic switching between SU and MU

Release 10 raises the downlink to eight layers and brings spatial multiplexing to the uplink with four layers. The key is a new split of reference signals. Sparse CSI-RS let the UE measure eight antenna ports, and UE-specific DM-RS travel with the data through the precoder.

TS 36.211 §6.10.3, §6.10.5 · TS 36.213 §7.1.9

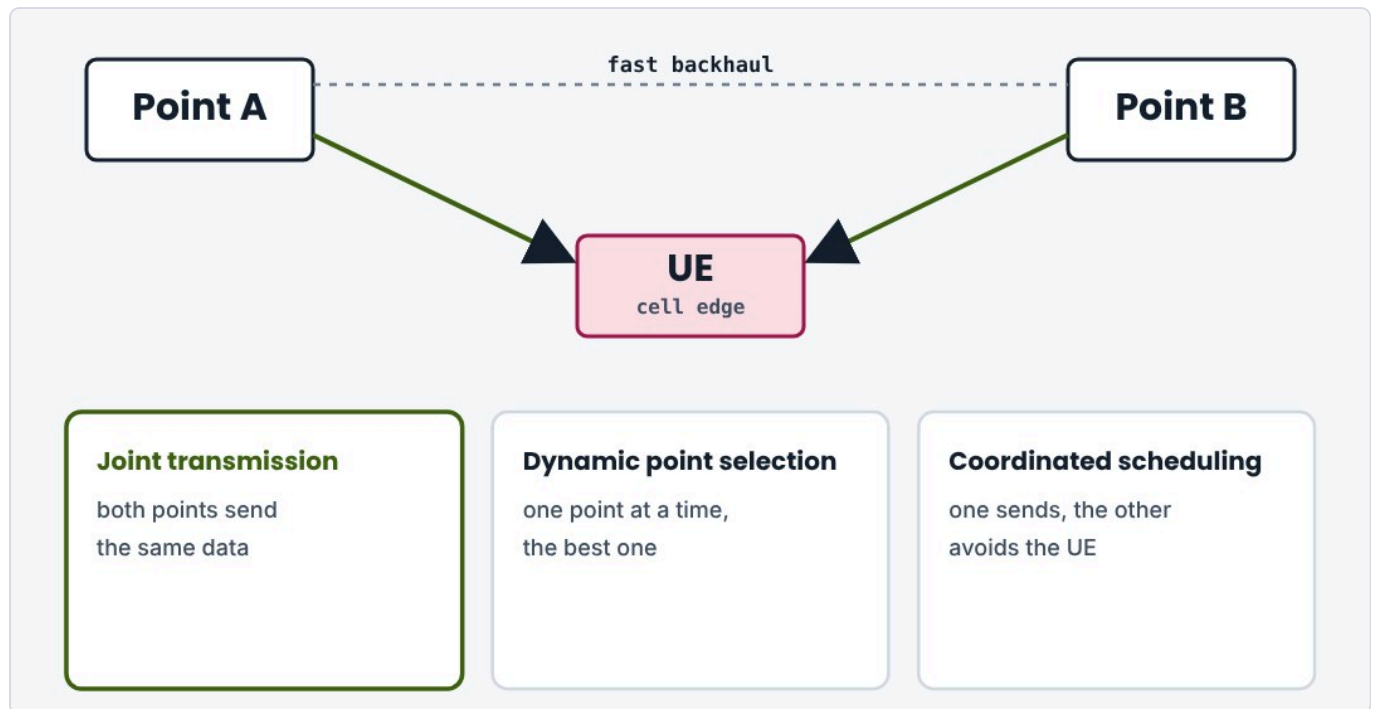
14.3 Small cells and eICIC



Adding low-power small cells inside a macro cell increases capacity. To push more UEs onto the small cell, its range is extended with a bias. Those UEs then suffer from the strong macro signal. The macro therefore leaves some subframes almost blank, and the small cell schedules its edge UEs in exactly those.

TS 36.300 §16.1.5 · TS 36.423 ABS Information

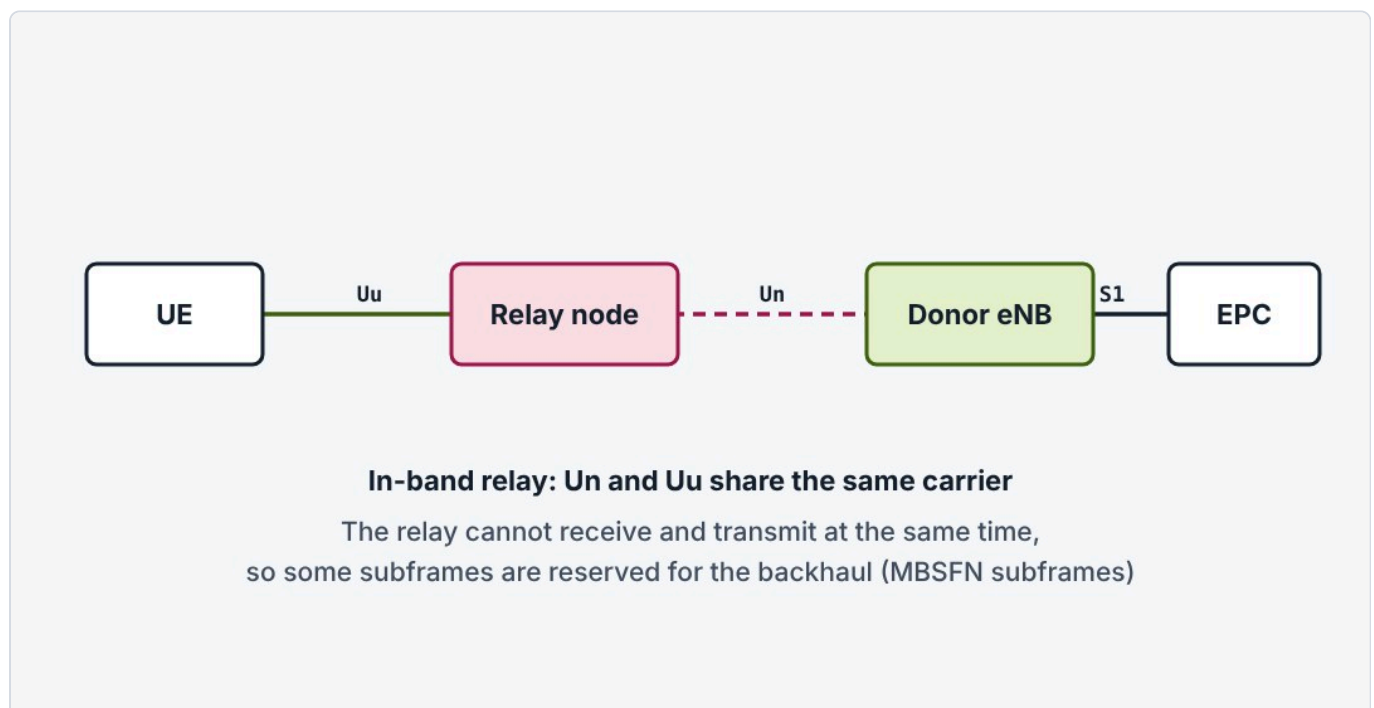
14.4 Coordinated multipoint (CoMP)



At a cell edge the neighbour is the strongest interferer. CoMP turns that into help. Several transmission points coordinate for one UE: they send together, pick the best point from subframe to subframe, or steer their beams and schedules away from each other.

TS 36.300 §5.1.5 · TR 36.819

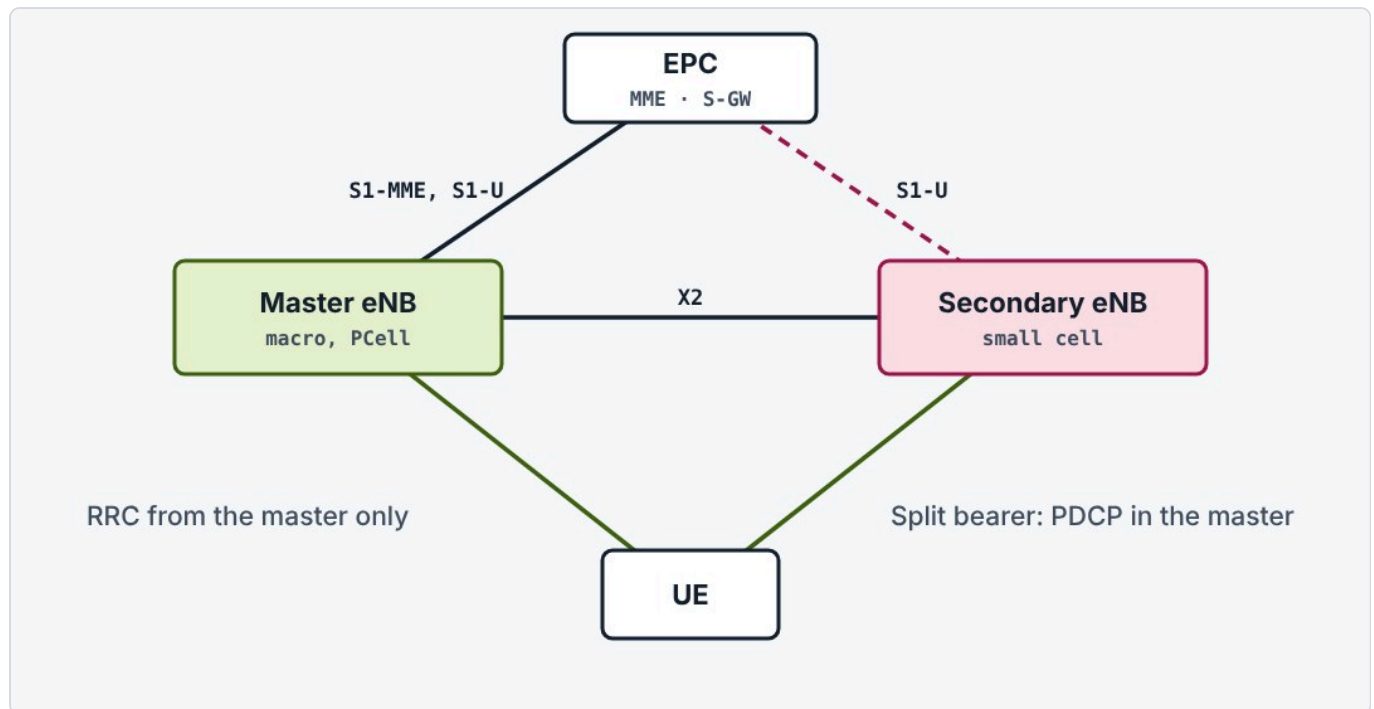
14.5 Relay nodes



A relay node extends coverage where no cable backhaul exists. To UEs it looks like a normal eNodeB. Towards the network it behaves like a UE and connects over the LTE radio to a donor eNodeB. This wireless backhaul link is called Un.

TS 36.300 §4.7 · TS 36.216

14.6 Dual connectivity



Release 12 lets a UE connect to two eNodeBs at once, a master on the macro layer and a secondary on a small cell, linked by a normal X2 backhaul. Each has its own scheduler. A bearer can use one node or be split over both. EN-DC in 5G reuses this design with an NR secondary node.

TS 36.300 §4.9, §6.5 · TS 36.423

14.7 LTE-Advanced Pro

LAA

Licensed assisted access
SCells in unlicensed 5 GHz
Listen before talk

32 carriers

Massive carrier aggregation
PUCCH on an SCell
New PUCCH formats 4, 5

LTE-M (Cat M1)

1.4 MHz, about 1 Mbit/s
Voice and mobility supported
Coverage extension

NB-IoT (Cat NB1)

180 kHz, tens of kbit/s
Up to 20 dB extra coverage
Battery life of years

Release 13 widened LTE in two directions. For more speed it added aggregation of up to 32 carriers and the use of unlicensed 5 GHz spectrum. For the internet of things it added two narrow, low-cost and low-power variants that reach deep indoors.

TS 36.300 §5.5, §5.7 · TS 36.306

14.8 Peak rates across the releases

RELEASE	DOWNLINK PEAK	UPLINK PEAK	ASSUMES
8	300 Mbit/s	75 Mbit/s	20 MHz, 4×4
10	3 Gbit/s	1.5 Gbit/s	100 MHz, 8×8 / 4×4
12	+33 %	–	256QAM downlink
13	more carriers	–	up to 32 carriers

Commercial LTE-Advanced Pro devices: about 1 to 2 Gbit/s
with 5 carriers, 4×4 MIMO and 256QAM

The headline rates grew with every step. They are upper limits: they assume the widest bandwidth, the highest layer count and perfect radio conditions. Real networks combine fewer carriers and layers, but the trend is the same.

TR 36.913 · TS 36.306

14.9 Three things to remember

1

More spectrum

Carrier aggregation, then unlicensed bands.

2

More antennas

8 layers down, 4 up, CSI-RS and DM-RS.

3

Denser networks

Small cells, eICIC, CoMP, dual connectivity.

LTE-Advanced starts with Release 10. Its tools are more spectrum, more antennas and denser cells. Many of them live on in 5G.

TR 36.912 · TS 36.300